

**TECHNISCHE
UNIVERSITÄT
DRESDEN**

<https://tu-dresden.de/cd>

TUD -Script v2.06o based on KOMA -Script

A LATEX bundle for documents in Corporate Design of the Technical University of Dresden User Guide*

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<https://github.com/tud-cd/tudscr>

*Online version

Foreword

This manual describes the tools required to create LATEX documents in corporate design of the Technical University of Dresden¹ developed classes and packages are described. These are based on the KOMA -Script classes that are often used in German-speaking countries and offer a wide range of setting options that go far beyond the possibilities of the LATEX standard classes. In addition, the TUD -Script bundle offers further selection options, particularly relating to the document layout.

It should be noted that the classes described here allow a deviation from the corporate design of the Technical University of Dresden, as this is certainly worth discussing, especially from a typographical point of view. In principle, it is possible to switch back to the layout of the KOMA -Script classes using the appropriate settings . Unless the user specifically uses these options, all of the corporate design specifications are implemented by default.

This manual contains a description of the options and commands that can be used *in addition* to the KOMA -Script classes . Basic knowledge of using LATEX is assumed. If you do not have this knowledge, reading the LATEX 2 ϵ short description `l2kurz.pdf`² highly recommended. For an in-depth introduction to LATEX, Nicola LC Talbot provides very detailed tutorials for LATEX novices³ and dissertations⁴ for free use. In addition , Part II of this manual offers minimal examples and somewhat more detailed tutorials.

Furthermore, every user should read the LATEX 2 ϵ sin register `l2tabu.pdf`⁵ to avoid typical errors. DanteFAQ⁶ provides answers to frequently asked questions If the user is inexperienced in using KOMA-Script , it is worth taking a look at the corresponding manual⁷ highly recommended, if not essential.

The current status of the classes and packages from the TUD script bundle in version v2.06o has been thoroughly tested to the best of our knowledge and belief. However, we cannot guarantee that errors will not occur. If a problem occurs, this should be reported in the TUD LATEX forum, as should incompatibilities with other packages. or in the GitHub repository `tud-cd/tudscr` under

<http://latex.wcms-file3.tu-dresden.de/phpBB3/> <https://github.com/tud-cd/tudscr/issues>

For quick and successful troubleshooting, please provide a minimum example⁸ provided. I may respond late or not at all to requests without this. *Questions, criticism and suggestions for improvement* - both regarding the bundle itself and the documentation - are also welcome in the forum mentioned . Since this bundle was created and is also maintained in my free time, I ask for your understanding if I am not able to respond immediately and/or correct any errors.

Falk Hanisch

Dresden, 9 August 2022

¹<https://tu-dresden.de/cd/>

²<http://mirrors.ctan.org/info/lshort/german/l2kurz.pdf>

³<http://www.dickimaw-books.com/latex/novices/>

⁴<http://www.dickimaw-books.com/latex/thesis/>

⁵<http://mirrors.ctan.org/info/l2tabu/german/l2tabu.pdf>

⁶<http://projekte.dante.de/DanteFAQ/>

WebHome ⁷[http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/](http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf)

`scrguide.pdf` ⁸<http://www.komascript.de/minimalbeispiel>

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Chapter 1

Introduction

To use TUD -Script v2.06o, both KOMA -Script v3.36 or later and the corporate design font **Open Sans** from the `opensans` package are required.

In addition, the LATEX distribution used must provide additional packages, which are listed in subsection 8.1. If one of the distributions TEX Live, MacTEX and MiKTEX in the latest versions is used, this should not be a problem.

However, if a distribution is used that does not provide TUD-Script in version v2.06o and an update is not possible, Appendix A should be consulted.

The TUD -Script bundle is primarily intended for the creation of scientific texts and papers and is *currently* not intended to replace the original classes⁹ from Klaus Bergmann's template package, but rather to complement them. An implementation of the corporate design for the beamer class as well as for letters and business correspondence based on the KOMA -Script letter class `scrletter` has unfortunately not yet been implemented with TUD -Script, but is planned for the long term. However, several styles already exist for the beamer class in the `tudmathposter` bundle. This is in the GitHub repository `tud-cd`¹⁰ To create letters with the TUD script classes, the `scrletter` package could be used.

1.1 Components and use of the TUD script bundle

v2.01
v2.02
v2.04
v2.06

The TUD script bundle is distributed via the Comprehensive TeX Archive Network (CTAN)¹¹ and can be used by LATEX distributions such as TEX Live, MacTEX or MiKTEX. It provides the three main classes `tudscrbook`, `tudscrrcpt` and `tudscrartcl` and the class `tudscrposter`, which are presented in Chapter 2 and Chapter 3 respectively. The `tudscrsupervisor` package - documented in Chapter 4 - can be used together with the TUD -Script classes to create assignments, notices or reports for student work. There are also independent packages, which are described in Chapter 5. To use the TUD -Script bundle - in addition to KOMA -Script at least in version v3.36 and the LATEX packages listed in subsection 8.1 - since version v2.06 only the **Open Sans** font is required, which is provided by the `opensans` package. A local user installation of the fonts - as in previous versions - is not necessary. Only if you want to use the old fonts **Univers** and **DIN Bold** do they have to be installed. Further information can be found in subsection A.4. To activate these, refer to `tudscrver=2.05` and `cdoldfont`.

Note on Windows Two different

LATEX distributions can be used for Windows. The advantages of TEX Live compared to MiKTEX are that it is maintained by several authors and that all updates are available earlier via CTAN. In addition to LATEX 2_ε, a *Perl interpreter* and *Ghostscript* are also included, which simplifies or improves the ad hoc use of some packages such as glossaries. For MiKTEX, these programs may have to be installed manually. In contrast, MiKTEX does not require the annual reinstallation that is necessary for TEX Live.

⁹`tudbook`, `tudbeamer`, `tudletter`, `tudfax`, `tudhaus`, `tudform`

¹⁰<https://github.com/tud-cd/tud-cd/>

¹¹<http://www.ctan.org/>

Note on Linux and OS X The

installation of one of the LATEX distributions TEX Live or MacTEX should be done directly via the packages offered (<https://tug.org/texlive/> or <https://tug.org/mactex/>) and not via **apt-get install** . This ensures that the latest version of the respective distribution is used.

1.2 How to use this manual

All newly defined options, environments and commands of the classes and packages of the TUD script bundle are listed and described in the manual. At the end of the document there are several indexes that should make it easier to look up or find previously unknown commands or options.

The options described below can – like most of the settings for KOMA -Script – in the syntax of the keyval package as key-value pairs when choosing the Document class must be specified: **`\documentclass[<key>=<value>]{<class>}`**

Furthermore, the KOMA -Script classes offer the possibility of late option selection. This means that options can not only be specified directly when loading as so-called class options, but can also be changed within the document after the class has been loaded. The KOMA -Script classes provide two commands for this.

With **`\KOMAoptions{<options list>}`** you can assign exactly one value to any number of keys, **`\KOMAoption{<option>}{<value list>}`** allows you to set several values for exactly one key at the same time. The commands **`\TUDoptions{<options list>}`** and **`\TUDoption{<option>}{<value list>}`** are defined in an equivalent way for the *additional* options provided by TUD -Script . This allows you to change the behavior of options in the document - even locally within a group.

The default setting for each option is indicated with "Default value: **`<value>`**" in its description . Some of these default settings are not always the same but are set depending on the user settings and options used. These conditional default settings are indicated with "Default value: **`<value>`** |Condition: **`<conditional value>`**" . If a key is used by the user *without* a value assignment, a predefined default value is set - if available - which is indicated in the description of all options by the *italicized* writing within the value list. In most cases, the default value of a key is **true**, which therefore corresponds to the specification **`<key>=true`**.

When the user explicitly assigns a value to a key, both simple and conditional default settings are always overwritten. The new commands and environments of the classes, which were introduced in addition to the options, are explained in the same style.

1.3 Quick start

The manual is divided into three parts. Part I contains the documentation of TUD -Script . All new options, environments and commands that go beyond the functionality of KOMA -Script are explained here. Part II contains simple minimal examples to demonstrate the basic handling and functionality of TUD -Script . It also offers detailed and documented tutorials, especially for LATEX newbies . The tutorial treatise.pdf in particular is worth a look if you want to write a scientific paper with LATEX 2_ε . Finally, various

Packages are presented that are not only used for the TUD script bundle itself but also for other LATEX 2 ϵ classes and could therefore be of interest to any user. In addition, some tips and tricks for working with LATEX are described here to solve small or large problems.

The classes tudscrbook, tudscrreprt and tudscrartcl are wrapper classes of the well-known KOMA -Script classes scrbook, scrreprt and scrartcl and can simply be used instead of them . Documents based on these can easily be converted to the corporate design of the Technical University of Dresden by changing the document class . If you have any questions about layout, fonts or similar, it is always advisable to take another look at this manual.



part One

The TUD script bundle

This is the main part of the TUD -Script bundle. Here the user will find all available options, environments and commands that go beyond the functionality of KOMA -Script.

Chapter 2

The main classes

The three new main classes

tudscrbook

tudscrreprt

tudscrartcl

introduced, which are based on the KOMA -Script classes and basically support all of their provided options - for example ***parskip=<method>*** for the paragraph settings or ***BCOR=<length value>*** for specifying the binding correction - as well as environments and commands . In addition to the KOMA -Script classes, further packages are required, which are listed in subsection 8.1 and are loaded by TUD-Script.

Please refer again to the user documentation of KOMA -Script¹² Many of the commands and options described below refer to the setting options presented therein. The adaptations and extensions of the KOMA -Script classes to the corporate design and the newly defined, modified or extended commands and options are explained below.

\TUDOptions{<options list>}
\TUDOption{<option>}{<value list>}

With these two commands it is possible to change the value of the options for most of the new class or package options after the class or package has been loaded . They are the equivalent of the two KOMA -Script commands ***\KOMAOPTIONS*** and ***\KOMAOPTION***.

You can optionally use the ***\TUDOptions*** command to change the values of several options at once, separating them with commas. In the options list, each option to be changed must be specified in the form ***<option>=<value>*** . Most of the TUD script options also have a default or default value. If these are used in the form ***<option>*** without specifying a value, the respective option is simply assigned the specified default value.

Some options can have several complementary values at the same time. For these, it is possible to use ***\TUDOption*** to assign a series of values to a single option , one after the other, whereby these values must be separated from one another by commas in the value list.

These two commands allow you to change the behavior of one or more options in the document if necessary . If these commands are used in an environment or group, the settings made remain locally limited within that environment.

2.1 The fonts of the corporate design

The corporate design of the Technical University of Dresden specifies the use of the **Open Sans** font family for body text, which is implemented in the standard configuration by TUD-Script . However, since the use of serif fonts is recommended for typographic reasons in longer texts, there is the option of not loading the fonts originally intended for the corporate design and instead using the LATEX standard fonts or another font package. The explanations for this can be found in section 2.1.1. To use the fonts with LATEX 2_ε classes that are not part of the TUD-Script bundle, the tudscrfonts package can be loaded.

¹²<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

v2.02
v2.04
v2.06

The corporate design does not specify any fonts for the mathematics typesetting. This is particularly unacceptable for mathematical treatises as well as engineering and scientific documents. In the mathematics mode, the Latin glyphs are therefore used with the help of the `mathastext` package and the Greek letters of **Open Sans**. The `mdsymbol` package can be used for additional mathematical symbols. Loading .

This setting can be deactivated, which allows the standard fonts or, if necessary, those of an additional package to be used for mathematical typesetting. The relevant settings are explained in section 2.1.2. Additional information on typographically clean mathematical typesetting can also be found in chapter 7.

The font formats Type1 and OpenType v2.02

The TUD script bundle supports the use of the corporate design fonts in both Type1 and OpenType format. Both formats are provided by the `opensans` package, whereby the fonts are loaded in Type1 format regardless of the typesetting system used. The use of the fonts in OpenType format is not mandatory even when using LuaLATEX or XL ATEX. However, this is easily possible by simply including the `fontspec` package.

However, it is not always possible to do without the Type1 fonts. On the one hand, they are required for compiling a document using the classic process path via `latex > dvips > ps2pdf` - as is necessary, for example, for creating graphics with `pstricks`. On the other hand, the glyphs for the math typesetting are taken from the Type1 fonts. Unfortunately, using the OpenType fonts for the math typesetting is not possible without considerable effort, which is why TUD-Script does not use them - particularly because of the lack of added value.

2.1.1 Fonts for typesetting

v2.02
v2.03
v2.05
v2.06

`cdfont=<setting>` Default: `true,normalbold,liningfigures`

With the `cdfont` option, the user can make all central font settings for the TUD script classes. This applies to the fonts for both the body text and the math typesetting. In particular, different combinations can be used

of normal and bold font weight for **Open Sans**. At the beginning of the document, these are decisive for the definition of the math fonts. The font weight in the characteristic horizontal bar of the header can also be set with this option.

`cdfont=false/off/no` The LATEX standard fonts are used and not the house font of the corporate design. The user can use any font package.¹³

`cdfont=true/on/yes/light/lightfont/noheavyfont` (default value)

The **Open Sans** font is used in the style of the corporate design. The headings of the top levels of the structure up to and including `\subsubsection` are set in either bold or extra-bold font (see **`headings=<setting>`** on page 25), those below¹⁴ always use the bold font style. This uses Open Sans Regular as standard, but can be made heavier with **`cdfont=heavybold`**. Open Sans Light is used in the body text.

¹³For the use of classic LATEX fonts, the `lmodern` package is highly recommended.

¹⁴`\paragraph` and `\subparagraph`

cdfont=heavy/heavyfont

This setting differs from **cdfont=true** in that the font weight of the house font is increased. The body text of the document is set in Open Sans Regular.

The bold font style is normally set to **Open Sans Semi-Bold**.

The thickness of the bold fonts can be adjusted using the following settings.

v2.05 **cdfont=normalbold** Open Sans

Regular is used for bold fonts, or **Open Sans Semi-Bold** for heavier fonts (**cdfont=heavy**). This is the default setting.

v2.05 **cdfont=heavybold/ultrabold**

The bold fonts are highlighted more strongly. **Open Sans Semi-Bold** or **Open Sans Bold** with increased font weight (**cdfont=heavy**) is used.

The following table illustrates the possibilities for combining the settings:

cdfont=normalbold		cdfont=heavybold	
lmdseries	lbseries	lmdseries	lbseries
cdfont=true Open Sans Light	OpenSans Regular	Open Sans Light	Open Sans Semi-Bold
cdfont=heavy Open Sans Regular	Open Sans Semi-Bold	Open Sans Regular	Open Sans Bold

Furthermore, the shape of the numbers can be customized, choosing between capital numbers and old-style numbers.

v2.06 **cdfont=liningfigures/normalfigures** With this

setting, equal-width/equidistant/proportional uppercase figures or majuscule figures (1234567890) are used. Old-style figures or lowercase figures can be set explicitly with **loldstylenums{<digit(s)>}**. This behavior is activated by default. v2.06 **cdfont=oldstylefigures/osf/oldnumbers**

For the body text, non-proportional old style numerals or lowercase numerals (0123456789) are used as standard.

In addition, the option **cdfont=<setting>** serves as an extended interface for the user to make additional settings for font usage, for which separate options are normally provided. These options are briefly explained below and reference is made to the actual underlying option.

The math fonts used can be influenced with the following values:

cdfont=nomath/nocdmath

This setting deactivates **Open Sans** for mathematical typesetting. The LATEX standard fonts are used and any font package can be used for the math mode, see **cdmath=false**. **cdfont=math/cdmath** Open **Sans** is used for the math fonts, see **cdmath=true**. **cdfont=upgreek/**

uprightgreek

Greek uppercase and lowercase letters are upright, see **slantedgreek=false**.

cdfont=slgreek/slantedgreek The

Greek glyphs are output in italics, see **slantedgreek=true**.

v2.06 **cdfont=texgreek/standardgreek**

The LATEX standard is used, in which Greek capitals are displayed upright and Greek capitals are displayed italic, see **slantedgreek=standard**.

v2.02

For the TUD script classes, the tudheadings page styles provide the following setting options for the font in the crossbar:

cdfont=nohead/noheadfont

If the house font is deactivated for the body text, it can also be deactivated in the horizontal bar, see

cdhead=false. cdfont=head/lighthead/

lightfonthead/noheavyfonthead

For the horizontal bar of the header, **Open Sans** is used in normal font weight, regardless of whether the house font is used in the body text, see **cdhead=true. cdfont=heavyhead/heavyfonthead**

For the horizontal bar, **Open Sans** is

used in increased weight, regardless of the document font, see **cdhead=heavy.**

v2.06

ttfont=<setting> Default: **true**

If the **cdfont** option is activated, you can also set the **typewriter font** used, which can be selected from one of the packages `roboto-mono` or `lmodern` is used and in particular the selected settings for the font weight of the **cdfont** option are observed. It is not necessary for the user to load the respective package directly.

ttfont=false/off/no

There is no adjustment of the typewriter font (**ttfamily**), possible changes can be made by the user.

ttfont=true/on/yes/roboto/roboto-mono (default

value)

If the package `roboto-mono` installed, the typewriter font defined therein will be used. This is sans serif and is available in a variety of different font styles, which is why it can be ideally combined with **Open Sans**.

ttfont=lmodern/lmtt/lm

The typewriter font from `lmodern` is explicitly activated.

Markups in text Regardless

of which font family is used in the document, the fonts of the corporate design can be used at any time within the document using one of the text switches or text commands listed here. A text switch has a global effect on the document, such as `\bfseries`, if it is not limited locally by a group or environment. With a text command, however, the font is only used for the argument specified below, such as `\textbf{<Text>}`.

v2.04

\cdfont{} \textcd{}
{<text>}

These two commands are used to specifically activate a font of the corporate design in weight and style. Here, **\cdfont** corresponds to a text switch and immediately changes the font used in the current scope to ****, whereas **\textcd** functions as a text command and sets the **<text>** given in the second argument to **** without changing the document font itself.

To select the font, the name of the desired font must be specified in the first argument. Possible values can be found in the table below. The prefix **Open Sans** for the **** in the argument is optional. Neither spaces nor the appropriate upper and lower case are necessary. To select the font weight, the name **Light/Regular/Semi-Bold/Bold/Extra-Bold** is sufficient.

Instead of the suffix ' **Italic** ', the use of **Oblique** or **Slanted** as an alias for the italic fonts possible.

Font/Description	Switch	Text command
Open Sans Light	<code>\cfontln \textcdln{<Text>}</code>	
Open Sans Regular	<code>\cfontnrn \textcdnrn{<Text>}</code>	
Open Sans Semi-Bold	<code>\cfontnsn \textcdnsn{<Text>}</code>	
Open Sans Bold	<code>\cfontnbn \textcdnbn{<Text>}</code>	
Open Sans Extra-Bold	<code>\cfontnxn \textcdnxn{<Text>}</code>	
<i>Open Sans Light Italic</i>	<code>\cfontlli \textcdlli{<Text>}</code>	
<i>Open Sans Regular Italic</i>	<code>\cfontlri \textcdlri{<Text>}</code>	
<i>Open Sans Semi-Bold Italic</i>	<code>\cfontlsi \textcdlsi{<Text>}</code>	
<i>Open Sans Bold Italic</i>	<code>\cfontlbi \textcdlbi{<Text>}</code>	
<i>Open Sans Extra-Bold Italic</i>	<code>\cfontlxi \textcdlxi{<Text>}</code>	

As an alternative to `\cfont` and `\textcd`, explicit text switches are also used for each font style. and commands are provided where the argument for the name of the font These are listed in the second and third columns of the table.

2.1.2 Fonts for mathematics typesetting

v2.03

`cdmath=<on-off value>` Default: **true** | **`cdfont=false: false`**

This option is used to customize the math fonts. If this is activated, the House font appropriate glyphs used in math mode. Normal and bold font weight are displayed at *the beginning of the document* depending on the setting active at that time for the fonts of the body text (see **`cdfont`**) . Bold math fonts can be switched off in the Document can be switched with **`\boldmath`** , an adjustment of the settings is possible by **`\TUOptions`** . Valid values for the **`cdmath`** option are:

`cdmath=false/off/no`

The normal LATEX serif fonts or the fonts of any usable packages for math typesetting.

`cdmath=true/on/yes` (default value)

In math mode, **Open Sans** is available for both Latin and Greek letters are used.

v2.06

`slantedgreek=<on-off value>` Default: **true**

The option changes the default slant of uppercase Greek letters in math mode when using standard Greek letter commands.

`slantedgreek=false/off/no`

The Greek glyphs are all set upright in mathematical typesetting.

`slantedgreek=true/on/yes` (slanted value)

All Greek letters are italicized in math mode.

`slantedgreek=standard/latex`

The edition, with upright majuscules and italic minuscules, corresponds to the Greek Letters follow the LATEX standard for mathematical typesetting.

<code>\Gamma</code>
<code>\Delta</code>
<code>\Theta</code>
<code>\Lambda</code>
<code>\Xi</code>
<code>\Pi</code>
<code>\Sigma</code>
<code>\Upsilon</code>
<code>\Phi</code>
<code>\Psi</code>
<code>\Omega</code>
<code>\alpha</code>
<code>\beta</code>
<code>\gamma</code>
<code>\delta</code>
<code>\epsilon</code>
<code>\varepsilon</code>
<code>\zeta</code>
<code>\eta</code>
<code>\theta</code>
<code>\vartheta</code>
<code>\iota</code>
<code>\kappa</code>
<code>\lambda</code>
<code>\mu</code>
<code>\nu</code>
<code>\xi</code>
<code>\pi</code>
<code>\varpi</code>
<code>\rho</code>
<code>\varrho</code>
<code>\sigma</code>
<code>\varsigma</code>
<code>\tau</code>
<code>\upsilon</code>
<code>\phi</code>
<code>\varphi</code>
<code>\chi</code>
<code>\psi</code>
<code>\omega</code>

Command (upright) symbol		Command (italic) Symbol	
<code>\upGamma</code>	Γ	<code>\litGamma</code>	Γ
<code>\upDelta</code>	Δ	<code>\litDelta</code>	Δ
<code>\upTheta</code>	Θ	<code>\litTheta</code>	Θ
<code>\upLambda</code>	Λ	<code>\litLambda</code>	Λ
<code>\upXi</code>	Ξ	<code>\litXi</code>	Ξ
<code>\upPi</code>	Π	<code>\litPi</code>	Π
<code>\upSigma</code>	Σ	<code>\litSigma</code>	Σ
<code>\upUpsilon</code>	Υ	<code>\litUpsilon</code>	Υ
<code>\upPhi</code>	Φ	<code>\litPhi</code>	Φ
<code>\upPsi</code>	Ψ	<code>\litPsi</code>	Ψ
<code>\upOmega</code>	Ω	<code>\litOmega</code>	Ω
<code>\upalpha</code>	α	<code>\litalpha</code>	α
<code>\upbeta</code>	β	<code>\litbeta</code>	β
<code>\upgamma</code>	γ	<code>\litgamma</code>	γ
<code>\updelta</code>	δ	<code>\litdelta</code>	δ
<code>\upepsilon</code>	ϵ	<code>\litepsilon</code>	ϵ
<code>\upvarepsilon</code>	ϵ	<code>\litvarepsilon</code>	ϵ
<code>\upzeta</code>	ζ	<code>\litzeta</code>	ζ
<code>\upeta</code>	η	<code>\liteta</code>	η
<code>\uptheta</code>	θ	<code>\littheta</code>	θ
<code>\upvartheta</code>	ϑ	<code>\litvartheta</code>	ϑ
<code>\upiota</code>	ι	<code>\litiota</code>	ι
<code>\upkappa</code>	κ	<code>\litkappa</code>	κ
<code>\uplambda</code>	λ	<code>\litlambda</code>	λ
<code>\upmu</code>	μ	<code>\litmu</code>	μ
<code>\upnu</code>	ν	<code>\litnu</code>	ν
<code>\upxi</code>	ξ	<code>\litxi</code>	ξ
<code>\uppi</code>	π	<code>\litpi</code>	π
<code>\upvarpi</code>	ϖ	<code>\litvarpi</code>	ϖ
<code>\uprho</code>	ρ		ρ
<code>\upvarrho</code>	ϱ	<code>\litvarrho</code>	ϱ
<code>\upsigma</code>	σ	<code>\litsigma</code>	σ
<code>\upvarsigma</code>	ς	<code>\litvarsigma</code>	ς
<code>\uptau</code>	τ		τ
<code>\upupsilon</code>	υ	<code>\litupsilon</code>	υ
<code>\upphi</code>	ϕ	<code>\litvarphi</code>	ϕ
<code>\upvarphi</code>	φ	<code>\litchi</code>	ϕ
<code>\upchi</code>	χ		χ
<code>\uppsi</code>	ψ	<code>\litpsi</code>	ψ
<code>\upomega</code>	ω	<code>\litomega</code>	ω

Regardless of the **slantedgreek** option, both italic and upright Greek glyphs can be used directly in math mode with these commands. This is useful to distinguish between *italic variables* and *upright constants*. Furthermore, instead of `\up{<Glyphe>}` or `\lit{<Glyphe>}`, the prefix `\other{<Glyphe>}` can also be used. This means that – depending on the option **slantedgreek** – the complementary Font position of the specified glyph.

Additional notes on the mathematics theorem

In some cases, additional symbols are required for the math set. The `amssymb` package is often used for this purpose. This provides numerous additional fonts for the LATEX standard fonts. However, in combination with **Open Sans**, these are not an ideal solution from a typographic point of view. The symbols from the `mdsymbol` package fit much better with the font family in question, which is why this package should be preferred if the fonts of the corporate design are used for mathematical typesetting.

Further information on good typographical mathematics typesetting can also be found in Chapter 7.

2.1.3 Vertical space depending on font size

In the TUD script classes, several lengths are normally dependent on the font size selected for the document - in contrast to KOMA script. This has the great advantage that if the font size is changed, the following commands or lengths do not have to be adjusted separately by the user in order to continue to be used effectively.

The adaptation to the font size is carried out using the expandable lengths `\bigskipamount`, `\medskipamount` and `\smallskipamount`, which are used by the commands `\bigskip`, `\medskip` and `\smallskip` respectively for inserting vertical spacing, as well as the two lengths `\abovecaptionskip` and `\belowcaptionskip` for the spacing between a floating object - for example an image or a table - and its description set with `\caption`. In addition, the length `\columnsep` is set sensibly in relation to the font size as a measure for the horizontal spacing of the individual text columns in a two- or multi-column layout, as can be created with the `multicol` package, for example.

The font size used can be specified by the user using the KOMA -Script option Attention! **`fontsize=<font size>`** .

Please note that this should always be specified as a class option. Further information on choosing the appropriate font size can also be found in section 3.1.1.

v2.05

`relspacing=<on-off value>` Default: **true**

With the option **`relspacing=<on-off value>`**, the previously described font size dependency can be adjusted both for vertical space between two paragraphs or for labels and for the horizontal distance between the text columns in a multi-column layout . **`relspacing=false/off/no/absolute/standard`** The lengths mentioned are not adjusted; suitable

values should be selected by the user when changing the font size.

`relspacing=true/on/yes/relative/fontsize` (relative value)

Depending on the selected font size, the previously mentioned lengths are determined automatically.

2.2 The layout of the corporate design

2.2.1 The type area

v2.03

v2.05

`cdgeometry=<setting>` Default: `true,restricted` | `cd=false: false`

This option is responsible for the division or calculation of the type area. The size of the page margins is fixed in the corporate design and is adhered to by default by the TUD -Script classes. However, the page margins can be adjusted to enable reasonable double-sided typesetting, for example.¹⁵ It is also possible to fall back on the KOMA -Script standard behavior and have the type area calculated by the `typearea` package. Here, the class option **`DIV=<setting>`** in particular has a significant influence on the type area.

v2.05

More information can be found in the documentation of KOMA -Script¹⁶ to be found. In addition, the user can set the type area (almost) as he likes with **`cdgeometry=custom`**. **`cdgeometry=false/off/no`** The type area is calculated via `typearea`,

the corporate design specifications regarding the page margins are ignored. **`cdgeometry=true/on/yes/asymmetric/cd`** (default value)

The page margins are defined in the asymmetrical style of the corporate design and are also used for double-sided typesetting (class option **`twoside=true`**).¹⁷ **`cdgeometry=symmetric/centred/centered`**

The type area is centered on the page in single-sided and double-sided typesetting.¹⁸

`cdgeometry=twoside/balanced`

This setting activates double-sided typesetting (**`twoside=true`**) and changes the type area so that the margins of the inner pages are narrower than those of the outer pages.¹⁹ *However, the type area created in this way is disadvantageous because the logo of the Technical University of Dresden is placed very close to the inner page margin of the document and consequently moves very far to the edge of the page on right-hand or odd-numbered pages.* This problem can in principle be solved - with `tudscrbook` and `tudscrreprt` - by activating the KOMA -Script option **`open=left`** so that titles, parts and chapters always start on a left-hand page, although this is rather unusual from a typographical point of view.

Danger!

v2.05 The following values can be used to set the variant in which the type area is displayed after the Corporate design of the Technical University of Dresden should be created.

`cdgeometry=restricted` The

type area corresponds to the explicit specifications of the corporate design.

`cdgeometry=adapted`

According to the corporate design manual, for paper formats between DIN A6 and DIN A4, "in the interest of maximum uniformity, the dimensional relationships are 'frozen' over a larger format range." However, this can lead to type areas that are difficult to use. With this setting, the fixation can be deactivated and the equivalent type area activated for the DIN A5 format, for example. This setting has no effect on formats outside the fixed range.

¹⁵The inner margin should be narrower than the outer one ¹⁶<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

¹⁷left: 30 mm, right: 20 mm, top: 20 mm, bottom: 30 mm ¹⁸left: 25 mm, right: 25 mm, top: 20 mm, bottom: 30 mm ¹⁹inside: 20 mm, outside: 30 mm, top: 20 mm, bottom: 30 mm

cdgeometry=calculated The

type area is scaled based on the reference dimensions for the DIN A4 format for the set paper format . The actually defined discrete dimensions for different design heights are ignored. v2.05 Since there are often very restrictive - although usually completely

arbitrary - specifications for the page margins, it is also possible to set these manually as far as possible.

cdgeometry=custom

The geometry package is primarily responsible for determining the page margins .

The user can define the type area himself using the two commands **lgeometry** and **lnewgeometry** provided by the package. For information on how to use the commands, the geometry documentation should be used.

v2.03

extrabottommargin=<height> Default: **0pt**

This option can be used to adjust the size of the bottom margin when the corporate design type area is used (see **cdgeometry**). In particular, if third-party logos are used in the footer area of pages in **the tudheadings** style either with **tfootlogo** and these have been enlarged beyond the standard height using the optional argument or via **tfootlogoheight** , or if oversized content has been specified using **tfootcontent** , the bottom margin may be a little too small. The **extrabottommargin** option enlarges the footer area using positive values, while negative values reduce it accordingly .

As an alternative to the **extrabottommargin** option, the setting **cdfoot=<height>** can also be used with a length value. For both options, it does not matter whether a LATEX length, a TEX distance or a TEX extension is used as the length when assigning the value.

Binding correction

In connection with the page margins or, more accurately, the type area, the KOMA script option **BCOR=<length value>** provided by the typearea package should be mentioned. This allows a binding margin or a binding correction to be taken into account when calculating the type area . The TUD script classes also pass the value specified with this option on to the geometry package , so that the user can use the **BCOR** option regardless of the type area design (**cdgeometry**) . For example, a binding correction of 5 mm can be specified with the *class option* **BCOR=5mm** .

Of course, an adjustment of the binding correction *always* results in a change in the available width of the text area and therefore necessarily leads to an adjustment of the type area. However, since the binding correction should be chosen depending on the height of the book block, which can ultimately only be determined when the finished document is printed , this must be estimated at the beginning.

Example: As a rule of thumb, the required binding correction should be approximately half the height of the book block. The height of the book block depends on the number of pages and the density of the paper used. If normal paper with a density of 80 g/m² is used, 100 sheets correspond to a height of approximately 10 mm, and for 100 g/m², approximately 12 mm. Accordingly, a binding correction of **BCOR=5mm** or **BCOR=6mm** would be selected in this example.

Header and footer in conjunction with the type area

Since the corporate design does not specify how the header and footer should be designed in a scientific paper, the user has a certain amount of freedom. Ideally, the `scrpage` package belonging to KOMA-Script should be used for this.

The `typearea` documentation also explains when headers and footers should be added to the margin or the body of the text when constructing the type area. This should be taken into account when creating your own headers and footers. The setting for this is made using the two KOMA-Script options **`headinclude=<on-off value>`** and **`footinclude=<on-off value>`**. These can be used independently of the setting for type area design using the **`cdgeometry`** option.

2.2.2 The design of the title, cover page, parts, chapters, etc.

If the option **`cd=<setting>`** described below is activated, some special pages are set in a concise style with the logo of the Technical University of Dresden and the corresponding header with a horizontal bar. This particularly applies to the cover page and the title from section 2.2.4, the part pages in section 2.2.5 and the chapter pages in section 2.2.6. Additional pages in this style can be created using the `tudheadings` page styles or the `tudpage` environment. If the `tudscrsupervisor` package is used and a task, a report or a notice is created using the commands or environments provided, these will also appear in the page style in question.

v2.03
v2.04

`cd=<setting>` Default: **true**

This option determines whether and how the corporate design of the Technical University of Dresden is used throughout the document. It influences the appearance of title, part and chapter pages as well as the headings of the other structure levels.

`cd=false/off/no`

This setting creates the standard behavior of the KOMA-Script classes; no corporate design is used.

`cd=true/on/yes/nocolor/`

monochrome (default value)

The layout for title, part and chapter pages is in the corporate design, black font is used for titles, part and chapter headings. The appearance of the page header depends on the **`cdhead`** option.

`cd=lightcolor/pale` The setting largely corresponds to the

`cd=true` option, but the primary house color **HKS41** is used for the head of the `tudheadings` page style and headings.

v2.04 **`cd=barcolor`**

In addition to the previous setting, the crossbar is also highlighted in color.

v2.03 **`cd=bicolor/bichrome`**

The header is set with a colored background in the house color, the crossbar is also given a colored background. The primary house color is used for the headings.

`cd=color`

The title as well as the part and chapter pages are all designed in color, the page header is set in the primary house color **HKS41**, the horizontal bar is bordered by lines. v2.03 **`cd=fullcolor/full`** Corresponds to the previous

setting, but the horizontal bar is not

bordered by lines but has a colored background.

v2.06

headings=<setting> Default: **heavy**

The option **headings=<setting>** is already defined by KOMA-Script and is used, among other things, to specify the size of headings. This is extended by TUD-Script in order to be able to adjust the font weight of the headings in the layout of the corporate design.

headings=light/normalbold The

headings are set in bold font.

headings=heavy/ultrabold

Extra-bold font weight is used for headings.

The settings apply to the outline levels up to and including **\subsubsection**.

Individual settings for individual elements of the layout

The behavior of all elements relevant to the layout is determined by the option **cd=<setting>** explained above. This affects both the title (**\maketitle**) and the cover page (**\makecover**) and all part pages (**\part**, **\addpart**) and chapter pages (**\chapter**, **\addchap**) as well as all subordinate structure levels.

If a specific element of the layout is to appear differently from the general setting for the entire document, one of the following options can be used to customize it and overwrite the defaults set with **cd=<setting>** for the respective element.²⁰ The valid value assignments for the individual elements correspond to the possible values for the option **cd=<setting>**.

Please note that the font used for the layout elements depends on the option **cdfont=<setting>**. For the title and cover pages, this can be changed directly using the optional argument of **\maketitle** and **\makecover** respectively.

v2.03

v2.04

cdtitle=<setting>

The general setting for the title can be overwritten with the **cdtitle** option. You can switch between the normal (**cdtitle=false**) and the one in the corporate design. The new title page supports all commands for the title defined by KOMA-Script.²¹ In addition, many new fields are defined which are particularly relevant for academic work. More details can be found in section 2.2.4. Regardless of the selected variant of the title page, it is always created with **\maketitle**.

v2.02

v2.03

v2.04

cdcover=<setting>

The TUD script classes also introduce the **\makecover** command, which can be used to create a separate cover page in addition to the title. This is very similar in appearance to the title page, but is normally set in a different type area than that of the book block. The **cdcover** option can be used to change the appearance of the cover page - regardless of **cd=<setting>**. If **cdcover=false** is selected, the cover page corresponds to the original KOMA script title. The use of the **\makecover** command and the associated parameters are explained in detail in section 2.2.4.

²⁰**cdtitle** for the title, **cdcover** for the cover, **cdpart** for parts, **cdchapter** for chapters and **cdsection** for all levels of structure below

²¹**\extratitle{<second title>}**, **\frontispiece{<frontispiece>}** for a page before the actual main title as well as **\titlehead{<header>}**, **\subject{<typing>}**, **\title{<title>}**, **\subtitle{<subtitle>}**, **\author{<author>}**, **\land**, **\thanks{<footnote>}**, **\publishers{<publisher>}** and **\date{<date>}** for the title itself as well as **\uppertitleback{<back of title header>}** and **\lowertitleback{<back of title footer>}** for the back of the title and finally **\dedication{<dedication>}** for an acknowledgment.

v2.03
v2.04**cdpart=<setting>**

For the parts pages, the value of the key **cd=<setting>** can be overwritten separately, thus influencing their layout or appearance, which is used when using the commands **\part** and **\addpart** and their starred versions. Further information on the parts page in the corporate design can be found in section 2.2.5.

v2.03
v2.04**cdchapter=<setting>**

For chapter pages, the key value **cd=<setting>** can also be adjusted to change the layout or appearance that is used when using **\chapter** or **\addchap** and the associated star versions. Further information on the chapter page in the corporate design can be found in section 2.2.6.

v2.05

cdsection=<setting>

Headings of the other structural levels **\section**, **\subsection**, **\subsubsection** as well as **\paragraph** and **\subparagraph** are output in the primary house color **HKS41** if a colored version of the layout has been set using the option **cd=<setting>**. If **cdsection=true** is specified, the headings of the above structural levels appear without any color.

Example: If the title page is to be in color but the rest of the document is to be in black
be set, this can be achieved as follows:

```
\documentclass[cd=true,cdtitle=color]{<document class>}
```

The vertical position of the headings

v2.05

pageheadingsvskip=<length value>
headingsvskip=<length value>

These two options can be used to change the vertical position of specific headings. The option **pageheadingsvskip** can be used to move the title on a title page (KOMA script option **titlepage=true**) as well as the headings of separate chapter pages (**chapterpage=true**) and parts vertically. In contrast, the second option **headingsvskip** can be used to adjust the vertical position of both the title of the title header (KOMA script option **titlepage=false**) and the chapter heading when the chapter page is deactivated (**chapterpage=false**).

The headings mentioned above are normally placed relatively deep in the text area in the layout. Negative values move the headings upwards, but care should be taken to ensure that they are still within the type area.

Positive values set the headlines lower accordingly.

2.2.3 Pages in the style of the corporate design

```
\faculty{<Faculty>}
\department{<Institution>}
\institute{<Institute>}
\chair{<Chair>}
\extraheadline{<Text line>}
```

The page style of the corporate design of the Technical University of Dresden is characterized by the header with the distinctive horizontal bar. In this, the text is shown in bold, if specified.

The faculty is printed in the first line, followed by the institution, the institute and the chair or professorship, separated by commas. If there is not enough space in the first line, a line break occurs automatically.

In exceptional cases, the corporate design allows the specification of an additional second or third line below the information about the department at the Technical University of Dresden, which contains additional, freely selectable information. This can be defined using the command **\extraheadline{<text line>}**.

v2.02

tudheadings (page style) **plain.tudheadings** (page style) **empty.tudheadings** (page style)

A central element of the corporate design of the Technical University of Dresden is the concise page header with the details of the faculty (**\faculty**), department (**\department**), institute (**\institute**) and chair (**\chair**) in the corresponding horizontal bar. By using the **scrlayer-scrpage** package, either individual pages or entire documents can be very easily set in this style. All you have to do is load one of the page styles with **\pagestyle{<page style>}**.

All page styles have the typical header with the characteristic crossbar in common, the shape of which can be adapted for *all* page styles using the **cdhead** option. Attention! The **\headlogo** command can be used to display an additional secondary logo in the header area. *For the special layout elements title and cover page as well as part and chapter pages, the **cdhead** setting is overwritten by using the **cd=<setting>** option.*

The appearance of the footer differs for each page style. It is always empty for the **empty.tudheadings** page style. The two styles - or rather the page style pair - **tudheadings** and **plain.tudheadings** take the settings for the footer from the **scrlayer-scrpage** user interface.

²² How to use the individual commands to individualize the footer can be found in the KOMA -Script-Instruction²³ be taken.

As an alternative to defining your own footer, you can also use the **cdfoot** option. In addition, you can use **\footcontent** to define free content for the footer area, and **\footlogo** can be used to output one or more logos in the footer area.

The font used in the footer area is determined by the font element **tudheadings**.

As soon as one of the defined styles has been activated with **\pagestyle{<page style>}**, the two page styles **tudheadings** and **plain.tudheadings** can also be used under the names **headings** and **plain** respectively. This has the advantage that with options or commands that automatically switch between the two page styles **headings** and **plain**, selecting one of the **tudheadings** styles once will switch between them.

However, the **empty** page style still produces a completely empty page. If a page is to appear with the concise header of the Technical University of Dresden and an empty page footer, **\pagestyle{empty.tudheadings}** must be called manually. To switch back to the normal behavior of KOMA-Script, one of the two page styles **scrheadings** or **plain.scrheadings** must be activated with **\pagestyle{<page style>}**.

v2.03

v2.04

v2.05

cdhead=<setting> Default: **true,nocolor,textwidth**

This option allows you to influence the shape of the logo and the crossbar as well as the font used in the **tudheadings** page styles. The following values

²²The commands **\leftfoot**, **\leftfoot** and **\leftfoot** as well as **\lfoot**, **\lfoot** and **\lfoot** or **\lfoot**, **\lfoot** and **\lfoot** can be used.

²³<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

can be used to adjust the font in the bar:

cdhead=false/off/no If the

use of the house font in the style of the corporate design of the Technical University of Dresden has been deactivated with **cdfont=false**, the header in the horizontal bar will be set in the sans serif of the active font. If the house font is activated, this setting has no effect.

cdhead=true/on/yes/light/lightfont/noheavyfont (default value)

In the crossbar, **Open Sans Semi-Bold** is used for **\faculty**, Open Sans Light is used for **\department**, **\institute**, **\chair** and **\extraheadline**. **cdhead=heavy/heavyfont** The content of **\faculty** is set in

Open Sans Bold, Open Sans Regular is used for the remaining fields.

There are several variants for the design of the head and the crossbar. On the one hand, the crossbar can be represented with two outer lines:

cdhead=nocolor/monochrome

The header and the lines of the crossbar appear in black color. **cdhead=lightcolor/pale** Both the header and the crossbar are set in the primary house color.

On the other hand, a display with more color is also possible, in which the horizontal bar and, if necessary, the entire page header is set off in color. The header area always extends over the entire width of the page, see **cdhead=paperwidth**.

v2.04 **cdhead=barcolor**

In contrast to the previous setting, the horizontal bar is used with a colored background, the part of the header above it is set without a colored background. v2.03 **cdhead=bicolor/bichrome** The header is set in color, with the background of the logo and the horizontal bar being different. The outer lines of the horizontal bar are omitted.

In case the crossbar is only displayed with two outer lines, its tracking can also be specified:

cdhead=textwidth/slim The

horizontal bar in the header only extends over the text area. This setting is particularly useful if borderless printing is not technically possible. **cdhead=paperwidth/wide** The horizontal extension of the horizontal bar extends over the entire page width to the edge of the page. This behavior cannot be deactivated in the colored layout of the header.

v2.05 In addition to the previously described options for designing the header area, the date specified with **\date** can be displayed right-aligned below the crossbar on all pages with the tudheadings page style activated.

cdhead=date/showdate The

set date is displayed above the text area on the right.

cdhead=nodate/hidedate No date

appears between the header and text area.

v2.03

cdfoot=<setting> Default: **false**

The TUD script classes are very flexible and individually adaptable when designing headers and footers, particularly due to the option of using the `scrlayer-scrpage` package. The form and content of these are not explicitly specified by the corporate design and can be selected and changed by the user as desired.

One possible way of designing them is shown in the manual for the corporate design of the Technical University of Dresden. This is set without a header and with a simple footer, which contains the current column title and the pagination. This feature can be activated with **cdfoot**, which is also what has been done for this user manual.

cdfoot=false/off/no

The headers and footers show standard behavior; for manual changes the KOMA-Script package `scrlayer-scrpage` should be used.

cdfoot=true/on/yes (default value)

The footers of the document are set in the same way as in the Corporate Design Manual of the Technical University of Dresden with a live column title and page number, whereby in double-sided typesetting (class option **twoside=true**) the pagination is placed on the outside.

If the **manualmark** option is not explicitly specified when loading the class or the `scrlayer-scrpage` package, the automatic column titles are activated with **cdfoot=true** via the KOMA-Script option **automark**, which uses the titles of the outline levels as marks. More details about this and the option of specifying the column titles manually can be found in the KOMA-Script manual²⁴ refer to.

v2.03

If one of the `tudheadings` page styles is activated and a colored layout is used on the generated page - for example the two-color header (**cdhead=bicolor**) or a colored page background - the footer can also have a colored background.

cdfoot=nocolor/monochrome

The foot is always placed without a colored background.

cdfoot=color/bicolor/bichrome

The footer is set off in color if either the header is used in a colored variant (see **cdhead**) or a page with a colored background in the house color (title or chapter page) is active.

Additional content can be defined for the footer area using the commands **\footlogo** on page 30 and **\footcontent** on page 30. If the available space is not sufficient for this, it can be enlarged.

cdfoot=<height>

If a length value is passed to the option, this is exactly the same as using the option **extrabottommargin=<height>**.

v2.04

tudheadings (font element)

The font element **tudheadings** is used in the footer area of pages in the `tudheadings` page style. This affects the page numbers, the running head and the content specified with **\footcontent**. This is used to choose the right font color depending on the page background and the settings for the **cdhead** and **cdfoot** options.

How to customize **tudheadings** can be found in section 2.2.8.

²⁴<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

`\headlogo[<parameter list>]{<file name>}`

In addition to the logo of the Technical University of Dresden, a second logo may be used in the header. This can be integrated with this command. It is usually placed on the height of the first logo is scaled. The optional argument can be used to pass additional formatting commands to the **`\includegraphics`** command used in the background, for example to adjust the size of the second logo. The documentation for the **graphicx** package tells you which parameters can be adjusted.

If the **`ddc`** option is activated, the DRESDEN concept logo is not placed in the header but automatically in the footer. The **`ddchead`** option always places it in the header and thus overwrites the secondary logo specified with **`\headlogo`**.

v2.03

`\footlogo[<parameter list>]{<file name list>} \footlogosep`

According to the corporate design guidelines, other logos may appear in the footer, for example those of cooperating companies or sponsors. The file names of the desired logos can be specified as a comma-separated list in the obligatory argument of the **`\footlogo`** command. If not just a file name but a list has actually been passed, the separator stored in **`\footlogosep`** - **`\hfill`** by default - is placed between the logos when the logos are output. This can be adjusted as desired by the user using **`\renewcommand*\footlogosep{. . .}`**. The separator is also set if only a comma was used in **`<file name list>`**. With **`\footlogo{<file name>,<file name>,<file name>}`**, for example, a logo can be placed centered in the footer. Please note that a DRESDEN concept logo set with the **`ddc`** or **`ddcfoot`** option could be overlaid in the footer - in contrast to using **`\footcontent`**. If in doubt, the user has to do some formatting work by

inserting separators – i.e. commas – in the argument of **`\footlogo`**.

The optional argument of **`\footlogo`** is passed on to **`\includegraphics`**. This happens equally for all files specified in the list. If individual settings are to be made for individual logos, the corresponding parameters must be passed in the obligatory argument after the respective file name with a colon ":" as a separator (**`\footlogo{<file name>:<parameter>}`**), whereby these are applied to all logos according to the general settings. The possible parameters and values for the optional arguments can be found in the documentation of the **graphicx** package.

v2.05

`footlogoheight=<length value>`

If no optional argument for the size is specified for **`\footlogo`**, all logos in the foot are scaled to the height of the logo of the Technical University of Dresden. This can be changed globally for all logos by setting the option **`footlogoheight=<length value>`**. If the height of the footer area is not sufficient to display all logos in the desired size, it can be adjusted using **`extrabottommargin=<height>`** or **`cdfoot=<height>`**.

v2.04

v2.05

`\footcontent[<instructions>]{<content>}[<content>]`

`\footcontent*[<instructions>]{<content>}[<content>]`

This command can be used to set any content in either one or two columns in the footer area of the tudheadings page styles. In the form **`\footcontent{<content>}`** the content is output across the entire text width in the footer. However, if the command is used in the second variant **`\footcontent{<left content>}[<right content>]`** with an optional

the obligatory argument is used, the footer appears in two columns, with the content from the first, obligatory argument in the left footer column and the content from the second, optional argument in the right footer column. Any DRESDEN concept logo that was set using the **ddc** or **ddcfoot** option is taken into account and the space available for the text in the footer is reduced. Possible settings of the `scrpage` package for the footer are not taken into account, as this can lead to the content overlapping. The same applies to using the values **true** and **false** for the **cdfoot** option .

v2.05

If only an asterisk ***** is passed to the argument for the left or right column , the content defined up to that point in this column is retained. For example, the left footer column can be adjusted with `\footcontent{<content>}[*]` without changing the content of the right column, or only the font used in the footer can be adjusted with `\footcontent[<instructions>][*]` while keeping the content the same .

In the footer, the font element **tudheadings** is used for the font. The font size is also adjusted, based on the header. In addition, the first optional argument of **\footcontent** can be used to execute additional font settings or **<instructions>** before the actual output of the content. If the content in the footer is to be defined *without* automatically adjusting the font size , the star version **\footcontent*** should be used. Here, too, the optional argument for font formatting can be used if necessary . Last but not least, when changing the content in the footer, previous font adjustments can remain unchanged by using an asterisk in the optional argument with **\footcontent[*]**

{<content>}[<content>] .

v2.02

ddc=<setting>
ddchead=<setting>
ddcfoot=<setting>

Default: **false**

This option inserts the DRESDEN -concept logo either in the header or footer of the pages with the **tudheadings** style . This is automatically placed either in the header or – if a secondary logo was specified with **\headlogo** – in the footer.

Alternatively, the options **ddchead** and **ddcfoot** can be used, which explicitly place the logo either in the header or footer. A secondary logo specified with **\headlogo** is definitely suppressed by **ddchead=<setting>** , **ddcfoot=<setting>** always places the DRESDEN -concept logo in the page footer. Using one of the three options therefore deactivates the other two.

The DRESDEN concept logo is set by default in both the header and footer at the same height as the logo of the Technical University of Dresden and cannot be changed, at least for the header. If it is used in the footer, the size can be changed using the **footlogoheight** option . If the height of the footer area is not sufficient after enlarging the display , it can be adjusted using **extrabottommargin=<height>** or **cdfoot=<height>** . The possible values for the three keys are:

ddc=false/off/no

The tudheadings page styles do not display the DRESDEN -concept logo.

ddc=true/on/yes (default value)

The DRESDEN -concept logo is used in the header and footer. The color of the logo is chosen to match the color scheme of the page itself.

The colour of the DRESDEN -concept logo depends on the colour of the

Layouts (option **cd=<setting>**) are automatically selected. This can be changed manually:

ddc=color

The eight-colour 4C variant of the DRESDEN -concept logo is used in the header or footer .

ddc=colorblack

The eight-colour logo with black DRESDEN -concept lettering is used instead of the grey one. For the foot, the green claim is replaced by a black one. This is particularly useful for small representations of the logo in the foot.

ddc=gray/gray

This edition of the DRESDEN -concept logo is in grayscale.

ddc=black

Use of the logo in greyscale with black lettering.

ddc=blue

Lettering and logo are set in shades of the house color **HKS41** . **ddc=white** The DRESDEN -concept logo and the

corresponding lettering are completely white.

v2.02
v2.03

```
\begin{tudpage}[<language> || <parameter list>]
. . .
\end{tudpage}
language=<language> (parameter)
columns=<number> ( parameter)
pagestyle=<setting> (parameter)
cdfont=<setting> (parameter)
cdhead=<setting> (parameter)
cdfoot=<setting> (parameter)
headlogo=<filename> (parameter)
footlogo=<filenamelist> (parameter)
ddc=<setting> (parameter)
ddchead=<setting> (parameter)
ddcfoot=<setting> (parameter)
```

(see **cdfont** on page 16)
(see **cdhead** on page 27)
(see **cdfoot** on page 29)
(see **\headlogo** on page 30)
(see **\footlogo** on page 30)
(see **ddc** on page 31)
(see **ddchead** on page 31)
(see **ddcfoot** on page 31)

The tudpage environment has its origins in an earlier version, when the **tudheadings** page styles were not yet available, but can now be used instead of this environment. Various parameters can be specified as optional arguments for the tudpage environment . If the babel package is used, the language used can be changed with **language=<language>** , which leads to the adaptation of the language-specific separation patterns and identifiers. If the multicol package has been loaded, the content of the environment is set to multiple columns with the parameter **columns=<number>** . The page style can be adapted with **pagestyle** , with **headings**, **plain** and **empty** being valid values.

The other parameters behave in the same way as the class options or commands of the same name, but only have an effect within the tudpage environment. The behavior and valid value assignments are documented on the specified pages.

2.2.4 The title and cover page

To create a title with the command **\maketitle**, the KOMA -Script option **titlepage=<on-off value>** is used to specify whether the title is to be created as a whole title page or just as a

Title header should appear. For the title in the corporate design of the Technical University of Dresden, all fields that are already defined by KOMA-Script are supported. In addition, further fields are provided for the TUD -Script classes that affect the design of the title. These are explained in this section. The title - consisting of a possible extra title (**\extratitle**) including the corresponding back cover (**\frontispiece{<frontispiece>}**), the actual title page or the title header and the subsequent elements - can be output with **\maketitle**. In addition, **\maketitleonecolumn** can be used in two-column typesetting, which enables a one-column addition after the title itself.

In addition to the title, a cover page can be created with **\makecover**. This can be used in particular for bound works. In comparison to the title, only a reduced number of fields are output on this.

v2.01
v2.02
v2.03
v2.06

\maketitle[<page number > || <parameter list>]

pagenumber=<page number>

(parameter) **cdgeometry=<setting>**

(parameter) **cdfont=<setting>**

(parameter) **cdhead=<setting>**

(parameter) **cdfoot=<setting>**

(parameter) **headlogo=<file name>**

(parameter) **footlogo=<file name list>**

(parameter) **ddc=<setting>**

(parameter) **ddchead=<setting>**

(parameter) **ddcfoot=<setting>** (parameter)

(see **cdgeometry** on page 22)

(see **cdfont** on page 16)

(see **cdhead** on page 27)

(see **cdfoot** on page 29)

(see **\headlogo** on page 30)

(see **\footlogo** on page 30)

(see **ddc** on page 31)

(see **ddchead** on page 31)

(see **ddcfoot** on page 31)

The command **\maketitle** sets the normal KOMA -Script title for **cdtitle=false**, otherwise the title page is created in the corporate design of the Technical University of Dresden. The latter variant has been expanded to include a large number of fields compared to the standard title and allows the specification of data for the cover page of an academic thesis. The individual fields are explained later in this section. If the document is set double-sided and with chapters opening to the right,²⁵ the option **cleardoublespecialpage** is also included. This is particularly important for the commands **\uppertitleback** and **\lowertitleback** for the back of the title.

The optional argument allows you to change the page number of the title page - just like with the KOMA -Script classes. This is not output, however, but only affects the count. You should definitely choose an odd number here, otherwise the entire count will be messed up. If a title page (KOMA -Script option **titlepage=true**) is set in the corporate design of the Technical University of Dresden (**cdtitle=true**), the parameters listed below can also be used in the optional argument. Their behavior corresponds to the options or commands of the same name, but they only have a local effect and only affect the title page. For example, the use of a DRESDEN-concept logo can be restricted to the title.

\maketitleonecolumn[<page number > || <parameter list>]{<one column text>}

In two-column typesetting (class option **twocolumn**), the title page itself is always set in one column using **\maketitle**. The two-column body text normally follows directly after the title.

With the command **\maketitleonecolumn**, additional content – for example a summary or an abstract – can be placed in one column after the title.

²⁵KOMA -Script options **twoside=true** and **open=right**, default for tudscrbook

If a title page is activated (KOMA script option **titlepage=true**), the argument `{<single-column text>}` is output directly after this on one or more new pages, also in one column. If the command **\maketitleonecolumn** is used with a title header instead of a title page (KOMA script option **titlepage=false**), the single-column text passage from the obligatory argument follows directly, with an automatic page break occurring if the content is appropriate. After that, the system switches directly to the two-column layout without any additional page breaks.

The optional parameter of **\maketitleonecolumn** can be used in the same way as **\maketitle** to change the page number, the font used and to adjust the header and footer. Please note that most of the parameters only have an effect if a title page (KOMA-Script option **titlepage=true**) is used.

v2.02

v2.03

\makecover[<page number > || <parameter list>]

pagenumber=<page number>

(parameter) **cdgeometry=<on-off value>**

(parameter) **cdfont=<setting>**

(parameter) cdhead=<setting>

(parameter) **cdfoot=<setting>**

(parameter) **headlogo=<file name>**

(parameter) **footlogo=<file name list>**

(parameter) **ddc=<setting>**

(parameter) **ddchead=<setting>**

(parameter) **ddcfoot=<setting>** (parameter)

(see **cdfont** on page 16)

(see **cdhead** on page 27)

(see **cdfoot** on page 29)

(see **\headlogo** on page 30)

(see **\footlogo** on page 30)

(see **ddc** on page 31)

(see **ddchead** on page 31)

(see **ddcfoot** on page 31)

A cover page is usually required for bound theses, for example to be used for embossing on the book cover. Therefore, the color of the cover page is deactivated even if it is active for the rest of the document (**cd=color**). However, this can be overwritten at any time with **cdcover=<setting>**.

If **cdcover=true** is selected, the cover page is set in the corporate design of the Technical University of Dresden. The title of the document, the typing using **\thesis** and/or **\subject**, the author or authors and, if applicable, the publisher specified using **\publishers** are displayed on this page. If **cdcover=false** is selected, only the normal KOMA-Script title is displayed as a separate cover page.

The title page itself always belongs to the book block and should therefore be set in the same type area. In contrast, the cover page usually appears in a different layout. Normally, the cover is set in the asymmetric type area of the corporate design - regardless of the **cdgeometry** option. The behavior can be changed with **cdgeometry=false** in the optional argument. In this case, the cover page also appears in the book block of the rest of the document. However, for this setting, the page margins can be freely adjusted by the user with the commands **\coverpagetopmargin**, **\coverpageleftmargin**, **\coverpagerightmargin** and **\coverpagebottommargin**. More information can be found in the KOMA-Script manual²⁶ to find.

In addition, the optional argument can be used to change the page number of the cover page. This is not output, however, but only affects the counting. You should definitely choose an odd number here, otherwise the entire counting will be messed up. The parameters listed below correspond in their behavior or function to the options or commands of the same name, but only have a local effect.

²⁶<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

and only on the cover page.

v2.01

\title{<title>}
\subtitle{<subtitle>}

The commands **\title** and **\subtitle** are self-explanatory. It should be noted that the font weight of the title depends on the **headings** option and the subtitle always appears in bold .

v2.02

v2.06

titlepage (font element)
thesis (font element)
\raggedtitle

The TUD -Script classes define these new font elements. **titlepage** is used on the title page for all fields that do not use a special font element that is provided by KOMA-Script anyway . The field specified with **\thesis** , in which the type of thesis is specified, uses the font element **thesis**.

v2.02

The fonts used can be customized for all fields of the title and cover page. Section 2.2.8 explains exactly how this works. The font elements **titlehead**, **subject**, **title**, **subtitle**, **author**, **publishers**, **date** and **dedication** already provided by KOMA-Script as well as the new v2.06 **titlepage** and **thesis** are supported for the title and cover page. The command **\raggedtitle** defines the alignment

of the title, which is left-aligned by default.

Example: To center the entries on the title page, the following definition is sufficient:

\let\raggedtitle\centering

v2.02

v2.05

v2.06

\author{<author(s)>}
\authormore{<author additional information>} **\emailaddress[<settings>]{<email address>}**
\emailaddress*{<email address>} **\dateofbirth**
{<date of birth>} **\placeofbirth{<place of birth>}**
\matriculationnumber{<matriculation number>}
\matriculationyear{<year of enrollment>} **\course{<course of study>}** **\discipline{<field of study>}**

The **\author** command is used to specify the author. Multiple authors can be listed within the argument , in which case they must be separated with **\and** .

All other commands presented here can be used in the argument of **\author** , which allows individual information for each author.

With **\authormore** a line is output under the author, which can be freely assigned by the user. With **\emailaddress** an email address can be specified for each author, which is defined as a hyperlink if the hyperref package has been loaded. The optional argument is passed to **\hypersetup** from said package and can thus be used for additional settings. With the star version **\emailaddress*** the entry in the argument is not formatted.

If the isodate or datetime2 package is loaded, the formatting of the date set by **\dateofbirth** – as with any other date field in the TUD script classes – is used for the date of birth on the title.

either `\printdate` from `isodate` or `\DTMDate` from `datetime2` is used. The command `\placeofbirth` can also be used to specify a place of birth.

The other commands as additional information are more or less self-explanatory. It should be noted that the information provided with the commands `\matriculationnumber`, `\matriculationyear`, `\course` and `\discipline` are also used by the `tudscrsvsupervisor` package within the `task` environment, if this is used.

`\and`

This command is used in both the LATEX standard classes and the KOMA-Script classes only on the title page to separate multiple authors in the argument of `\author`.

In the TUD script classes, however, this command has been expanded in its function to allow the specification of collaborative authorship for theses within the `\author` command. It can also be used in the argument of `\supervisor`, `\referee` and `\advisor` to specify multiple supervisors or reviewers and subject specialists. It is not restricted to use for the title alone, but can also be used to specify people in the corresponding fields of the `task`, `evaluation` and `notice` environments from the `tudscrsvsupervisor` package.

Example: Assuming that a thesis is to be written by two different authors in a collaborative community, the author information can be structured as follows: `\author{% Mickey Mouse% \matriculationnumber{12345678}% \dateofbirth{2.1.1990}%`

`\placeofbirth{Dresden}% \and%`

`Donald Duck% \matriculationnumber{87654321}% \dateofbirth{1.2.1990}% \placeofbirth{Berlin}%`

`} \matriculationyear{2010}`

All additional information outside the argument of `\author` is applied equally to both authors. Information within the argument of `\author` is assigned to the respective authors, separated by `\and`. More on this can be found in the minimal example in subsection 6.3.

v2.05

```
\date[<suffix> || <parameter list>]{<date>} \date*[<suffix> ||
<parameter list>]{<date>} before=<prefix> (parameter)
after=<suffix> (parameter)

place=<place> (parameter)
\defensedate{<defense date>}
```

v2.05 The date can be specified using the command `\date`. The two parameters `before` and `after` can be used via the optional argument to output additional information before or after the actual date. The star version `\date*` sets the

place specified with **\place** in front of the date. This also happens for the normal version of **\date** when the **place** parameter is used.

In normal documents, the date is printed directly after the author or authors . In the case of theses – activated by using **\thesis** or **subjectthesis** in conjunction with **\subject** – this appears at the end of the title page as the submission date. In this case, the date of the defense can also be specified using the **\defensedate** command , as is common when printing dissertations, for example .

If one of the packages isodate or datetime2 is loaded, **\printdate** or **\DTMDate** will use the output formatting of the date set by the respective package for all date fields of the document and consequently also for the two fields **\date** and **\defensedate** .

\thesis{<typification>}
\subject{<typification>}

These two commands can be used to specify the type of document or thesis. While the **\thesis** command outputs the content of the field below the title, vertically centered on the title page, the content of the **\subject** command appears above the title. Both commands can also be used in parallel with different contents.

The command **\thesis** is also intended for the TUD -Script document classes to recognize theses, since special fields are provided for these and the title page is also set slightly differently.

Furthermore, both commands allow you to use special values as arguments for typing the document. These are decoded and set according to the selected document language - either German or English. The possible values can be found in Table 2.1. It should be noted that it is possible to set a special value for *either* **\thesis** or **\subject** . Using one of the values mentioned always results in the document being recognized as a thesis and the extended title page being activated. At the same time, this influences the **subjectthesis** option . If the user has not defined an explicit behavior for **subjectthesis** , using **\thesis{<value>}** results in **subjectthesis=false** and **\subject{<value>}** results in **subjectthesis=true**.

subjectthesis=<on-off value> Default: **false** | **\subject{<table 2.1>}**: **true**

The **\thesis** command is used by the TUD script main classes to distinguish between two different versions of the title page and is specifically intended for theses. In addition, the behavior for theses can also be activated by using special values from Table 2.1 within the argument of **\subject** , whereby the setting **subjectthesis=true** is automatically made.

In the standard case – **subjectthesis=false** – the type of thesis specified by **\thesis** and the desired degree, if applicable , specified by **\graduation**, are set in large letters and very centrally on the title page. The use of **\subject** is still possible here. If the option is activated with **subjectthesis=true** , the designation specified by **\thesis** is not displayed below the title but above it, in place of **\subject** . The degree specified by **\graduation** is still displayed below the title, but in a slimmer font. Any use of the **\subject** command is ignored in this case.

Table 2.1 Special values for typing the document for **\thesis** and **\subject**

Value	Identifier	German	English
habilitated	\habilitationname	Habilitation	Habilitation
diss	\dissertationname	dissertation	dissertation
phd	\dissertationname	dissertation	dissertation
diploma	\diplomathesisname	Diploma thesis	Diploma Thesis
master	\masterthesisname	Master thesis	master thesis
Bachelor	\bachelorthesisname	Bachelor thesis	Bachelor Thesis
University student	\studentthesisname	Term paper	Student Thesis
evidence	\studentresearchname	Large document	Student Research Project
	\projectpapername	Project work	Project Paper
projectseminar	\seminarpapername	Seminar paper	Seminar Paper
term	\termpapername	House work	Term Paper
research	\researchname log	Research Report	Research Report
\logname		protocol	log
report	\reportname	Report	report
internship	\internshipname	Internship Report	Internship Report

subjectthesis=false/off/no

The output of the type of thesis (**\thesis**) itself as well as the intended

Graduation **appears** in large letters centrally on the title page.

subjectthesis=true/on/yes (default value)

The type of the thesis (**\thesis**) is placed above the title in the subject line.

The desired degree (**\graduation**) is issued centrally.

v2.02

\graduation[<short form>]{<degree>}

This command will print the desired academic degree on the title page.

Since this can only be achieved with a thesis, the issue is only made if either **\thesis** or **\subject** was used, where in the latter command the argument

A value from Table 2.1 must be used.

When issuing the academic degree, the **subjectthesis** option influences the Output on the title page. If **subjectthesis=false**, the conclusion – similar to the type of the thesis – centrally and in relatively large letters. For **subjectthesis=true** the output is smaller and in less bold letters.

\supervisor{<name(s)>}

\referee{<name(s)>}

\advisor{<name(s)>}

\professor{<name>}

\supervisor, **\referee** and **\advisor** are used to indicate the supervisors of a thesis or the reviewers and subject referees of a dissertation. In addition,

with **\professor** the supervising university lecturer or the supervising professors for student work are specified. The specification of several people is done as follows: in the command **\author** by separating with **\and**.

v2.06

titlesignature=<on-off value> Default: **false**

Some chairs at the Technical University of Dresden require a signature of the Author of a thesis directly on the title page. By activating this option, a

such a field is created at the bottom of the title.

```
\titledelimiter{<separator>}
```

The TUD script classes provide a number of additional fields for the title and cover page . Some of these fields are preceded by a description (see subsection 2.4). A separator is inserted between them in the output. A colon followed by a space (**\nobreakspace**) is the default setting.

With the command **\titledelimiter** this separator can be adapted to the individual needs of the user.

```
\extratitle{<frontispiece>}
\frontispiece{<frontispiece>}
\titlehead{<header>}
\publishers{<publisher>}
\thanks{<footnote>}
\uppertitleback{<back of title header>}
\lowertitleback{<back of title footer>}
\dedication{<dedication>}
```

These commands correspond in their behavior to the original counterparts of the KOMA-Script classes and are mentioned here for the sake of completeness.

The output of the half-title defined with **\extratitle** – which can be designed and formatted as desired – and the back cover defined with **\frontispiece**, if applicable, is done **as** part of the frontispiece with **\maketitle** before the actual title page.

The **\titlehead** command can be used to output additional, freely formattable text above the typography and title. Since the vertical position of the document title is fixed by the corporate design, it can happen – in contrast to the KOMA script classes – that the head of the main title itself extends into the header. This is not checked by the TUD script classes and must be checked by the user if necessary. The content defined with the **\publishers** command does not necessarily have to identify a publisher but can also contain other information that should be output at the end of the title page.

Footnotes are not created on the title with **\footnote**, but with the **\thanks** command . These are usually used for comments on the title or the authors. Symbols are used as footnote symbols instead of numbers. The **\thanks** command can only be used within the argument of one of the instructions for the title page, such as **\author** or **\title** .

In double-sided printing, the back of the main title page can be used for additional information. The user can freely design both the back title header and the back title footer using **\uppertitleback** and **\lowertitleback** .

With **\dedication** you can set a separate dedication page centered and in a slightly larger font . The back page is always blank, as is the back of the half-title. The dedication is printed with the rest of the title page and must therefore be specified before using **\maketitle** .

2.2.5 The parts page

v2.02 If the layout of the corporate design is used for the part pages, the page style (`\partpagestyle`) is set **to** `plain.tudheadings` . If you want to use a different

If you use a different page style, you can adjust it with `\renewcommand*{\partpagestyle}{<page style>}` .

v2.06

`\setpartsubtitle{<subtitle>}`
partsubtitle (font element)

With `\setpartsubtitle{<subtitle>}` a subtitle can be set for parts after the heading itself. The command must be specified in the same way as `\setpartpreamble` before using `\part` or its derived variants. With the font element **partubtitle** the font for the given subtitle can be changed. Section 2.2.8 explains how it can be customized.

`parttitle=<On-Off-Value>` Default: **false**

This option allows the title of the document itself, given with `\title` , to be printed in large font on a part page. In this case, the name of the part created with `\part{<name>}` is placed directly below it as a subtitle. This layout variant can be found in the manual for the corporate design of the Technical University of Dresden. This option is not available for the classes `tudscrartcl` and `tudscrposter`.

parttitle=false/off/no The name of the part appears in large font, the title of the document is not printed.

parttitle=true/on/yes (default value)

The specified title is placed in large font on the parts page, the name of the part itself as a subtitle.

2.2.6 The chapter page

`chapterpage=<on-off value>` Default: **false** | **cd=color: true**

With this setting, the heading of a chapter can be output separately on a page . The following text is output on the next page or, in the case of double-sided typesetting and chapters opening on the right²⁷, on the page after that. The back page created in this case is determined in its appearance - as are partial pages - by the setting of **cleardoublespecialpage** . In the colored layout, this option is activated by default. This option is not available for the classes `tudscrartcl` and `tudscrposter`. **chapterpage=false/off/no** There is no special position for chapters; the following text is output directly under the heading or after the chapter

preamble created with `\setchapterpreamble` on the same page.

chapterpage=true/on/yes (default value)

The chapter heading and, if applicable, the chapter preamble are set on a separate page. The following text appears on the next or next but one page, *see also* **cleardoublespecialpage**.

v2.02 With `\renewcommand*{\chapterpagestyle}`

`{<page style>}` you can adjust the page style of chapters – regardless of the **chapterpage** option . When using chapter pages with **chapterpage=true**, it is also recommended to activate the KOMA -Script option **chapterprefix=<on-off value>** . This will display the chapter headings with

²⁷KOMA -Script options **twoside=true** and **open=right**, default for `tudscrbook`

a frontispiece line. If a numbered chapter is created, “Chapter” followed by the current chapter number is first output in one line, and the actual heading is then output in the next line in left-aligned flush text. More information can be found in the KOMA-Script documentation²⁸ refer to.

v2.06

\setchaptersubtitle{<subtitle>}
chaptersubtitle (font element)

With **\setchaptersubtitle{<subtitle>}** a subtitle can be set for chapters after the heading itself. The command must be specified in the same way as **\setchapterpreamble** before using **\chapter** or its derivatives.

The font element **chaptersubtitle** can be used to change the font for the given subtitle . See section 2.2.8 how to customize it.

2.2.7 Vacant pages

Automatically generated blank pages – also called intentionally empty pages – can be found at the beginning of parts and chapters in documents depending on the KOMA-Script options **twoside=<setting>** and **open=<method>**.²⁹ The page style for the blank pages can be set using the KOMA-Script option **cleardoublepage=<setting>** .

v2.06

cleardoublespecialpage=<setting> Default: **true**

This option only has an effect when double-sided typesetting is activated and only right-hand pages are used for parts or chapters.³⁰ In this case, the style of the following left-hand page - i.e. the back page - can be influenced. The normal behavior is that after a part, the back page that follows is always set as a completely blank page without headers or footers, regardless of the setting for **cleardoublepage** .

This option allows you to deactivate the normal behavior and to use the page style of the **cleardoublepage** option for the page after the part page - and depending on **chapterpage** also after the beginning of a chapter on a separate page. You can also choose another, already defined page style. In addition, in the colored layout, the back page can be set in the same color as the front page of the title, part or chapter . This option is not available for the **tudscrartcl** and **tudscrposter** classes.

cleardoublespecialpage=false/off/no

The backs are completely blank pages, regardless of the **cleardoublepage** option.

cleardoublespecialpage=true/on/yes (default value)

The page style of the back of parts and, if applicable, chapters corresponds to the setting of **cleardoublepage** for blank pages.

cleardoublespecialpage=current The

currently defined page style (**lpagestyle**) is used for the generated back.

cleardoublespecialpage=<page style>

By specifying **cleardoublespecialpage=<page style>** , any previously defined page style can be used for the back page according to parts and chapters. **cleardoublespecialpage=color**

In the coloured layout, the back of parts and chapters is also coloured, see **cd**. The

²⁸<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

²⁹Defaults **tudscrbook**: **twoside=true**, **open=right**; **tudscrartcl**, **tudscrreprt**: **open=any**, **twoside=false**

³⁰KOMA-Script option **twoside=true** and **open=right**

Setting also affects the back of the title.³¹

cleardoublespecialpage=nocolor

White backs are created for titles, parts and, if applicable, chapters.

2.2.8 Use of font elements

The TUD -Script bundle defines additional font elements in addition to those already provided by KOMA -Script . These are **titlepage**, **thesis**, **tudheadings** as well as **partsubtitle** and **chaptersubtitle**. Both the font elements defined by KOMA -Script and all of the font elements mentioned here and explained below can be adjusted if necessary using the command **\addtokomafont{}{<settings>}** . More information can be found in the KOMA -Script manual³² can be found within the *Text Markup* section .

2.2.9 Use of field contents

v2.06

\getfield{<field name>}

This macro can be used to access the contents of a previously specified field.

For example, the date specified for this document is "August 9, 2022". It can be output as follows:

\getfield{date}

Other possible arguments include **faculty**, **department**, **institute** and **chair** as well as **title**, **author**, **thesis**, **professor**, **supervisor** or **place**.

2.2.10 The colours of the corporate design

The tudscrcolor package is used to use the colors of the corporate design . If this is not loaded in the preamble - for example to call up additional options - the TUD script classes include it automatically. Detailed information can be found in the tudscrcolor documentation in subsection 5.1 on page 71.

2.3 Additional options and extensions

In addition to the commands for adapting the layout to the corporate design of the Technical University of Dresden, the TUD script classes provide additional commands and environments to facilitate the application, especially for scientific work.

2.3.1 Summary/Executive Summary

v2.02

v2.04

abstract=<setting>

This option is already provided by KOMA-Script for the classes scrartcl and scrreprt as standard . This is not the case for the class scrbook . The manual states:

³¹see **\uppertitleback** and **\lowertitleback** in the KOMA-Script manual

³²<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

For books, a different type of summary is usually used. There, a corresponding chapter is placed at the beginning or end of the work. Often this summary is linked either to the introduction or to a further outlook. Therefore, there is generally no abstract environment in scrbook . For reports in a broader sense, such as a study or thesis, a summary in this form is also recommended.

The TUD -Script classes extend the **abstract option** . In addition to the selection options that KOMA-Script already offers for the tudscrartcl and tudscrreprt classes, the heading for the summary can also be output in the form of a subchapter or, for tudscrreprt and tudscrbook, in the form of a chapter. **abstract=false/off/no** (tudscrartcl and tudscrreprt only)

No heading is output for the abstract environment.

abstract=true/on/yes (default value) (tudscrartcl and tudscrreprt only)

As with the KOMA -Script classes, a centered heading with the identifier **abstractname** is placed before the actual summary.

abstract=section/addsec

The heading (**abstractname**) uses the outline command **section**.

abstract=chapter/addchap (default value for tudscrbook) (tudscrreprt and tudscrbook only)

The command **chapter** is used to set the heading (**abstractname**) . **abstract=heading** The highest possible level of

structure is used. For tudscrartcl this corresponds to **abstract=section**, for tudscrreprt and tudscrbook this corresponds to **abstract=chapter**.

The behavior for placing an entry in the table of contents is determined depending on the selected structure level of the heading . Without or with a centered heading, no entry is created by default. However, if the heading is selected in the form of a structure level, the summary usually appears in the table of contents at the top level. The default behavior for the entries in the table of contents can be overwritten by the user at any time with the following values.

abstract=notoc/nototoc

The summary will definitely not be included in the table of contents. **abstract=toc/totoc** An unnumbered entry for the

summary will be placed in the table of contents at the highest level of the currently used document class (**chapter** or **section**) .

v2.02 **abstract=tocleveldown/leveldown/totocleveldown**

The table of contents entry is created one level below the top level of the structure (**section** or **subsection**) .

v2.04 **abstract=tocmultiple/totocmultiple/tocaggregate/totocaggregate**

A *single* table of contents entry is created for *all* summaries.

v2.02 In addition, the behavior of the running headers can be influenced by the user.

These are normally set automatically if they have been activated using the KOMA -Script option **automark** and depend on the current structure level of the heading. However, if manual column titles are used using the KOMA -Script option **manualmark** , these must normally also be set by the user for the summary . However, with **abstract=markboth**, the setting of the column titles can be forced independently of this.

abstract=markboth

Regardless of the use of manual or automatic running titles, these are set on right and left pages with **\abstractname** .

abstract=nomarkboth

The settings for manual or automatic column titles are taken into account and set depending on the structure level of the heading used.

With the optional parameter **markboth** of the abstract environment, the running head can be set with any content.

For theses, it is often required that an English summary be written in addition to the German summary. The setting **abstract=multiple** allows multiple summaries to be printed on one page - provided there is enough space. In addition, the standard vertical centering of the **abstract environment** on a page can be suppressed. These settings for positioning the summaries within the abstract environment only take effect if a title page (KOMA -Script option **titlepage=true**) and *no* headings in the form of chapters (**abstract=chapter**) are used.

abstract=single/one/simple

Each summary is output on a separate page or in a two-column set in a new column. v2.02 **abstract=multiple/multi/all/aggregate** Summaries separated with

\nextabstract are output one after the other on the same page if there

is enough space on it. If the class option **twocolumn** is activated, all summaries are output without column breaks.

abstract=fill/fill/vfil/vfill

All summaries are centered vertically on one page when output. This setting is not available for two-column layout (class option **twocolumn**) . **abstract=nofill/nofil/novfil/novfill**

The output is the same as in normal running text.

v2.02

```
\begin{abstract}[<language> || <parameter list>]
. . .
\end{abstract}
\nextabstract[<language> || <parameter list>]
language=<language> (parameter)
markboth=<setting> (parameter)
pagestyle=<page style> (parameter)
columns=<number> (parameter)
option=<setting> (parameter)
```

(see **abstract** on page 42)

The abstract environment is used specifically for outputting a summary, either at the beginning of a document or, for example, before a part or chapter. If a title header and no title page is used (KOMA -Script option **titlepage=false**), a summary is output in a quotation environment - identical to the KOMA -Script classes - whereby the heading is *not* set in the form of an outline level.

However, this environment has the disadvantage that the KOMA script option **parskip=<method>** is not respected. To fix this, the package quoting can be loaded, which

the **quoting** environment is used instead .

The **abstract** option explained above can be used to specify the form in which the summary should be output. Furthermore, each abstract environment can be individually customized using additional parameters as an optional argument. This allows the global settings of the **abstract** option to be changed locally and specifically customized for a specific abstract environment .

If the **babel** package is loaded by the user, the language within the abstract environment can be changed using the optional parameter **language=<language>** . To do this, the desired language must have already been specified when loading **babel**, either as a package option or, better still, as a class option. This means that the name **abstractname** and the separation patterns within the environment are adapted to the language . The desired language can also be passed directly as an optional argument without using the **language** parameter.

v2.02

With **markboth** you can influence the column headings that are set. If **markboth=false** is specified, automatic or manual column headings are used. The setting **markboth=true** sets these to **abstractname for the left and right pages**.

In addition, the column title can be specified directly using **markboth=<column title>** . For example, the columns can also be deleted using **markboth={}** . If **markboth** is activated, the page style **headings** is automatically used in the environment - if a title page (KOMA script option **titlepage=true**) is used.

This can also be specified manually using the **pagestyle** parameter ; the **tudheadings** page styles will also be supported.

If the **multicol** package has been loaded, the summary can be set to multiple columns using the **columns=<number>** parameter. All valid values of the abstract option, as already explained, can be passed to the option parameter. The settings made with this **only** have a local effect on the abstract environment used - in contrast to the late option selection variant³³ or when specifying a class option .

v2.02

If several summaries are to be created and the settings **abstract=single** or **abstract=multiple** as well as **abstract=fill** or **abstract=nofill** are to be observed , the abstract environment should only be used once. Within this, the individual summaries must be separated from each other using **\nextabstract** . The command accepts all parameters in the optional argument that are also supported by the abstract environment itself. The minimal example in subsection 6.2 on page 84 shows the necessary procedure for this.

If the abstract environment is used within the argument of the commands **\setpartpreamble** or **\setchapterpreamble** , the heading is *always* set in text size and centered – unless **abstract=false** is selected .

Renaming the abstract The KOMA -Script

command **\renewcaptionname** can be used to change the identifier – i.e. the wording – of the heading used for the abstract environment. More information can be found in subsection 2.4.

Example: The title of the abstract environment should be renamed from “Summary ” to “Summary” for the language **ngerman** . The macro **\renewcaptionname** expects the three obligatory arguments **{<language>}{<macro>}{<content>}**:

33\TUDOption{abstract}{<setting>} or \TUDOptions{abstract=<setting>}

`\renewcaptionname{ngerman}{\abstractname}{Abstract}`

2.3.2 Declaration of independence and blocking notice

v2.02
v2.04

`declaration=<setting>` Default: **true**

With **declaration**, the design of the declaration of independence and blocking notice can be customized in the same way as the **abstract** option. The **declarations** environment and the **\declaration**, **\confirmation** and **\blocking** commands are provided to output the declarations.

The two options **abstract** and **declaration** are very similar. All possible value assignments for **declaration** have already been explained in detail in the description of **abstract**. This is why this is done here in a somewhat shorter version. If you find the explanation a little lacking, a look at the explanation of the option **abstract** in section 2.3.1 on page 42 will certainly help.

The possible values for the design of the heading are listed below. In contrast to the **abstract** option, the two settings **declaration=true** and **declaration=false** are also available for the tudscrbook class.

declaration=false/off/no

No heading is displayed above the declaration itself. **declaration=true/on/yes** (default value)

A centered heading with the identifier **\confirmationname** before the declaration of independence or **\blockingname** before the blocking notice is set.

declaration=section/addsec

The heading uses the **\section** command. **declaration=chapter/addchap** (default value for tudscrbook) (tudscrreprt and tudscrbook only)

The command **\chapter** is used to set the heading. **declaration=heading** The highest possible level of structure is

used. For tudscrtcl this corresponds to **declaration=section**, for tudscrreprt and tudscrbook this corresponds to **declaration=chapter**.

The behavior for placing an entry in the table of contents is determined depending on the selected structure level of the heading. Normally, an entry for the declaration is only created in the table of contents for headings in the form of a structure level; for **declaration=true** and **declaration=false**, this does not happen by default.

The following values can be used to override the default behavior.

declaration=notoc/nottotoc

The declaration will definitely not be included in the table of contents. **declaration=toc/totoc**

Regardless of the choice of heading, each explanation receives an unnumbered entry in the table of contents at the highest level of the document class used (**\chapter** or **\section**).

v2.02 **declaration=tocleveldown/leveldown/totocleveldown**

The table of contents entry is created one level below the topmost level of the outline (**\section** or **\subsection**). v2.04 **declaration=tocmultiple/**

totocmultiple/tocaggregate/totocaggregate A *single* table of contents entry is created for *all* declarations.

v2.02 Normally, automatic column titles are set depending on the structure level of the heading, if they are activated via the KOMA -Script option **automark** . If manual column titles are used, these must also be set manually for the explanations.
With **declaration=markboth** you can force the column titles to be placed on the left and right pages independently , using the title of the heading.

declaration=markboth

Regardless of the use of manual or automatic running headings, these are set on right and left pages with the identifiers **\confirmationname** or **\blockingname** .

declaration=nomarkboth

The settings for manual or automatic running titles are taken into account.

For **\declaration** , **\confirmation** and **\blocking** as well as the **declarations environment**, any column title can be set using the **markboth** parameter.

The following settings for positioning the explanations only have an effect if the heading of the explanation *is not* output in the form of a chapter and a title page is used with the KOMA-Script option **titlepage=true** .

declaration=single/one/simple Each

declaration is printed on a separate page or in a two-column set in a new column. v2.02 **declaration=multiple/multi/all/aggregate** Declarations that

were set in the **declarations environment** with the commands

\confirmation, **\blocking** and **\declaration** or outside of it with **\declaration** are printed one after the other on the same page if there is enough space on it. If the class option **twocolumn** is activated, all declarations are printed without column breaks.

declaration=fill/fil/vfil/vfill

All declarations on an output page are centered vertically. This setting is not available for two- column typesetting (class option **twocolumn**) . **declaration=nofill/nofil/novfill/novfill**

The output is the same as in normal running text.

v2.02
v2.04

```
\begin{declarations}[<language> || <parameter list>]
. . .
\end{declarations}
\nextdeclaration[<language> || <parameter list>]{<heading>}{<explanation>} language=<language>
(parameter) markboth=<setting>
(parameter) pagestyle=<page style>
(parameter) columns=<number>
(parameter)
option=<setting> (parameter)
supporter=<supporter> (parameter)
place=<place> (parameter)
closing=<end> (parameter)
company=<company> (parameter)
```

For declaration of independence and blocking notice, **\declaration** or **\confirmation** and **\blocking** should be used in the simplest case. However, as soon as these are in

The **declarations environment** offers the necessary freedom to create declarations in a different order, multiple executions, different languages or to add additional declarations.

Within this environment, the declaration of independence and blocking notice can be output directly one after the other using the **\declaration** command , or separately using **\confirmation** and **\blocking** . This can be done in any order and even multiple times in v2.04 , for example to make it multilingual. Furthermore, **\nextdeclaration** offers the **option** of writing a declaration completely freely. This command can *only* be used within the **declarations environment** , whereby the desired heading must be specified in the first argument and the content or text of the declaration itself in the second .

The parameters described below can be used as an optional argument for the **declarations** environment itself as well as for the previously mentioned commands. Like the options of the same name, the **abstract** and **declarations** environments are very similar to each other. The explanations are therefore kept relatively short. If an explanation for a parameter is somewhat incomprehensible, you can read it for the **abstract** environment in section 2.3.1 on page 44.

If the babel package has been loaded, the language – provided it was specified as a package option or, better still, as a class option – can be changed for the **declarations environment** using the **language=<language>** parameter. This will adapt the identifiers – including **\confirmationname** and **\blockingname** – as well as the separation patterns within the environment to suit the language.

The column titles can be influenced with **markboth** . With **markboth= true** for these, **\confirmationname** and **\blockingname** are used on the left and right sides . Automatic (**automark**) and manual (**manualmark**) running headings are used with **markboth=false** . These can be set directly with **markboth=<running title>** . If **markboth** is used in any form, the page style **headings** is set automatically. This can also be specified manually for the environment using the **pagestyle** parameter. If the multicol package has been loaded, the contents of the environment are set in multiple columns using **columns=<number>** . For **option**, all valid values of the **declaration** option can be specified . The use of the other parameters **supporter** , **place** and **closing** can be found in the documentation for the **\confirmation** command ; the **company** parameter is explained for **\blocking** .

v2.02
v2.05

```
\confirmation[<supporter> || <parameter list> ]
\confirmation*[<parameter list>]
supporter=<supporter> (parameter)
place=<place> (parameter)
closing=<end> (parameter)
language=<language> ( parameter)
markboth=<setting> (parameter)
pagestyle=<page style> (parameter)
columns=<number> (parameter)
option=<setting> (parameter)
```

This command outputs a language-specific standard text for a declaration of independence, which is stored in **\confirmationtext** . How this can be adapted or changed can be found in subsection 2.4. It can be used both within the **declarations environment** and outside of it directly in the document.

If **\confirmationtext** is not changed, it can be adjusted using the optional argument of **\confirmation** and the declared parameters. The standard text of the declaration of independence lists both the title and the type of the thesis - if this was specified with **\thesis**, **\subject{<Table 2.1>}** or with the **subjectthesis** option. Other people involved in the work can be specified using the **supporter** parameter or *previously* with the **\supporter** command. Here, too, multiple people to be named must be separated by **and**. The supporters field can also be adjusted in v2.05 using the optional argument alone without specifying a parameter. With the star version **\confirmation***, the supervisors of the thesis defined with **\supervisor{<Name(s)>}** are specified as supporters.

After the actual text of the declaration of independence, the location specified with **place** or **\place** and the date set with **\date** are output. The default **location** is "Dresden". This is followed - with some vertical space for the necessary signature - by the author or authors, specified by the **\author** command. If something else is to be placed after the text of the declaration of independence instead, this can be adjusted using the **closing** parameter or beforehand using the **\confirmationclosing** command. The **language**, **markboth**, **pagestyle**, **columns** and **option** parameters behave in the same way as the **declarations environment**.

v2.02

```
\blocking[<company> || <parameter list>]
company=<company>
(parameter) language=<language>
(parameter) markboth=<setting>
(parameter) pagestyle=<page style>
(parameter) columns=<number> (parameter)
option=<setting> (parameter)
```

The blocking notice is equivalent to the declaration of independence. The standard text stored in **\blockingtext** is output in the selected language. This can be changed as described in subsection 2.4. The **\blocking** command can be used both within the **declarations** environment and directly in the document.

In its original definition, it can be adjusted in the optional argument using the declared parameters. In the standard text of the embargo notice, both the title and the type of the thesis are listed - if this was specified with **\thesis**, **\subject{<Table 2.1>}** or with the **subjectthesis** option. With **company** or *previously* with **\company**, a company or similar to be named in the embargo notice can also be specified. This field can also be defined directly in the optional argument without using a parameter. The other parameters **language**, **markboth**, **pagestyle**, **columns** and **option** behave in the same way as in the **declarations environment**.

v2.02

v2.05

```

\declaration[<parameter list>]
\declaration*[<parameter list>]
language=<language> (parameter)
markboth=<setting> (parameter)
pagestyle=<page style> (parameter)
columns=<number> (parameter)
option=<setting> (parameter)
supporter=<supporter> (parameter)
place=<place> (parameter)
closing=<end> (parameter)
company=<company> (parameter)

```

This command outputs the declaration of independence and the blocking notice one after the other. The settings for positioning the individual declarations, which are made using the assignments **declaration=single** or **declaration=multple** as well as **declaration=fill** or **declaration=nofill**, are taken into account. It can be used both inside the **declarations environment** and outside of it directly in the document and accepts all parameters described for the **declarations environment** v2.05 in the optional argument. The star version forces the supervisors defined with **\supervisor{<Name(s)>}** to be specified in the declaration of independence.

```

\supporter{<supporter>}
\place{<place>}
\confirmationclosing{<end>}
\company{<company>}

```

In contrast to the parameters of **\confirmation** and **\blocking**, these macros change the corresponding field values globally. This avoids specifying a parameter *multiple times*, for example when generating an explanation in different languages.

2.3.3 Bookmarks

```

tudbookmarks=<on-off value>
\tudbookmark[<level>]{<text>}{<anchor name>}

```

Default: **true**

This option takes effect when hyperref has been loaded. Bookmarks or outline entries are created in the PDF document for the cover and title page, the table of contents and – when using the tudscrsupervisor package – the task.

tudbookmarks=false/off/no

No additional bookmarks are entered. **tudbookmarks=true/on/yes** (default value)

Additional bookmarks are automatically added.

The command **\tudbookmark** works in principle in the same way as **\pdfbookmark** from hyperref. However, the bookmarks are only generated if the **tudbookmarks** option is activated.

2.4 Language-dependent identifiers

KOMA -Script provides commands that can be used to create or change language-dependent identifiers. These are provided by the TUD -Script bundle.

to provide localized terms for the *English* and *German* languages . A large part of these are names for fields on the title page (section 2.2.4).

`\providecaptionname{<language>}{<macro>}{<content>}` is used, where `<language>` is the loaded language package – usually the babel package – must be known.

Should the user use the identifiers explained below or other identifiers provided by provided by any (language) package , use the command `\renewcaptionname{<language>}{<macro>}{<content>}` . Of course, a `<language>` should be specified, which is represented in the document by babel or another language pack is used, for example `ngerman` or `english`.

The macros of the identifiers and their use are briefly described below and In this table, an attempt was made to list all the commands of the identifiers for certain Terms should end with `\. . .name` and descriptive texts with `\. . .text` .

v2.02	<code>\refereename</code> (identifier)
v2.05	<code>\refereeothername</code> (identifier)
	<code>\advisorname</code> (identifier)
	<code>\advisorothername</code> (identifier)
	<code>\supervisorname</code> (identifier)
	<code>\supervisorothername</code> (identifier)
	<code>\professorname</code> (identifier)
	<code>\professorothername</code> (identifier)

These language-dependent terms are the identifiers for the title page fields of Supervisor (**\supervisor**), assessor (**\referee**) and subject consultant (**\advisor**). Should within one of these If more than one person is specified in one of the fields, the individuals are each identified with the command `\and` . In this case, all characters after the first following persons are supplemented by the identifier `\. . .othername` .

The designation distinguishes whether one or more persons are specified If only one person is mentioned, a distinction is not necessary and the singular is used. However, if two or more persons are specified, it is checked whether the corresponding identifier for the second person (**\. . .othername**) is defined. If this so, the alternative name for the first named person will be used, otherwise the plural of the identifier is used. This applies to all fields that are accessed via `\referee`, `\advisor`, `\supervisor` or `\professor` were specified.

Identifier	German	English
Supervisor	(Primary)	(First) Supervisor(s)
<code>\supervisorname</code> <code>\supervisorothername</code> <code><empty></code>		<code><empty></code>
<code>\refereename</code>	(First) Reviewer	(First) Referee(s)
<code>\refereeothername</code>	Second reviewer	Second referee
<code>\advisorname</code>	(First) subject consultant(s)	(First) Advisor(s)
<code>\advisorothername</code>	<code><empty></code>	<code><empty></code>
<code>\professorname</code>	Supervising professor(s)	
<code>\professorothername</code> <code><empty></code>		<code><empty></code>

`\dissertationname` (identifier)
`\diplomathesisname` (identifier)
`\masterthesisname` (identifier)
`\bachelorthesisname` (identifier)
`\studentthesisname` (identifier)
`\studentresearchname` (identifier)
`\projectpapername` (identifier)
`\seminarpapername` (identifier)
`\termpapername` (identifier)
`\researchname` (identifier)
`\logname` (identifier)
`\internshipname` (identifier)
`\reportname` (identifier)

These identifiers are used for typing specifically for a thesis. How these are used can be explained in the explanation of **subjectthesis** or **\thesis** and **\subject** can be found in section 2.2.4 on page 37.

Identifier	German	English
<code>\dissertationname</code>	dissertation	dissertation
<code>\diplomathesisname</code>	Diploma thesis	Diploma Thesis
<code>\masterthesisname</code>	Master thesis	master thesis
<code>\bachelorthesisname</code>	Bachelor thesis	Bachelor Thesis
<code>\studentresearchname</code>	Major document	Student Research Project
<code>\projectpapername</code>	Project paper	Project Paper
<code>\seminarpapername</code>	Seminar paper	Seminar Paper
<code>\researchname</code>	Research Report	Research Report
<code>\logname</code>	protocol	log
<code>\internshipname</code>	Internship Report	Internship Report
<code>\reportname</code>	report	report

v2.02

`\graduationtext` (identifier)

If it has been recognized that the current document is a thesis, the degree to be obtained academic degree can be specified with the command **\graduation**. When output to The corresponding text is given in the title.

Identifier	German	English
<code>\graduationtext</code>	to obtain the academic degree	to achieve the academic degree

`\datetext` (identifier)
`\defensedatetext` (identifier)

If **\date** is used to specify the (submission) date and **\defensedate** is used to specify the date of the defense for If a thesis is specified, these fields are also described by text.

Identifier	German	English
<code>\datetext</code>	Submitted on	
<code>\defensedatetext</code>	Defended on	Defended on

v2.02

\dateofbirthtext (identifier)
\placeofbirthtext (identifier)
\matriculationnumbername (identifier)
\matriculationyearname (identifier)
\coursename (identifier)
\disciplinename (identifier)

If the date of birth of the author or authors is specified using the appropriate command (**\dateofbirth**), the place of birth (**\placeofbirth**), the course of study (**\course**), the field of study (**\discipline**) or the matriculation number (**\matriculationnumber**) and/or the year of enrollment (**\matriculationyear**) are indicated on both the title page and

on the task created with tudscrsupervisor, if necessary, the corresponding identifiers. On the title, these are separated by the **\titledelimiter** separated from the actual field by the given separator.

Identifier	German	English
\dateofbirthtext	Born on in	Born on
\placeofbirthtext		in
\matriculationnumbername	Matriculation number	Registration number
\matriculationyearname	Matriculation year	
\coursename	Study program	Course
\disciplinename	Field of study	Discipline

\coverpagename (identifier)
\titlepagename (identifier)

These two identifiers are used to enter bookmarks in a PDF document when **tudbookmarks** is activated.

Identifier	German	English
\coverpagename	Cover page	
\titlepagename	title page	Title page

\abstractname (identifier)

This identifier is defined for the class tudscrbook, since it is used by KOMA-Script for the book class is not provided.

Identifier	German	English
\abstractname	Summary Abstract	

v2.02

\confirmationname (identifier)
\blockingname (identifier)

The terms for declaration of independence and blocking notice are defined for the corresponding headings.

Identifier	German	English
\confirmationname	Declaration of authorship	
\blockingname	Blocking notice	Restriction note

v2.02

\confirmationtext (identifier)**\blockingtext** (identifier)

The texts of the declarations themselves are structured in such a way that they are carried out differently depending on the information provided. Within the declaration of independence (**\confirmation**), the fields for the title (**\title**) and the type of the thesis as well as the specified sponsors³⁴ are taken into account where appropriate. For the blocking notice (**\blocking**), the field for the external company³⁵ is optionally used in addition to the title (**\title**). For the sake of completeness, the texts for the declaration of independence and the blocking notice are listed below - but only the German version. All possible fields are displayed.

Identifier **\confirmationtext**

I hereby certify that I have written this document entitled **<Title>** independently and without unauthorized assistance from third parties. No resources or sources other than those specified in this document were used.

I have marked the quotations that have been used verbatim and in essence as such. During the preparation of this document, I was only supported by the following people:

<First Name Last Name>

<First Name Last Name>

No other persons were involved in the intellectual creation of this document. I am aware that failure to comply with this declaration may result in the subsequent withdrawal of my university degree.

Identifier **\blockingtext**

This document entitled **<Title>** contains confidential information disclosed by **<Company>**.

Publication, reproduction and inspection - even in part - are not permitted without the express permission of **<Company>**, nor are publications of the contents of this document.

It is to be made available only to the supervisor at the Technical University of Dresden, the examiners and the members of the examination board.

If one of the declarations is to be changed and the content of a field is to be used, **\getfield** can be used.³⁶ If necessary, the definition of **\and** must be adapted.

\listingname (identifier)**\listlistingname** (identifier)

If a package is used to include external source code - for example the listings package - these names are used for source code excerpts and the source code directory.

Identifier	German	English
\listingname	Source code	Listing
\listlistingname	Source directory	List of Listings

³⁴**\confirmation[supporter=<supporter>]** or **\supporter{<supporter>}**

³⁵**\blocking[company=<company>]** or **\company{<company>}**

³⁶Title: **\getfield{title}** ÿ **<Title>**, Type of thesis: **\getfield{thesis}** ÿ **<Thesis>**,

Author: **\getfield{author}** ÿ **<First name Last name>**, Company: **\getfield{company}** ÿ **<Company>** and also

Supporter: **\getfield{supporter}** ÿ **<Firstname Lastname>**

2.5 Compatibility settings for previous versions

When developing TUD-Script, it is not always possible to avoid improvements and corrections to classes and packages leading to changes in the output, especially in terms of pagination and layout. However, for documents that have already been archived and were created with an earlier version, this may be undesirable when recompiling .

v2.03

tudscrver=<Version> Default: **last**

This option makes it possible to switch to the (page-wrapping) behavior of an older or earlier version of TUD-Script in order to obtain the expected result after compilation . New options that do not affect page-wrapping or layout are also available in the event that compatibility with an older version has been selected via option.

If an unknown version is specified as a value, a warning is issued and **tudscrver=first** is assumed. With **tudscrver=last**, the currently available version is selected and future compatibility of the document with the currently used version is therefore not ensured. This behavior corresponds to the default setting. Please note that **tudscrver** can only be used as a class option.

v2.03 **tudscrver=first/2.02**

The type area in the corporate design layout (see **cdgeometry**) was slightly changed in version v2.03. The upper page margin was reduced and the lower one enlarged by the same amount. The available text area is therefore identical. When activating the DRESDEN concept logo in the footer of the page (see **ddcfoot**), the same type area is used as in version v2.02. This behavior can be deactivated with **tudscrver=first** .

v2.04 **tudscrver=2.03**

Since version v2.04, several lengths are defined depending on the selected font size. By selecting **tudscrver=2.03** this functionality is deactivated, whereby only the TUD script option **relspacing=false** is activated.

v2.05 **tudscrver=2.04**

With version v2.05, the default settings for the type area are made based on the B ISO/DIN series . This means that the page margins are identical for all paper formats of a specific ISO/DIN class . If you select **tudscrver=2.04**, the type area depends on the A ISO/DIN series , so that the B and C paper formats of the same class have larger page margins than the D and A formats.

v2.06 **tudscrver=2.05**

The change of the house font to **Open Sans** resulted in some changes to the appearance of the corporate design. This compatibility setting activates the old font families **Univers** and **DIN Bold** (**cdoldfont=true**, **headings=light**, **ttfont=lmodern**) . Furthermore, for these font families in math mode, the option **slantedgreek=true** only affects the Greek majuscules.

tudscrver=2.06

This is a compatibility setting for TUD-Script v2.06o and is already kept for future changes. If a document created with the current version should also produce the same output result after a LATEX run with a later version of TUD-Script , this must be specified with **tudscrver=2.06** .

tudscrver=last

No compatibility settings are made for the document. With a later version of TUD-Script, a different pagination behavior within the document may occur. This is the default setting.

Chapter 3

The poster class

v2.05 In addition to the main classes offered for typesetting documents in the corporate design of the Technical University of Dresden, the class `tudscrposter` is provided. This enables the creation of posters in the same layout with LATEX .

The basis for this is `tudscrartcl` and *almost* all commands and options offered by this class can be used equally with `tudscrposter` . A minimal example of using the class can be found in subsection 6.7 on page 95.

The biggest difference to the main classes is a simplified title. A cover page is not available for `tudscrposter` , the corresponding option **`cdcover`** and the associated command **`\makecover`** are not defined. The title itself can only be set as a title header with **`\maketitle`** ; a separate title page does not exist. For this reason, only a reduced number of commands are available for the title: **`\title`**, **`\subtitle`**, **`\subject`** and **`\titlehead`** , which can be used as usual.

In principle, the two commands for the half-title **`\extratitle`** and **`\frontispiece`** can also be used, although their use for a poster is rather questionable.

All other commands and options presented in section 2.2.4 are not defined for `tudscrposter` . This applies to the **`thesis`** font element as well as the commands **`\titledelimiter`**, **`\thesis`**, **`\referee`**, **`\advisor`**, **`\graduation`** and **`\defensedate`**. The macros for additional author information **`\dateofbirth`**, **`\placeofbirth`**, **`\matriculationyear`** and **`\matriculationnumber`** as well as the **`subjectthesis`** option are *not* available. The commands **`\date`**, **`\publishers`** and **`\dedication`** as well as **`\uppertitleback`** and **`\lowertitleback`** provided by KOMA-Script for a title page have no functionality in the `tudscrposter` class . The commands **`\author`** and **`\authormore`** still exist , but are not used for the title but for the special footer of a poster, which is described in more detail in subsection 3.2.

In addition to the significant simplification of the title, a number of other commands and environments are no longer required for the `tudscrposter` class. These include the **`tudpage environment`**, the **`headingsvskip`** and **`pageheadingsvskip`** options , and all elements relating to declaration of independence and blocking notices, such as the **`(declaration)`** option , the **`(declarations)`** environment , and the commands **`(\declaration, \confirmation, \blocking)`**. The **`abstract`** environment can still be used, but the **`abstract`** option can only be used to adjust the structure level of the heading.

3.1 Layout and formatting of a poster

The most obvious settings for designing a poster are certainly the paper format used and the color, as well as the choice of font and its size. The default setting for the `tudscrposter` class is to activate the fonts of the corporate design of the Technical University of Dresden. These can be customized - as with the main classes. For more information, see the explanation of the **`cdfont`** option in section 2.1.1 on page 16.

The following briefly explains how to customize the general formatting of a poster created with `tudscrposter` . The content can be designed completely freely; there are currently no predefined commands or options from TUD-Script that translate the content into a specific layout.

Table 3.1 Recommended combinations for selecting paper size (**paper**) and Font **size**

a) Single column layout

Class								
6	5	4	3	2	1	0		
Row min. . .max	min. . .max	min. . .max	min. . .max	min. . .max	min. . .max	min. . .max	min. . .max	min. . .max
D 05. . .07pt	06. . .09pt	10. . .14pt	14. . .20pt	20. . .29pt	28. . .40pt	40. . .60pt		
A 06. . .08pt	07. . .10pt	11. . .16pt	16. . .23pt	23. . .33pt	32. . .46pt	45. . .66pt		
C 07. . .09pt	08. . .11pt	12. . .18pt	18. . .26pt	26. . .37pt	36. . .52pt	50. . .72pt		
B 08. . .10pt	09. . .12pt	13. . .20pt	20. . .29pt	29. . .41pt	40. . .58pt	55. . .78pt		

b) Two-column layout

Class			
3	2	1	0
Row min. . .max	min. . .max	min. . .max	min. . .max
D 07. . .10pt	10. . .14pt	14. . .19pt	19. . .28pt
A 08. . .11pt	11. . .16pt	16. . .22pt	22. . .32pt
C 09. . .12pt	12. . .18pt	18. . .25pt	25. . .36pt
B 10. . .13pt	13. . .20pt	20. . .28pt	28. . .40pt

c) Three-column layout

Class	
1	0
Row min. . .max	min. . .max
D 07. . .10pt	10. . .14pt
A 08. . .11pt	11. . .16pt
C 09. . .12pt	12. . .18pt
B 10. . .13pt	13. . .20pt

3.1.1 Choosing paper format and font size

Specifying the *paper format* and *font size* is essential for creating a poster and should **always be** done. Both *settings must be made as class options*. When selecting the font size, it is important to consider whether the poster should be set in one or more columns. For the latter, the multicols environment from the multicols package is highly recommended.

To specify the paper format, use the KOMA - Script option ***paper=<setting>***. With ***paper=<format>***, you can select the common classes of the ISO/DIN series A to D as well as landscape or portrait format. Any format can be selected with the setting ***paper=<height>:<width>***. For additional information, see the KOMA-Script manual³⁷ to consult.

To match the selected paper format and the desired number of text columns on the poster, the font size should be specified using ***fontsize=***.

To select the right paper format and font size, use Table 3.1 as a reference. If you receive one or more warnings of the type

Font shape 'T1/cmr/m/n' in size <...> not available

please note the information in subsection 9.24.

³⁷<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

3.1.2 The design of a poster

The color of a poster is determined using the **cd option**, which is described below . You can choose from several color options. If none of these predefined values provide the desired layout, you can subsequently adjust it more precisely using the **cdhead** and **cdfoot** options as well as **cdtitle**, **cdpart** and **cdsection** .

cd=<setting> Default: **bicolor**

Equivalent to the TUD script main classes, this option specifies the use of the corporate design of the Technical University of Dresden for the poster. It influences the color design of the outline headings and the page style, which is set to **empty.tudheadings** by default.

cd=false/off/no

This completely deactivates the corporate design and no specific settings are made for a poster. Only the page style is set to **empty** . **cd=true/on/yes/nocolor/monochrome** (default value)

Black font is used for headings and the page header. The footer area is not accentuated in color.

cd=lightcolor/pale The setting

is largely the same as

the option **cd=true**, but the primary house color **HKS41** is used for the header, footer and headings.

cd=barcolor

In addition to the previous setting, the crossbar is also highlighted in color.

cd=bicolor/color/fullcolor

The header is set with a colored background in the primary house color **HKS41** , the horizontal bar is set off in color. The house color is also used for all headings , the footer area also has a colored background.

backcolor=<setting> Default: **true**

This option can be used to define the background color of a poster.

backcolor=false/off/no/nocolor

No color is specified, the background appears white. **backcolor=true/on/yes/color** (default value)

The page background is chosen in the primary house color **HKS41** .

backcolor=<color>

The specified **<color>** will be used as the background for the poster.

bleedmargin=<length value> Default: **0.2in**

If the poster is to be printed in a paper format that is then cut to the target format because, for example, borderless printing is not possible, this option can be used to enlarge the colored elements of the layout into the area of the bleed or overprint. This makes "cutting to the color" very easy and can be achieved without any major problems.

The elements that depend on the setting **bleedmargin=<length value>** are the header and footer areas, influenced by the options **cdhead** and **cdfoot**. If these are set in color, they are extended by the specified **<length value>** beyond the desired

The page background, which may have been set with **backcolor** , is also enlarged. Subsection 9.21 on page 124 shows an example of how a poster can actually be designed in a specific target format on an oversized sheet of paper.

3.2 Fields for the footer area

The footer area of a poster can be defined individually and freely using **\footcontent** . If this is not done, a predefined footer is set by default, which outputs information from certain fields that are intended in particular as contact information. What these are in detail is explained below. The color of the footer is determined by the **cdfoot** option .

```
\faculty[<footer>]{<faculty>}
\department[<footer>]{<institution>}
\institute[<footer>]{<institute>}
\chair[<footer>]{<chair>}
```

The information provided with these commands is not only displayed in the header but also in the left part of the footer. If you want to adapt this for the footer, you can use the optional argument for this, whereby specifying an empty optional argument suppresses the field for the footer completely. The identifier **\contactname** is displayed in bold before all information.

```
\professor{<name>}
```

In addition to the information about the institution, the current holder of the professorship can be specified with **\professor** in the left footer.

```
\author{<author(s)>}
\contactperson{<name(s)>}
\authormore{<author additional
information>}
\course{<course of study>}
\discipline{<field of study>}
\office{<address/ building>}
\telephone{<telephone number>} \telefax{<fax number>}
```

The author(s) specified with **\author** are output in the right part of the footer (one after the other); multiple authors are separated with **and** . The commands **\authormore**, **\course** and **\discipline** as well as **\office**, **\telephone**, **\telefax** and **\emailaddress** can be used for additional information about each author within the argument of **\author** . Before all author information is output, the identifier **\authorname** is printed in bold. Without the specification of **\author**, no output is given.

This is followed by all the information provided with **\contactperson** . Here, too, **and** is used to separate multiple people, whereby only the commands **\emailaddress**, **\office**, **\telephone** and **\telefax** are to be used , but not **\authormore** , **\course** and **\discipline** for additional information. Before the contact persons are output, the identifier **\contactpersonname** is set in bold. It is of course also possible to enter only author(s) or contact persons.

```
\webpage[<Settings>]{<URL>}
\webpage*{<URL>}
```

Finally, a homepage can be specified for the right column of the footer area. If the `hyperref` package has been loaded, this will be converted into a hyperlink. the optional argument can be any settings to **`\hypersetup`** from said package. If the formatting of the entry is to be done manually, the star version **`\webpage*`** can be used, whereby all desired settings are within the Arguments – if necessary in a group – must be discussed.

3.3 Language-dependent identifiers for the footer area

```
\contactname (identifier)
\authorname (identifier)
\contactpersonname (identifier)
```

As already explained, these identifiers are displayed in the left and right columns in the Foot set before the output of the actual fields.

Identifier	German	English
<code>\contactname</code>	Contact	Contact
<code>\authorname</code>	Author	Author
<code>\contactpersonname</code>	Contact Counterpart	

Chapter 4

Supervision of student work

This package provides the user with suitable environments and commands for creating tasks and reports for scientific papers as well as official notices in the corporate design. It is therefore primarily aimed at employees at the Technical

University of Dresden, but can of course also be used by students.

4.1 Task for a scientific paper

v2.05

```
\begin{task}[<heading> || <parameter list>]
. . .
\end{task}
headline=<heading> (parameter)
style=<style> (parameter)
```

(**tudpage** in section 2.2.3)

The task environment can be used to output a task for a scientific paper. This is based on the **tudpage** environment and therefore accepts all parameters in the optional argument that were explained in the description of **tudpage** in section 2.2.3 on page 32.

A heading is usually set for the task, which consists of **\taskname** and – if the type of thesis was specified – **\tasktext** and **\thesis**. The **headline** parameter can be used to customize this automatically generated heading.

At the beginning of the task, a table appears with the specified information on the author or authors of the thesis. The only mandatory information that needs to be provided is the name of the author (s) (**\author**) and the title of the thesis (**\title**), which is listed in bold at the end of the table. Optionally, the fields for the course of study (**\course**), the subject area (**\discipline**), the matriculation number (**\matriculationnumber**) and the year of enrollment (**\matriculationyear**) are also output; fields not specified are ignored during output. The actual content of the environment - i.e. the task itself - is output after the generated header. The content of the task is followed by a line-by-line list of the defined reviewer or examiner (**\referee**) and supervisor (**\supervisor**). The corresponding identifier (**\refereename**, **\refereeothername** v2.05 or **\supervisorname**, **\supervisorothername**) is placed in front of the respective name. This is the default behavior and can be activated by selecting the parameter **style=table**. If, however, the parameter **style=inline** is set, the output of several examiners and supervisors is shown in one line. The identifiers can be easily adapted, see subsection 2.4.

The issue date (**\issuedate**) and the mandatory submission date (**\duedate**) then appear. Finally, the signature line for the chair of the examination board (**\chairman**) and the supervising professor (**\professor**) is entered. For named persons, the identifiers are displayed under the name itself (**\chairmannname** and **\professorname**).

```
\taskform[<parameter list>]{<goals>}{<focus>}
```

In addition to the freely configurable **task** environment for creating a task, a separate command is provided for a standardized output. This structures the task into two areas: *objectives* and *focus* of the work with corresponding headings (**\objectivesname**, **\focusname**).

All parameters of the **task** environment can be used in the optional argument. The first obligatory argument should contain a text with a brief thematic classification and the actual goal of the work, and the second argument should name the thematic focus in bullet points. The content of the second required argument is set in an itemize environment. Therefore, each bullet point *must* be preceded by **litem**.

Example: The recommended usage of the **taskform** command is as follows:

```
\taskform{%
  Motivation for the work in the first paragraph. .
  .
  Aims of the work in the second paragraph. . }% litem
  .
  Focus 1 litem Focus 2
}
```

Please also refer to the minimal example in subsection 6.4.

```
\chairman{<Chairman of the Audit Committee>}
```

If this field is used, the chair of the examination board will be listed at the end of the assignment in addition to the supervising professor (\professor). This is usually required **for** final papers such as master's theses or diploma theses.

```
\issuedate{<issue date>} \duedate{<due
date>}
```

These two commands should be used to specify the date of the assignment and the latest possible submission date. If the isodate or datetime2 package has been loaded, the output formatting of the date set with **\printdate** or **\DTMDate** for **\duedate** and **\issuedate** is used.

4.2 Reports for scientific papers

```
\begin{evaluation}[<heading> || <parameter list>]
. . .
\end{evaluation}
headline=<Headline> (parameter) grade=<Grade>
(parameter)
```

(tudpage in section 2.2.3)

This environment is provided for creating a report of a scientific paper. It also supports all parameters that were described for the **tudpage** environment in section 2.2.3 on page 32.

For an evaluation, a heading is usually generated from **levaluationname** and – if the thesis type was specified – **levaluationtext** and **lthesis**. This automatically generated heading can be replaced with the **headline** parameter. At the end of the evaluation, the **grade** given is highlighted in bold.

At the beginning of the **evaluation environment**, the same table with author information is displayed as is the case with the task environment. After the table header, the actual content follows here too, i.e. the assessment of the final thesis. The environment is completed

with the given grade – which is output within **\gradetext** – as well as the location and date (**\place**, **\date**) and the following signature line for the referee(s) (**\referee**), which in turn are supplemented with the corresponding language-dependent identifiers (**\refereename**, **\refereeothername**) .

```
\evaluationform[<parameter list>]{<task>}{<content>}{<evaluation>}{<grade>}
```

In addition to the individually usable **evaluation** environment , a separate command is provided for creating a standardized assessment. This structures the output into the four areas of *task*, *content*, *assessment* and *grade* and provides each with the corresponding heading or text output (**\taskname**, **\contentname**, **\assessmentname** and **\gradetext**). The optional argument supports all parameters of the evaluation environment.

Example: The recommended use of the **\evaluationform** command is as follows: **\evaluationform**{%

Short description of

the task. . }% Summary of content and structure. . }

Evaluation of the written final paper. . }% Number (grade) .

}

Please also refer to the minimal example in subsection 6.5.

```
\grade{<Grade>}
```

In addition to specifying the grade for an assessment using the **grade** parameter of the **evaluation** environment , this global command can also be used.

4.3 Notice

```
\begin{notice}[<heading> || <parameter list>]
. . .
\end{notice}
headline=<headline> (parameter)
```

(**tudpage** in section 2.2.3)

This environment can be used to create a poster. It is again based on the **tudpage** environment and supports all of its parameters.

If a date is specified, it is displayed right-aligned above the text area by default (see **cdhead**). The headline is then displayed, which usually corresponds to the content of **\noticename** and can be changed using **headline** .

The content of the environment follows the heading. If one or more contact persons were specified with **\contactperson** , this information is placed at the end of the environment.

```
\noticeform[<parameter list>]{<content>}{<focus>}
```

There is also a command for a standardized form for this environment. This is primarily intended for use in posting student work topics. All parameters that the notice environment supports can be used for the optional argument.

The first obligatory argument is used for a short description of the content. In addition to the text, a thematically appropriate illustration should be inserted here if possible (**\includegraphics**). The second argument is used - as with **\taskform** - to list some key points. Here too, an itemize environment is used after the structuring heading (**\focusname**) , and all key points must be preceded by **\item** .

Example: The recommended usage of the **\noticeform** command is as follows:

```
\noticeform{%
  Short description of the content of the student work. .

  Image (optional), to be included with:
  \includegraphics[<settings>]{<file>}
}%
  \item center of gravity
  1 \item center of gravity 2
}
```

Please also refer to the minimal example in subsection 6.6.

v2.02
v2.05

\contactperson{<contact person(s)>}
\office{<office address>}
\telephone{<telephone number>}
\telefax{<fax number>}

At the end of a notice, contact information for one or more contacts can be specified using **\contactperson** . If more than one contact person is to be named , they must be separated within the **\contactperson** command using the **\and** command. For each person, the office location (**\office**), the work telephone number (**\telephone**) and fax number (**\telefax**) and the work email address (**\emailaddress**) can be specified within the argument of **\contactperson** .

4.4 Additional language-dependent identifiers

For the tudscrsupervisor package, additional identifiers are defined for the additional commands and environments . For any adaptation of these, please refer to subsection 2.4.

\taskname (identifier)
\tasktext (identifier)

The name of the task itself is contained in **\taskname** . This is used to generate a heading. If the type of the thesis38 was also specified with **\thesis** or **\subject** , the heading is extended with the typification together with the identifier **\tasktext** . If desired, the automatically generated heading can be overwritten with the **headline** parameter of the **task** environment .

Identifier German	English
\taskname Task \tasktext for the preparation of a	Task

38subjectthesis or specific value from Table 2.1

v2.04

\namesname (identifier)**\titlename** (identifier)

These two identifiers are used in the author information table at the beginning of the task is used.

Identifier	German	English
\namesname	Name	Name
\titlename	Title	Title

\issuedatetext (identifier)**\duedatetext** (identifier)

At the end of the task, the supervisor will provide the issue date and Submission date (**\issuedate**, **\duedate**) of the thesis is explained with the following identifiers.

Identifier	German	English
\issuedatetext	Issued on	
\duedatetext	Due date for submission	

\chairmannname (identifier)

If the chairperson of the audit committee (**\chairman**) has been specified, the name even the output of the identifier.

Identifier	German	English
\chairmannname	Chairman of the Audit Committee	

\focusname (identifier)**\objectivesname** (identifier)

The standard forms for task (**\taskform**) and notice (**\noticeform**) use these identifiers for the headings.

Identifier	German	English
\focusname	Focus of work	
\objectivesname	Objectives of the work	Objectives of work

\evaluationname (identifier)**\evaluationtext** (identifier)

The name of the report itself is contained in **\evaluationname** . For the generation The title uses the identifier **\evaluationtext** and the type of thesis specified with **\thesis** or, if applicable, with **\subject** . This automatically generated headline can be changed with the parameter **headline** of the environment **evaluation** by be overwritten by the user.

Identifier	German	English
\evaluationname	Report Evaluation	
\evaluationtext	for the	for the

\contentname (identifier)

\assessmentname (identifier)

In the standardized form of the report (**\evaluationform**), the information contained therein – for a structured outline – generated headings with the identifiers **\taskname**, **\contentname** and **\assessmentname** are set.

Identifier	German	English
\taskname	Task	
\contentname	Contents	Content
\assessmentname	assessment	Assessment

\gradetext (identifier)

For the assessment of a scientific paper, the grade achieved is either given as

Command **\grade{<Grade>}** or alternatively with the parameter **grade=<Grade>** of the environment **evaluation** is specified, it will be used within **\gradetext** .

Identifier	German	English
\gradetext	The work is graded <Grade> rated.	The thesis is evaluated with a grade of <Note>.

v2.02

\noticename (identifier)

\contactpersonname (identifier)
--

The name of the notice itself is contained in **\noticename** . For the generation This is used for a heading. If desired, this can be modified with the parameter **headline** of the environment **notice** can be overwritten. If a contact person has been assigned to the Command **\contactperson** is specified, the header of the contact data is the identifier **\contactpersonname** is used.

Identifier	German	English
\noticename	Notice	Notice
\contactpersonname	Contact Counterpart	

Chapter 5

Additional packages in TUD script bundle

In addition to the classes and packages presented so far in the user manual, further packages are provided in the TUD -Script bundle. These are not necessarily dependent on the use of one of the TUD -Script classes but can in principle be used with any LATEX 2 ϵ class.

5.1 The tudscrcolor package – colors in corporate design

The tudscrcolor package is used to use the colors of the corporate design . If this is not loaded in the preamble - for example, to call up additional options - the TUD script classes include it automatically.

Several colors are provided for the corporate design. The most striking of all is the house color **HKS41**, followed by the colors for awards in the first (**HKS44**) and second category (**HKS36**, **HKS33**, **HKS57**, **HKS65**) and an exception color (**HKS07**). These colors can be used both in their basic form and in lighter tones with a gradation in 10 % increments. The xcolor package, which is used anyway, provides exactly this functionality. Each of the colors can be used both with **HKS<number>** and via a pseudonym **cd<color>** .

Example: The basic color **HKS44** is to be used in a lighter shade reduced to 20% . Within a command that expects a valid color as an argument, only **HKS44!20** must be specified. This is demonstrated here with the following box `\colorbox{HKS44!20}{Box}` .



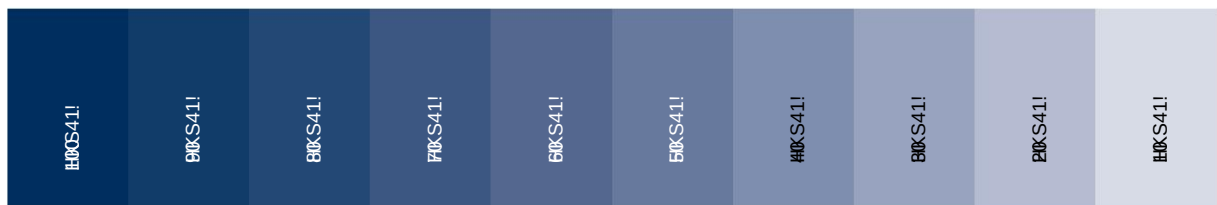
In the color design of the corporate design (**cd=color**), the background of the cover page, title and parts is in **HKS41** and the font on the same is in **HKS41!30** . The background of chapter pages appears in **HKS41!10**, the font in **HKS41**. If less color is used, only the fonts on the outline pages are set to **HKS41** .

If certain options are to be passed to the xcolor package , there are two ways to do this . These can either be passed directly to xcolor before loading the class39 or tudscrcolor is loaded with the corresponding option.40

5.1.1 General color definitions

Primary house color

HKS41 (cddarkblue) (color)



Secondary house colour (business equipment)

HKS92 (cdgray) (color)



39\PassOptionsToPackage{<package options>}{xcolor} before loading the used document class with
\documentclass[<class options>]{<class>}

40\usepackage[<package options>]{tudscrcolor}; tudscrcolor passes <package options> to xcolor

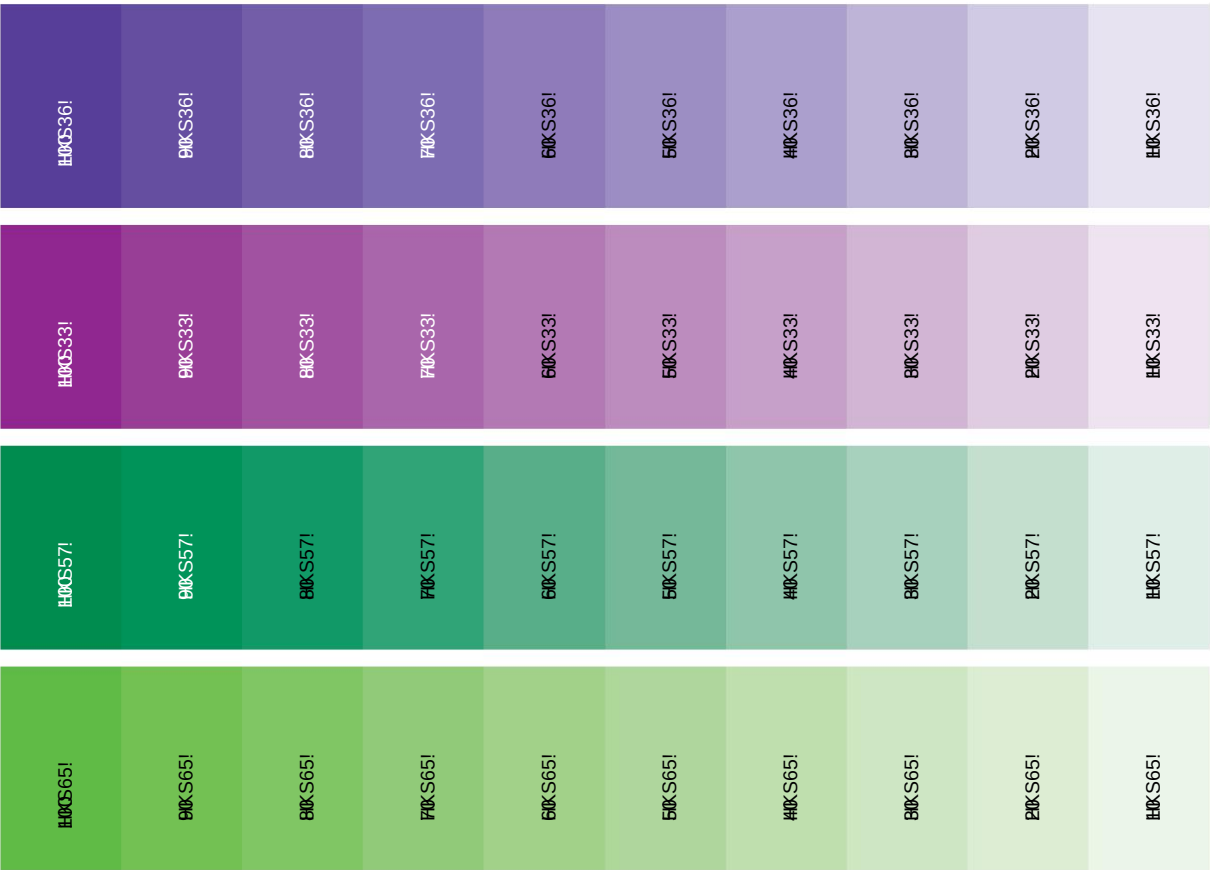
Award color 1st category

HKS44 (cdblue) (color)



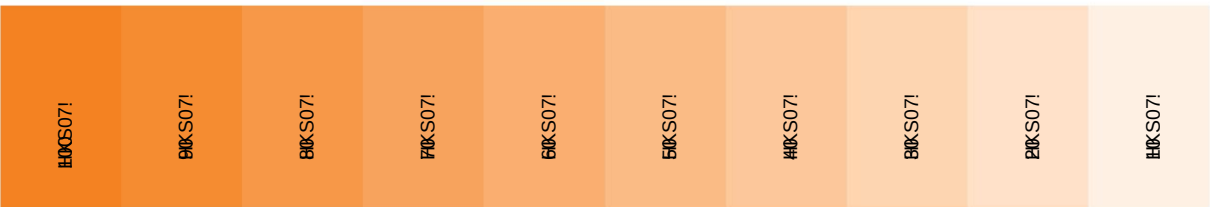
Award color 2nd category

HKS36 (cdindigo) (color)
HKS33 (cdpurple) (color)
HKS57 (cddarkgreen) (color)
HKS65 (cdgreen) (color)



Exceptional color

HKS07 (cdorange) (color)



5.1.2 Additional color definitions

The tudscrcolor package normally only defines the previously described basic colors **HKS41**, **HKS92**, **HKS44**, **HKS36**, **HKS33**, **HKS57**, **HKS65** and **HKS07**. All other color gradations can be generated using the options described in the xcolor package.

oldcolors

In recent years, many different classes and packages have been created for the corporate design of the Technical University of Dresden. Within these, there are different

Color definitions. To enable migration from the named classes and packages to TUD-Script, the package option **oldcolors** exists. If this is used, additional colors are defined according to the scheme **HKS41K<number>** and **HKS41-<number>**, whereby the numerical value at the end must come from the 10s series.

5.1.3 Changing the color model

Normally, tudscrcolor uses the CMYK color model. In addition, the RGB color space is supported. Switching the color model is necessary for certain functions of the tikz package, for example.

RGB

With this option, when loading the tudscrcolor package, the colors are defined globally in the RGB color space rather than according to the CMYK color model.

\setcdcolors{<color model>}

This command can be used to adjust the color model used within the document. This makes it possible to change the color mode locally within an environment and thus only switch from the CMYK color model to the RGB color space in certain situations, for example. Supported values for **<color model>** are **CMYK** and **RGB** or **rgb**.

Attention! *The representation of the colors in the respective color mode (**CMYK** or **RGB**) can vary depending on the screen, printer and software configuration used. The color values come from the corporate design manual and are only approximate values. Deviations from the printed HKS color register and self-determined values are technically unavoidable.*

5.2 The tudscrfonts package – fonts in corporate design

v2.02 This package provides the fonts of the corporate design for LATEX 2_ε classes that are *not* part of the TUD -Script bundle. The package supports most of the options and commands for font selection that are normally provided for the TUD -Script classes. To avoid duplication in the documentation, the options and commands available in the tudscrfonts package will not be explained again here. They are simply mentioned again below; the corresponding explanations can be found in subsection 2.1 on page 15.

The package options that can be used are **cdfont** for the body text – without the setting options for the corporate design horizontal bar (**cdhead**) – and **cdmath** and **slantedgreek** for the mathematical fonts. **ttfont** is available for the selection of typewriter fonts. The **relspacing** option is also provided. All of the options mentioned can be used either directly as class options or as package options in the optional argument

of `\usepackage[<package options>]{tudscrfonts}`. In addition, after loading the package, the options can be selected with `\TUOption` or `\TUOptions`.

The text switches and commands described in subsection 2.1 for the explicit selection of individual styles of the corporate design font as well as the commands for Greek v2.06 letters are also provided. If you do not want to use **Open Sans** but the old **Univers** and **DIN Bold** font families, you will need to install them. Information on this can be found in subsection A.4. To activate these, refer to **tudscrver=2.05** and **cdoldfont**.

v2.04
v2.05

Originally, the tudscrfonts package was intended for use with one of the classes tudbook, tudbeamer, tudletter, tudfax, tudhaus, tudform and, since version v2.04, also tudmathposter and tudposter. However, some minor issues arose when using the package with one of these classes. Therefore, since version v2.05, it is recommended to use the fix-tudscrfonts package for these classes.

5.3 The fix-tudscrfonts package – font compatibility

v2.05

This package is intended for use solely with one of the following classes:

- tudbook
- tudbeamer
- tudletter
- tudfax
- tudhaus •
- tudform •
- tudmathposter •
- tudposter

v2.06

For the corporate design of the Technical University of Dresden, the TUD script bundle uses **Open Sans** as the default font family. This differs greatly from the font families originally used by the classes just mentioned. If the fix-tudscrfonts package is used together with one of these classes, this has the advantage that the current corporate font is also used for these and the greatly improved math typesetting

Attention! come into play. *In this case, the output result may differ compared to the version with the old fonts. To set documents with the old **Univers** and **DIN Bold** font families, these must be installed. Information on this can be found in subsection A.4.*

To activate this, see **tudscrver=2.05** and **cdoldfont**.

In order to make all necessary settings correctly and without unnecessary warnings, the fix-tudscrfonts package must be loaded before the document class, using the same package options as for the tudscrfonts package: `\RequirePackage[<package options>]{fix-tudscrfonts}` \documentclass[<class options>]{<document class>}

. . .

`\begin{document}`

. . .

`\end{document}`

The tudscrfonts package is loaded at the end of the preamble at the latest. Alternatively, this can also be done by the user in the document preamble.

5.4 The mathswap package

The use of decimal and thousand separators in mathematical typesetting varies greatly from region to region. In most English-speaking countries, the period is used as the decimal separator and the comma is used to group digits; in the rest of Europe, the opposite is the case. This package is intended to be used to adapt any formatted numbers in their output. To do this, the period (.) and comma (,) characters are defined as active characters in math mode.

Similar functionalities are already provided by the `icomma` and `ziffer` packages .

However, with `icomma` , the author must consider when writing the document whether the comma used corresponds to a decimal separator ($t = 1,2$) or a normal comma in math typesetting ($z = f(x, y)$), where a certain distance after the comma is definitely desired. The `ziffer` package provides the desired functionality,⁴¹ but is somewhat inflexible when it comes to handling the separators. `ionumbers` can also be used as an alternative to this package.

The `mathswap` package ensures that separators directly before a number are recognized and replaced according to certain specifications. However, if there is space between the separator and the number, this is also set as such. For an example of how to use the package, see the `mathswap.pdf` tutorial in Chapter 7.

```
\commaswap{<separator>}
\dotswap{<separator>}
```

The two commands `\commaswap` and `\dotswap` are the central user interface of the package. The macro `\commaswap` defines the separator or the content that should replace a comma that is immediately followed by a number. LATEX normally adds an additional horizontal space after a comma in mathematical sentences. When replacing with `\commaswap`, this is omitted. The default setting for `\commaswap` is therefore a comma (,). The macro `\dotswap` can be used to define what should replace the dot in mathematical sentences when it is immediately followed by a number. Since the dot is used in German-speaking countries to group numbers, a half-breaking space is defined by default (`\,.`).

v2.02

```
\mathswap_off
```

The functionality of `mathswap` can be turned on and off within the document using these two commands . It is enabled by default when the package is loaded.

5.5 The twocolfix package

The LATEX 2_ε kernel contains a bug that places chapter headings higher in the two-column layout than in the single-column layout. The bug⁴² has been known for some time, but has not yet been incorporated into the LATEX 2_ε kernel. The `twocolfix` package fixes the problem.

An integration of this bug fix into KOMA-Script has already been requested from Markus Kohm, but he has not pursued it further so far⁴³ .

⁴¹no space after comma if a digit follows immediately after it
⁴²<http://www.latex-project.org/cgi-bin/ltbugs2html?pr=latex/3126>
⁴³<http://www.komascript.de/node/1681>

Future work

These things will be incorporated into the TUD script bundle in the long term:

The package `tudscrletter` for letters in the corporate design of the Technical University Dresden. Classes for fax and internal communications are also expected to be eliminated.

With `tudscrbeamer` A package is to be created with which LATEX beamer presentations can be created in the style of the corporate design of the Technical University of Dresden with the setting options of TUD-Script . Currently, the packages from the GitHub repository `tud-cd44` can be used for this purpose. be used.

There is also a package `tudscrlayout` It is conceivable that this activates the type area specified by the corporate design without using the page style itself, in order to be able to use paper that has already been printed with the header of the Technical University of Dresden, for example.

It would also be conceivable to make the `tudheadings` page styles available for other classes without implementing the type area of the corporate design. However, this has only a low priority.

⁴⁴<https://github.com/tud-cd/tud-cd/>



Part II

Additions and notes

It is not necessary to read this part to use the TUD -Script bundle . Chapter 6 shows several simple examples, especially for LATEX newbies and new users of the TUD -Script bundle. In addition , Chapter 7 contains more extensive tutorials for creating academic papers. The recommendations given there are not limited to TUD-Script but can also be implemented with other LATEX classes. Chapter 8 briefly introduces packages that I think are recommended for beginners and experienced LATEX users.

Application notes and one or two general tips for using LATEX 2_ε are given in Chapter 9. These are certainly interesting for using both the TUD script bundle and other LATEX classes . I am always open to suggestions, tips, advice or recommendations for other packages as well as tips.

Chapter 6

Minimal examples

This chapter is intended to make it easier to get started and use TUD -Script for the first time . A few minimal examples are given to illustrate individual functions . These are so reduced that they should be immediately apparent to the user .

6.1 Document

This shows how the preamble of a minimal LATEX 2_ε document should be designed.

In principle, this excerpt can be used as the basis for a new document . Only the inclusion of the blindtext package with `\usepackage{blindtext}` and the use of the resulting command `\blinddocument` can be omitted .

```
\documentclass[ngerman]{tudscrreprt} \iftutex

\usepackage{fontspec} \else

\usepackage[T1]{fontenc}
  \usepackage[ngerman=ngerman-x-latest]{hyphsubst} \fi

\usepackage{babel}
\usepackage[math]{blindtext}
\begin{document}
\faculty{Faculty of Law}
\department{Department of Criminal Law}
\institute{Institute of Criminology} \chair{Chair of
Criminal Prognosis} \date{18.02.2015} \author{Mickey
Mouse} \title{The Money
Vault in Duckburg} \maketitle

\blinddocument
\end{document}
```



Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

The money vault in Duckburg

Mickey Mouse

18.02.2015

1 heading at level 0 (chapter)

This is dummy text for testing text output. If you read this text, you only have yourself to blame. $\sin^2(\ddot{y}) + \cos^2(\ddot{y}) = 1$. The text simply indicates the gray value of the font $E = mc^2$. Is that really the case? Does it make no difference if I write: "This is dummy text" or "Huardest gefburn"?

Kjift – not at all! Dummy text provides me with important information. $\ddot{y}n\ a \cdot \ddot{y}n\ b = \ddot{y}n\ ab$. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to each other and check how wide or narrow it is. $\ddot{y}n\ a$ Dummy text should contain as many different letters as possible and be set in the original language. $a\ \ddot{y}n\ b = \ddot{y}n\ a\ nb$. It doesn't have to make sense, but it should be readable. $d\ddot{y} = \sin\ \ddot{y}d\ddot{y}d\ddot{y}$. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression.

1.1 Heading on level 1 (section)

This is dummy text for testing text output. If you read this text, you only have yourself to blame. $\sin^2(\ddot{y}) + \cos^2(\ddot{y}) = 1$. The text simply indicates the gray value of the font $E = mc^2$. Is that really the case? Does it make no difference if I write: "This is dummy text" or "Huardest gefburn"?

Kjift – not at all! Dummy text provides me with important information. $\ddot{y}n\ a \cdot \ddot{y}n\ b = \ddot{y}n\ ab$. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to each other and check how wide or narrow it is. $\ddot{y}n\ a$ Dummy text should contain as many different letters as possible and be set in the original language. $a\ \ddot{y}n\ b = \ddot{y}n\ a\ nb$. It doesn't have to make sense, but it should be readable. $d\ddot{y} = \sin\ \ddot{y}d\ddot{y}d\ddot{y}$. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression.

1.1.1 Heading at level 2 (subsection)

This is dummy text for testing text output. If you read this text, you only have yourself to blame. $\sin^2(\ddot{y}) + \cos^2(\ddot{y}) = 1$. The text simply indicates the gray value of the font $E = mc^2$. Is that really the case? Does it make no difference if I write: "This is dummy text" or "Huardest gefburn"?

Kjift – not at all! Dummy text provides me with important information. $\ddot{y}n\ a \cdot \ddot{y}n\ b = \ddot{y}n\ ab$. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to each other and check how wide or narrow it is. $\ddot{y}n\ a$ Dummy text should contain as many different letters as possible and be set in the original language. $a\ \ddot{y}n\ b = \ddot{y}n\ a\ nb$. It doesn't have to make sense, but it should be readable. $d\ddot{y} = \sin\ \ddot{y}d\ddot{y}d\ddot{y}$. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression.

1 heading at level 0 (chapter)

Heading at level 3 (subsubsection)

This is dummy text for testing text output. If you read this text, you only have yourself to blame. $\sin^2(\tilde{y}) + \cos^2(\tilde{y}) = 1$. The text simply indicates the gray value of the font $E = mc^2$. Is that really the case? Does it make no difference if I write: "This is dummy text" or "Huardest gefburn"? — Kjift – not at all! Dummy text provides me with important information. $\tilde{y}n\ a \cdot \tilde{y}n\ b = \tilde{y}n\ ab$. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to each other and check how wide or narrow it is. $\tilde{y}n\ a$ Dummy text should contain as many different letters as possible and be set in the original language. $a\ \tilde{y}n\ b = \tilde{y}n\ a\ nb$. It doesn't have to make sense, but it should be readable. $d\tilde{y} = \sin\ \tilde{y}d\tilde{y}d\tilde{y}$. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression.

Heading on level 4 (paragraph) This is dummy text for testing text output. It's your own fault if you read this text. $\sin^2(\tilde{y}) + \cos^2(\tilde{y}) = 1$. The text simply gives the gray value of the font $E = mc^2$. Is that really so? Does it matter whether I write: "This is dummy text" or "Huardest gefburn"? Kjift – not at all! Dummy text offers me important information. $\tilde{y}n\ a \cdot \tilde{y}n\ b = \tilde{y}n\ ab$. I use it to measure the legibility of a font, its appearance, how harmoniously the figures relate to one another and check how wide or narrow it is. $\tilde{y}n\ a$ Dummy text should contain as many different letters as possible $\tilde{y}n\ b$ and be set in the original language. $a\ \tilde{y}n\ b = \tilde{y}n\ a\ nb$. It doesn't have to make sense, but it should be legible. $d\tilde{y} = \sin\ \tilde{y}d\tilde{y}d\tilde{y}$. Foreign language texts such as "Lorem ipsum" do not serve the intended purpose as they give a false impression.

1.2 Lists

1.2.1 Example of a list (itemize)

- First list item, level 1
- Second list item, level 1
- Third list item, level 1
- Fourth list item, level 1
- Fifth list item, level 1

Example of a list (4*itemize)

- First list item, level 1
 - First list item, level 2
 - * First list item, level 3
 - First list item, level 4
 - Second list item, level 4
 - * Second list item, level 3
 - Second list item, level 2
- Second list item, level 1

1 heading at level 0 (chapter)

1.2.2 Example of a list (enumerate)

1. First list item, level 1
2. Second list item, level 1
3. Third list item, level 1
4. Fourth list item, level 1
5. Fifth list item, level 1

Example of a list (4*enumerate)

1. First list item, level 1
 - a) First list item, level 2
 - i. First list item, level 3
 - A. First list item, level 4 B.
Second list item, level 4
 - ii. Second list item, level 3
 - b) Second list item, level 2
2. Second list item, level 1

1.2.3 Example of a list (description)

- First list item, level 1
- Second list item, level 1
- Third list item, level 1
- Fourth list item, level 1
- Fifth list item, level 1

Example of a list (4*description)

- First list item, level 1
- First list item, level 2
- First list item, level 3
- First list item, level 4
- Second list item, level 4
- Second list item, level 3
- Second list item, level 2
- Second list item, level 1

6.2 Dissertation

A thesis or similar could be started as shown here. `\documentclass[`

```
    english,german,%
    BCOR=6mm,cdgeometry=no,DIV=13%
][tudscrreprt] \iftutex

\usepackage{fontspec} \else

\usepackage[T1]{fontenc}
    \usepackage[ngerman=ngerman-x-latest]{hyphsubst} \fi

\usepackage{babel}
\usepackage{isodate}
\usepackage{blindtext}
\begin{document}
\faculty{Faculty of Law} \department{Criminal
Law} \institute{Institute of Criminology} \chair{Chair
of Criminal Prognosis} \date{18.02.2015} \author{Mickey
Mouse} \title{% Development of an optimal procedure
for conquering the
money vault in Duckburg

} \thesis{diss}
\graduation[Dr. jur.]{Doctor of Law} \dateofbirth{2.1.1990}
\placeofbirth{Dresden}
\defensedate{15.03.2015}
\referee{Scrooge McDuck \and Mac
Moneysac} \maketitle \TUOption{abstract}{section,multiple}
\begin{abstract}

[pagestyle=empty.tudheadings] \blindtext
\nextabstract[english] \blindtext \end{abstract} \end{document}
```



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

Development of an optimal Procedure for the conquest of the Money vault in Duckburg

Mickey Mouse

Born on: January 2, 1990 in Dresden

dissertation

for obtaining the academic degree

Doctor of Laws (Dr. jur.)

First assessor

Scrooge McDuck

Second reviewer

Mac Moneybag

Filed on: 18 February 2015 Defended

on: 15 March 2015



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

Summary

This is dummy text for testing text output. If you read this text, it's your own fault. The text simply indicates the gray value of the font. Is that really the case? Does it make no difference whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! Dummy text provides me with important information. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to one another and to check how wide or narrow it is. Dummy text should contain as many different letters as possible and be set in the original language. It doesn't have to make sense, but it should be legible.

Foreign language texts such as "Lorem ipsum" do not serve the intended purpose as they give a false impression.

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will not receive any information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in the original language. There is no need for special content, but the length of words should match the language.

6.3 Final thesis (collaborative)

All additional information outside the argument of `\author` is applied equally to both authors.⁴⁵ The information within the argument of `\author` is assigned to the respective authors, separated by `\and`.⁴⁶ Without using `\and`, only one author can of course be listed. Also note the use of `\subject` instead of `\thesis` with a special value from Table 2.1.

```
\documentclass[ngerman,BCOR=4mm]{tudscrreprt} \iftutex \usepackage{fontspec} \else \usepackage[T1]{fontenc}
\usepackage[ngerman=ngerman-x-latest]{hyphsubst} \fi \usepackage{babel} \usepackage{isodate}
\begin{document} \faculty{Faculty of Law} \department{Criminal
Law}
\institute{Institute of
```

```
Criminology} \chair{Chair of Criminal
Prognosis} \date{18.02.2015} \title{% Development of an optimal
```

```
procedure for conquering
the money vault in Duckburg
```

```
} \subject{master}
\graduation[M.Sc.]{Master of Science} \author{%
Mickey
Mouse%
\matriculationnumber{12345678}%
\dateofbirth{2.1.1990}%
\placeofbirth{Dresden}% \and%

Donald Duck%
\matriculationnumber{87654321}%
\dateofbirth{1.2.1990}%
\placeofbirth{Berlin}%

} \matriculationyear{2010}
\supervisor{Scrooge McDuck \and Mac Moneysac} \professor{Prof.
Dr. Karlo the Cat} \maketitle \end{document}
```

⁴⁵In this example `\matriculationyear`

⁴⁶In this example `\matriculationnumber`, `\dateofbirth` and `\placeofbirth`



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

Master thesis

Development of an optimal Procedure for the conquest of the Money vault in Duckburg

Mickey Mouse Born

on: January 2, 1990 in Dresden Registration

number: 12345678 Year of

enrollment: 2010

Donald Duck

Born on: 1 February 1990 in Berlin Registration

number: 87654321 Year of

enrollment: 2010

18 February 2015

supervisor

Scrooge McDuck

Mac Moneybag

Supervising professor Prof. Dr. Kater

Karlo

6.4 Task (collaborative)

A task for a scientific paper can be presented as follows using the **task** environment or the **\taskform** command from the `tudscrsvsupervisor` package . `\documentclass[ngerman]{tudscrreprt}` `\iftutex \usepackage{fontspec} \else \usepackage[T1]`

```
{fontenc} \usepackage[ngerman=ngerman-x-latest]
{hyphsubst}
```

```
\fi \usepackage{babel}
```

```
\usepackage{isodate}
```

```
\usepackage{tudscrsvsupervisor} \usepackage{enumitem}
```

```
\setlist{noitemsep}
```

```
\begin{document}
```

```
\faculty{Faculty of Law}
```

```
\department{Department of Criminal Law} \institute{Institute
of Criminology}
```

```
\chair{Chair of Criminal Prognosis} \title{% Development of an optimal procedure for conquering
the money vault in Duckburg}
```

```
} \thesis{master}\graduation[M.Sc.]{Master of Science} \author{% Mickey
```

```
Mouselmatriculationnumber{12345678}%
```

```
\dateofbirth{2.1.1990}\placeofbirth{Dresden}% \course{Clinical
Prognostics}\discipline{Individual Prognosis}% \and% Donald
```

```
Duck\matriculationnumber{87654321}%
```

```
\dateofbirth{1.2.1990}\placeofbirth{Berlin}% \course{Statistical
Prognostics}\discipline{Macrosociological Forecasts}%
```

```
} \matriculationyear{2010} \issuedate{1.4.2015} \duedate{1.10.2015} \supervisor{Scrooge McDuck
```

```
\and Mac Moneysac} \professor{Prof. Dr. Tomcat Karlo} \referee{Prof. Dr. Tomcat Karlo} \chairman{Prof. Dr. Primus
by Quack} \taskform[pagestyle=empty]{%
```

At the moment, this topic is on everyone's lips. In particular, it is being discussed in many, if not all, media.

It is currently not possible to predict whether and when this situation will change. A short-term shift from the public spotlight is not expected.

The aim of this work is to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods must be developed to avoid a similar process in the future. }

\item Research \& Analysis \item Development of a concept \& Application of the developed methodology \item Documentation and graphical presentation of

the

results

```
} \end{document}
```



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

Task for the preparation of a master's thesis

Course of study:	Clinical prognosis	Statistical forecasting
Field of study:	Individual prognosis	Macrosociological forecasts
Surname:	Mickey Mouse	Donald Duck
Registration number:	12345678	87654321
Year of enrollment:	2010	2010
Title:	Development of an optimal procedure for conquering the Money vault in Duckburg	

Goals of work

At the moment, the topic in question is on everyone's lips. In particular, it is currently being discussed in many - if not all - media. It is currently not possible to predict whether and when this situation will change. A short-term shift from the public focus is not expected.

The aim of this work is to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods are to be developed to prevent a similar process from occurring in the future.

Main focus of the work •

- Research & analysis •
- Development of a concept & application of the developed methodology •
- Documentation and graphical preparation of the results

Reviewer:	Prof. Dr. Karlo the Cat
Supervisor:	Scrooge McDuck Mac Moneysac 1.
Issued on:	April 2015 1.
Submitted on:	October 2015

Prof. Dr. Primus von Quack
Chairman of the Examination Committee

Prof. Dr. Kater Karlo
Supervising Professor

6.5 Expert opinion

After loading the tudscrsupervisor package , a report for a scientific paper can be created using the **evaluation environment** or the `\evaluationform` command. `\documentclass[ngerman]{tudscrartcl}` `\iftutex \usepackage{fontspec} \else`

`\usepackage[T1]{fontenc} \usepackage[ngerman=ngerman-x-latest]`

`{hyphsubst} \fi`

`\usepackage{babel}`

`\usepackage{isodate} \usepackage{tudscrsupervisor}`

`\usepackage{enumitem}`

`\setlist{noitemsep}`

`\begin{document} \faculty{Faculty of Law}`

`\department{Criminal Law} \institute{Institute of Criminology}`

`\chair{Chair of Criminal`

`Prognosis} \date{16.10.2015} \author{Mickey Mouse} \title{% Development of an optimal procedure
for conquering the money vault in Duckburg}`

`} \thesis{diploma} \graduation[Dipl.-Ing.]{Diplomingenieur}`

`\matriculationnumber{12345678} \matriculationyear{2010}`

`\dateofbirth{2.1.1990} \placeofbirth{Dresden}`

`\referee{Scrooge McDuck}`

`\evaluationform[pagestyle=empty]{%`

The aim of this work was to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods were to be developed with which a similar process could be avoided in the future. }% The work is divided into several chapters on countless pages.

In the first two chapters this and that is discussed. In the following chapter also that. In the penultimate chapter everything is considered and in the last one there is a summary and outlook. }% The results presented in this work meet the work

priorities formulated in the task. The form, expression and terminology as well as the structure satisfy the requirements of an academic work. The individual chapters can be considered balanced in terms of their depth of content in relation to the required scope. The graphics and tables used are very well prepared and contribute to a better understanding overall. Technical terms are used very cleanly and there are no spelling errors. }{1.0 (very good)} \end{document}



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

Report for the diploma thesis

Surname: Mickey Mouse
Registration number: 12345678
Year of enrollment: 2010
Title: Development of an optimal procedure for conquering the
Money vault in Duckburg

Task The aim of this

work was to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods were to be developed with which a similar process could be avoided in the future.

Contents

The work is divided into several chapters on countless pages. In the first two chapters this and that is discussed. In the following chapter also that. In the penultimate chapter everything is considered and in the last one there is a summary and outlook.

Assessment

The results presented in this work meet the work priorities formulated in the task. The form, expression, terminology and structure all meet the requirements of an academic work. The individual chapters can be considered balanced in terms of their depth of content in relation to the required scope. The graphics and tables used are very well prepared and contribute to a better understanding overall. Technical terms are used very cleanly and there are no spelling errors.

The work is rated with the grade 1.0 (very good).

Dresden, 16 October 2015

Scrooge McDuck
Expert

6.6 Notice

The tudscrsupervisor package provides the **notice** environment for creating general notices and the **\noticeform** command for advertising scientific papers. `\documentclass[ngerman]{tudscrreprt}` `\iftutex \usepackage{fontspec} \else`

`\usepackage[T1]{fontenc} \usepackage[ngerman=ngerman-x-latest]`

`{hyphsubst} \fi`

`\usepackage{babel} \usepackage{isodate}`

`\usepackage{tudscrsupervisor} \usepackage{enumitem}\setlist{noitemsep}`

`\begin{document}`

`\faculty{Faculty of Law}`

`\department{Department of Criminal Law}`

`\institute{Institute of Criminology}\chair{Chair of Criminal`

`Prognosis} \title{%`

`Development of an optimal procedure for conquering the money vault in Duckburg } \date{16.10.2015}`

`\contactperson{% Dagobert Duck\emailaddress{dagobert.duck@tu-dresden.de} \office{Dingens-Bau,`

`Zimmer~08}\telephone{+49 351 463-12345} \and% Mac`

`Moneysac\emailaddress{mac.moneysac@tu-`

`dresden.de} \office{Dingens-`

`Bau,`

`Zimmer~15}\telephone{+49 351 463-54321}`

`} \noticeform[Offer for a thesis,pagestyle=empty]{% At the moment, this topic is on everyone's lips. In particular, it is being discussed in many--- if not all--- media.`

It is currently not possible to predict whether and when this situation will change. A short-term shift from the public spotlight is not expected.

The aim of this work is to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods must be developed to avoid a similar process in the future.

`\begin{center} \medskip\includegraphics[width=.7\linewidth]{DDC-21} \renewcommand*{\figureformat}`

`{\figurename} \captionof{figure}{Thematically appropriate image}`

`\end{center} }{%`

`\item Research \& Analysis \item Development of a concept \& Application of`

`the developed methodology \item Documentation and graphical`

`presentation of the results`

`} \end{document}`



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

16 October 2015

Offer for a thesis/diploma thesis

Development of an optimal procedure for conquering the money store in Duckburg

At the moment, the topic in question is on everyone's lips. In particular, it is currently being discussed in many - if not all - media. It is currently not possible to predict whether and when this situation will change. A short-term shift from the public focus is not expected.

The aim of this work is to identify why the topic is so omnipresent and how this effect could be mitigated. In addition, methods are to be developed to prevent a similar process from occurring in the future.

DRESDEN
concept

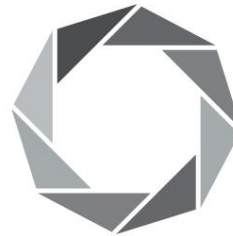


Illustration: Thematically appropriate image

Main focus of the work • Research

& analysis • Development of a
concept & application of the developed methodology • Documentation and graphical preparation
of the results

contact person

Scrooge McDuck
dagobert.duck@tu-dresden.de Dingens-
Bau, Room 08 +49 351 463-12345

Mac Moneysac
mac.moneysac@tu-dresden.de Dingens-
Bau, Room 15 +49 351 463-54321

6.7 Poster

The `tudscrposter` class can be used to create a poster in the corporate design of the Technical University of Dresden. The desired paper format and the appropriate font size must be specified.

```
\RequirePackage{fix-cm}
\documentclass[
  ngerman,%
  paper=A1,fontsize=22pt,%
  cdfoot=5ex,ddcfoot% ]
{tudscrposter} \iftutex

\usepackage{fontspec} \else

\usepackage[T1]{fontenc}
\usepackage[ngerman=ngerman-x-latest]{hyphsubst} \fi

\usepackage{babel}
\usepackage{blindtext}
\usepackage{multicol}
\begin{document}
\faculty{Faculty of Law}\department[]{\Department of Criminal Law} \institute{Institute of Criminology}
\chair[]{\Chair of Criminal Prognosis} \title{% Development of an optimal procedure for conquering the money vault
in Duckburg }
\date{18.02.2015} \contactperson{% Mickey Mouse\emailaddress{mickey.mouse@tu-
dresden.de}% \office{Office
58}\telephone{463--xxxxx}
%

} \professor{Prof. Dr. Karlo the Cat} \maketitle
\begin{abstract}
%[columns=2] \noindent\blindtext
\end{abstract}

\begin{multicols}{2}[\section*{Test}] \blindtext[3]
\end{multicols}
\bigskip

\begin{center}
\includegraphics[width=.7\linewidth]{TUD-black} \end{center}

\begin{multicols}{2}[\section*{Test}] \blindmathtrue
\blindtext[1]
\end{multicols}
\end{document}
```



TECHNISCHE
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Faculty of Law , Department of Criminal Law, Institute of Criminology, Chair of Criminal Prognosis

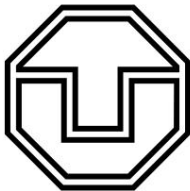
Development of an optimal procedure for Conquest of the money vault in Duckburg

This is dummy text for testing text output. If you read this text, you only have yourself to blame. The text simply indicates the gray value of the font. Is that really the case? Does it make no difference whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! Dummy text provides me with important information. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to one another and check how wide or narrow it is. Dummy text should contain as many different letters as possible and be set in the original language. It does not have to make sense, but it should be legible. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression.

test

This is dummy text for testing text output. If you read this text, it's your own fault. The text simply indicates the gray value of the font. Is that really the case? Does it make no difference whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! Dummy text provides me with important information. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are arranged in relation to one another and check how wide or narrow it is. Dummy text should contain as many different letters as possible and be set in the original language. It doesn't have to make sense, but it should be legible. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give the wrong impression. This is dummy text for testing text output. If you read this text, it's your own fault. The text simply indicates the gray value of the font. Is that really the case? Does it matter whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! Dummy text offers me important information. By it I measure the legibility of a text, its appearance, how hard-

monically the figures are in relation to each other and check how wide or narrow it is. Dummy text should contain as many different letters as possible and be set in the original language. It doesn't have to make sense, but it should be legible. Foreign language texts such as "Lorem ipsum" do not serve the actual purpose, as they give a false impression. This is dummy text for testing text output. Anyone who reads this text has only themselves to blame. The text simply indicates the gray value of the font. Is that really so? Does it matter whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! Dummy text provides me with important information. I use it to measure the legibility of a font, its appearance, how harmoniously the figures are in relation to each other and check how wide or narrow it is. Dummy text should contain as many different letters as possible and be set in the original language. It does not have to make sense, but it should be readable. Foreign language texts such as "Lorem ipsum" do not serve the intended purpose, as they give a false impression.



TECHNISCHE
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test

This is dummy text for testing text output. If you read this text, you only have yourself to blame. $\sin^2(\hat{y}) + \cos^2(\hat{y}) = 1$. The text simply indicates the gray value of the font $E = mc^2$. Is that really so? Does it matter whether I write: "This is dummy text" or "Huardest gefburn"? Kjift - not at all! A dummy text offers me important information. $\hat{y}n a \cdot \hat{y}n b = \hat{y}n$. I use it to measure the legibility of a font, its appearance, how harmonious the

Figures are placed in relation to each other and check how wide or narrow it is. $\hat{y}n a$ A dummy text should contain as many different $\hat{y}n b$ letters and be set in the original language. $\hat{a}\hat{y}n b = \hat{y}n a nb$. It doesn't have to make sense, but it should be readable. $d\hat{y} = \sin \hat{y}d\hat{y}d\hat{y}$. Foreign language texts such as "Lorem ipsum" do not serve the intended purpose, as they give a false impression.

Contact
Technical University of Dresden Faculty
of Law Institute of
Criminology Prof. Dr. Kater
Karlo

Contact Mickey Mouse
mickey.mouse@tu-
dresden.de Office 58 463-xxxxx



Chapter 7

Tutorial

This chapter provides further application examples. These tutorials are not included directly in the manual but are provided as external

Files are available which can be opened directly via hyperlink.

Guide for a scientific paper

Most users of the TUD script classes are students or members of the Technical University of Dresden who are taking their first steps with LATEX 2 ϵ when writing an academic paper or similar. During the introductory phase in LATEX 2 ϵ , a beginner can quickly become overwhelmed by the large number of recommended packages and the sometimes diametrically opposed instructions. The `treatise.pdf` tutorial is intended to shed some light on the subject. However, it makes no claim to be complete or perfect. Some of the options presented can certainly be solved differently, more simply and/or better. Nevertheless, it is intended as a guide for newcomers - and perhaps also for one or two LATEX veterans - for writing an academic paper.

A contribution to mathematical typesetting in LATEX 2 ϵ

The `mathtype.pdf` tutorial is aimed at all users who want to type mathematical formulas in their LATEX document. It explains in detail how to create typographically clean mathematical typesetting in just a few steps.

Changing the separators in math mode

If, when writing a LATEX document, data is imported in a number format that does not correspond to the customs of the document language, the output will usually be unsightly. The simplest example is data in which a point is used as the decimal separator, as is the case in English-speaking countries.

If these are to be included in a German language document, they would normally all have to be adapted and the original decimal separator replaced by a comma. This step is automated using the TUD script package `mathswap`. How this works exactly is explained in the `mathswap.pdf` tutorial.

Chapter 8

Required, supported and recommended packages

8.1 Necessary and additional packages for TUD -Script

These packages are required and included by the TUD script classes . If you want to use one of the packages listed below with certain options, these should be passed on to the package *before* loading the document class, unless otherwise stated in the following description of the package.

Example: Passing options to packages must be done as follows:

```
\PassOptionsToPackage{<package options>}{<package>}
\documentclass[<class options>]{tudscr. . .}
```

v2.02 koma-script, typearea, scrbase, sclayer-scrpage, scrextend

The KOMA -Script bundle is the central basis for TUD-Script. In addition to the classes scrbook, scrreprt and scrartcl, the scrbase package is required. This allows the definition of class options in the style of KOMA-Script, which can also be changed after the class has been loaded using the commands **\TUOption** and **\TUOptions** . The sclayer-scrpage package is required to provide the tudheadings page styles . If it is not loaded by the user – with any options – it is done automatically by TUD script at the end of the preamble.

v2.06 opensans, iwona

The opensans package provides the **Open Sans** font family for both body text and mathematical typesetting. It contains all the necessary font styles in both Type1 and OpenType format. Since the font family does not provide any mathematical glyphs in the current version , the mathastext and iwona packages are also used to enable at least a halfway acceptable mathematical typesetting with **Open Sans** . If additional symbols are required, it is recommended to forego the amssymb package and load mdsymbol instead . v2.06 mweights In LATEX 2_γ there are font families for serif fonts (**\rmfamily**), sans serif fonts (**\sffamily**) and typewriter fonts (**\ttfamily**). Their font weight is usually set uniformly using the two commands **\mddefault** and

\bfdefault . When using

different font packages , problems with the font weights can sometimes arise. This package allows the individual definition of the font weight for each of the three font families. geometry The package is used to define the page margins or the type area.

Danger!

Passing additional options to the package is strongly discouraged. graphicx This is the de facto standard

package for including graphics. **\includegraphics** is used to set the logo of the Technical University of Dresden in the header as well as all other images and logos . xcolor This defines the colors of the corporate design for use in the document.

More details can be found in the description of tudscrcolor in subsection 5.1.

kvsetkeys

The keyval package loaded by scrbase allows defining class and package options and parameters according to the key-value principle. This package can be used to specify the behavior for unknown keys.

etoolbox, xpatch, letltxmacro, etexcmds

These packages provide many functions for testing and process control. Furthermore

allows the manipulation of existing macros. environ

An improved declaration of environments is enabled, where the passed parameters can be accessed even when the environment is closed. trimspaces If there are multiple

input fields for the user, this package removes any unnecessary spaces from the arguments.

Packages directly supported by TUD-Script Some

of the packages described in subsection 8.2 are directly supported by TUD-Script and extend its functionality. These are namely hyperref, multicol, quoting, ragged2e and crop as well as isodate and datetime2 . Further information can be found in the following description of the respective package.

8.2 Recommended packages

In this subsection, a large number of packages are presented - mostly briefly - which I have personally found useful when working with LATEX . Some of them are also described in the tutorial treatise.pdf . For detailed information and if you have any questions about the individual packages, you should consult the relevant documentation⁴⁷ ; reading the brief description given here is in no way a substitute for this.

8.2.1 Packages to use in each document

In my opinion, the packages presented here belong in the preamble of every document. The document language should always be defined with babel or polyglossia - even if this is English. For German-language documents, the hyphsubst package must be used to ensure acceptable word separation when using pdfLATEX .

v2.02 fontenc, fontspec The

character set encoding of the output font should always be specified. For pdfLATEX, the output is selected as a 7-bit encoded font by default, which means, among other things, that no real umlauts are used in the generated PDF document.

To switch to 8-bit fonts, use `\usepackage[T1]{fontenc}` .

For the Unicode typesetting systems LuaLATEX or XL ATEX , the fontspec package should be used instead. This allows system fonts in OpenType format and any character set encoding to be integrated, which greatly expands the selection of fonts that can be used. The package is supported by TUD script.

microtype This package takes care of the optical margin alignment⁴⁸ and the leveling of the word spaces⁴⁹ in the

document. It does not work with the classic TEX engine, but it does work with pdfTEX or LuaTEX as well as X TEX.

E

⁴⁷Command line/terminal: `texdoc <package name>`

⁴⁸English: protrusion, margin kerning

⁴⁹English: font expansion

babel, polyglossia

The babel package is used to set the language(s) used in the document.

If several languages are specified, the last one loaded is the main language of the document. The desired languages should not be set as a package option but as a class option so that other packages can also access the language settings. For German-language documents, the option **ngerman** should be used for the new German spelling or **german** for the old German spelling.

When loading babel and the associated languages, both the separation patterns and the language-dependent identifiers are adjusted. Using the obsolete packages german or ngerman instead of babel is not recommended.

For LuaLATEX and XL^E ATEX the package polyglossia can be used.

hyphsubst, dehyph-exptl

LATEX calculates the possible hyphenation points of words using an algorithm. This is significantly improved for German texts using the hyphsubst package. LuaLATEX and XL^E ATEX use these hyphenation patterns automatically, for pdfLATEX they must be included as follows:

\usepackage[ngerman=ngerman-x-latest]{hyphsubst}

Subsection 9.3 discusses the interaction between hyphsubst and babel as well as fontenc in more detail; a look at this is highly recommended. In addition, further tips for improved word separation are given there.

8.2.2 Packages for situational use

The following packages should not necessarily be loaded in every document, but only if this is actually necessary. For a better overview, an attempt has been made to group them thematically. This does not imply any evaluation of their usefulness or my personal appreciation.

Typography and layout

In addition to the previously described microtype package, which is responsible for microtypographic details, there are other packages that primarily address macrotypography. setspace The increase in line spacing is:

- I. requested far too frequently and completely
- unnecessarily and II. finally chosen to be too large.

The requirement to increase line spacing - known in typography as leading - comes from the days of word processing with a typewriter. A single line spacing here meant that the descenders of the top line were exactly at the height of the ascenders of the following line. A one-and-a-half line spacing thus achieved acceptable leading. Increasing the leading when using LATEX is not actually necessary. This only makes sense if sans serif fonts are used in the body text in order to improve the associated poor readability.

If the increase in spacing is really necessary, the setspace package should be used. This provides the command **\setstretch{<factor>}** with which

with which the leading or line spacing can be adjusted. The value of the factor is set to 1 by default and should be increased to a maximum of 1.25.

The command **lonehalfspacing** from this package sets this value to exactly 1.25.

However, it should be noted here that the increase in line spacing - as I have read it - is not a gimmick from a typographer's point of view but rather serves solely to make the text easier to read and should be as small as possible.

The aim is to make it easier to find the next line when reading after the end of a line. With serifs, this is very easy to do by emphasizing the baseline. With sans serif fonts - such as **Open Sans** used in the corporate design of the Technical University of Dresden - this is more difficult and increased line spacing can be helpful here. However, the motto "more is better" should not be followed. For this document, **\setstretch{1.1}** was chosen for the line spacing.

Further tips can be found in subsection 9.7 and subsection 9.8. multicol

This allows any content to be output in two or more columns, whereby – in contrast to the normal two-column typesetting via the KOMA script option **twocolumn** – column balancing is ensured. The package is supported within the environments **abstract** and **tudpage**. balance This package enables column balancing in two-column typesetting on the last document page.

Alternatively, multicol can be used.

isodate, datetime2 With

\printdate{<date>} the isodate package automatically formats the output of a date in a specified format. The date can be specified in German, English or ISO format. Alternatively, the datetime2 package can be used, which requires input in ISO format. If one of the two packages is loaded, all date fields defined by the TUD script classes⁵⁰ are output in the output format defined by the respective package. quoting LATEX offers *two* different environments - **quote** and **quotation** - for quotations and the like. However, both environments ignore the KOMA-Script option **parskip=<method>**. This problem can be avoided with the **quoting**

environment from the package of the same name. If the package is loaded, it is used within the abstract environment. csquotes The package provides, among other things, the command **\enquote{<quote>}**, which sets quotation marks depending on the selected language. In addition, other commands and options are offered for the specific requirements of citing in academic papers. It is also supported by biblatex and should at least be loaded when using it.

v2.02 noindentafter

This package can be used to suppress automatic paragraph indentation for user-defined commands and environments. ragged2e This package improves

ragged left-

left text by activating word separation for it.

50\date, **\dateofbirth** and **\defensedate** as well as from tudscrsupervisor **\duedate** and **lissuedate**

v2.05 fnpct

 This

package ensures that the correct order of punctuation marks and footnotes is maintained and supports the typographically correct setting of several consecutive footnotes. `xspace`, `xpunctuate` With `xspace` you can use the **`\xspace`** command to define your own macros .

This automatically inserts any necessary spaces. The definition of such a command is given as an example in subsection 9.13. The `xpunctuate` package extends `xspace` to take punctuation into account. `ellipsis` In LATEX, the commands for ellipsis (**`\ldots`** and **`\textellipsis`**) are *always* followed by a space.

This can

be undesirable in some circumstances. With the `ellipsis` package , the following space is only set – in contrast to the standard behavior – if a punctuation mark and not a letter follows, see *also* subsection 9.15.

DeLig, `selnolig`

This is a Java script that uses a dictionary to automatically remove incorrect ligatures within a document. These would have to be resolved manually for a good typesetting – especially in German texts due to the large number of compounds.⁵¹ With LuaLATEX as a typesetting system, `selnolig` can also be can be used alternatively .

spelling, orthography

The editor used is normally responsible for spell checking.

However, there are a few packages that are dedicated to this topic. However, these can only be used if LuaLATEX is used as the typesetting system. `lua-check-hyphen` This allows you to use LuaLATEX

to mark hyphenation

points at the end of a line for checking. For information on *correct word hyphenation* , see subsection 9.3.

spelling If LuaLATEX is used as a typesetting system, this package extracts the pure text part from

the

LATEX document - removing macros and active characters - and writes it to a separate text file. This file can then be analyzed using external spell checking software such as *GNU Aspell*, *Hunspell* or *LanguageTool* and misspelled words can be highlighted in the PDF document.

Fonts and special characters

`lmodern`

If you want to work with the classic LATEX standard fonts, we recommend using the `lmodern` package . This improves the display of Computer Modern both on the screen and in the final print. `cfr-lm`

This experimental package provides additional font styles for the `lmodern` package.

⁵¹These are “ff”, “fi”, “fl”, “ffi”, and “ffl” in most LATEX fonts.

newtx, newtxmath

Some alternative fonts are provided for both the body text (*Times* and *Helvetica*) and the math mode. `libertine`

The package provides the fonts Linux Libertine and Linux Biolinum. To be able to use this font for math mode, `newtxmath` from the `newtx` bundle should be included with `\usepackage[libertine]{newtxmath}` in the preamble. The `libgreek` package contains Greek letters for Linux Libertine. `relsize` The size of a text markup can be set relative to the current font size. `textcomp` Additional symbols and special characters such as the per mille

or euro sign as well as arrows for the body text are provided.

There are also some useful packages for (sans serif) math fonts. If you are using the fonts of the corporate design, please refer to the `cdmath` option. `sansmathfonts`, `sansmath` If the normal LATEX fonts Computer Modern are used,

this package can be used for sans serif typesetting of mathematical expressions. An alternative

Package with the same objective is `sansmath` `sfmath`

This package has a similar goal, but unlike `sansmath`, it can be used not only for Computer Modern but also for Latin Modern, Helvetica and Computer Modern Bright with the appropriate option. `mathastext` The aim of the package is to extract all the necessary characters for mathematical typesetting

from the font used for the body text.

Mathematics theorem

These are packages that provide environments and commands for math typesetting and unit and number typesetting in general. `mathtools`, `amsmath` This package provides bug fixes and extensions to the

de facto standard package `amsmath` for math environments.

`bm`

The package offers `\bm` as an alternative to `\boldsymbol` in the mathematics set⁵². If errors occur when using the package, please refer to its documentation and in particular to the `\bmmax` macro.

Correctly formatting numbers is often a problem when using LATEX.

Problems arise, especially when data in English format is used in a German document. The `mathswap` package is provided in the TUD script bundle for this purpose. However, there are alternatives to this. `ionumbers` I actually only became aware of this package when working on `mathswap`. It offers more functionality and can be considered

as an alternative.

⁵²<http://tex.stackexchange.com/questions/3238>

icomma

If a space is placed after the comma in math mode, this will be taken into account in the output. The author must therefore always take care of the typographically correct output. digit

For German language documents, the comma is defined as the decimal separator between two digits. If the comma is not followed by a digit, the obligatory space is always set, which in my opinion is better than the behavior of icomma.

There are two useful packages for typographically correct setting of units – a half space between the number and *the upright* unit.

units

This is a simple and very useful unit typesetting package and is perfectly adequate for most needs. siunitx

This package is significantly more comprehensive than the others. In addition to units, numbers can also be set correctly in typographic terms. The output can be adapted to individual needs in many ways. The localization should be specified for German-language documents. More on this in subsection 9.16.

Further information on mathematical typography is given in Chapter 7.

Lists

enumite

The enumitem package extends the rudimentary functionalities of the LATEX standard lists **itemize**, **enumerate** and **description** and enables individual customization of these by providing many optional parameters according to the key-value principle. For example, a function that I use very frequently is removing the additional space between the individual entries of a list with **\setlist{noitemsep}**.

Directories of all kinds

In addition to creating the actual document, a scientific paper usually also requires all kinds of lists. A fixed component of this is the bibliography, and a list of abbreviations and formula symbols or symbols is also often required. A glossary may also be required. The appropriate packages are presented here. If complete source texts or just excerpts from them are to appear in the document and a corresponding list is to be generated for these, please refer to the listings package in subsection 8.2.2 on page 113. biblatex The package can be seen as a legitimate successor to BIBTEX. Similarly, biblatex offers the possibility to integrate literature databases and to select different styles of referencing and presentation of the bibliography.

With biblatex, adapting a specific style is much easier to implement than with BIBTEX. If *biber* is used to sort the bibliography, it is no problem to use a UTF-8 encoded literature database. In conjunction with biblatex, the additional use of the csquotes package is highly recommended.

glossaries, nouns This is

a very powerful package for creating a glossary and a list of abbreviations and symbols. The large number of options may be a bit daunting at first. This package should be considered especially if you need lists of abbreviations and *symbols*.

Alternatively, a manually set table can be used for a symbol directory. In my opinion, the `nomencl` package, which is often recommended for this purpose, offers no advantages over this.

acro, acronym

If you only want to create a list of abbreviations, this package is the first choice.

It provides commands for defining abbreviations, for using them in text and for sorting a list. Alternatively, the acronym package can be used. However, the user must sort the list of abbreviations manually.

Sliding objects

Packages for influencing the appearance, labeling and positioning of floating objects are presented. In subsection 9.23 you will also find information on manually manipulating floating object placement. With the option **`captions=<setting>`**, the KOMA-Script classes already offer

some

options for formatting the captions for floating objects. This package is therefore usually only necessary in certain exceptional cases for special instructions, but is also harmless when used. This package can be used to easily set subfigures or tables with the corresponding captions. The alternative package `subfig` should be avoided because it is no longer

maintained

and problems often arise when used in conjunction with other packages. If the range of functions provided by `subcaption` is not sufficient, the package `floatrow` can be used instead, which provides similar functionality to `subfig`. With this package, global settings and formatting can be made for *all* floating objects in a document using the package options or with `\floatsetup{. . .}`. Among other things, the font used within the **figure** and **table** environments can be set with `\floatsetup{font=. . .}`. With `\floatsetup[figure]{captionposition=bottom}` `\floatsetup[table]{captionposition=top}`

The typographically correct placement of *figure captions* and *table headings* can be automatically enforced - regardless of the position of the caption command `\caption` within the respective floating object environment. If the behavior is set as recommended, the KOMA-Script option **`captions=tableheading`** should also be used for good vertical placement of the table headings. This package can be used to force the output of floating objects before chapters and optionally subchapters.

flutter

This package allows floating objects to be placed as early as possible in the output document, at the point where they appear in the source code. They will therefore never appear before they are defined at the beginning of the page.

Tables

The environments **tabbing** and **tabular** or **tabular*** are provided by default for table typesetting, but their functionality is usually not sufficient for high-quality table typesetting. Therefore, additional packages are presented here.

array

This package allows you to create new column types and to define table columns in an extended way using **\newcolumntype** ($\langle \{ . . \} \langle \text{column type} \rangle \langle \{ . . \} \rangle$), whereby so-called "hooks" can be used to set specific instructions before and after entries within a column. The row height can be changed using **\extrarowheight**.
booktabs

There are already numerous tips for good table typesetting with LATEX53 on the Internet. Two rules should definitely be followed:

- I. no vertical lines
- II. no double lines

The booktabs package (German documentation booktabs-de) is a great help for typesetting high-quality tables and provides the commands **\toprule**, **\midrule** as well as **\cmidrule** and **\bottomrule** for different horizontal lines.

widetable

The standard LATEX environment **tabular*** can be used to set a table with a defined width. This package provides the environment **widetable**, which can be used as an alternative and creates a symmetrical table.

tabularx

This package can also be used to specify the total width of a table. For this, the column type **X** is defined, which can be specified as often as you like as an argument of the tabularx environment (**\begin{tabularx}\langle width \rangle\{\langle columns \rangle\}**). The X columns are similar to those of type **p\langle width \rangle**, whereby the width of these is automatically calculated from the desired total table width minus the space required for any standard columns that may be present. tabulary This is another package for automatically calculating column widths. The available

space -

the desired total width minus the required width for standard columns - is not distributed evenly across all columns as with **tabularx**, but is allocated in the tabulary environment for the **LCRJ** column types based on their cell content. (**\begin{tabulary}\langle width \rangle\{\langle columns \rangle\}**). longtable, ltxtable, ltablex If you want to create multi-page tables with page breaks, this package is the tool of choice. To combine multi-page tables with a **tabularx** environment, you can use the ltablex or, even better, ltxtable packages.

Alternatively, you can also use tabu.

53<http://userpage.fu-berlin.de/latex/Materialien/tabsatz.pdf>

multirow

The **\multirow** command is defined, which allows multiple rows to be combined into one column – similar to the **\multicolumn** macro for columns.

tabularborder

In tables, a horizontal space (**\tabcolsep**) is automatically placed between columns - or rather, before and after each column. This also happens *before* the first and *after* the last column. This additional space on the outer edges can be distracting, especially if the table is placed across the entire text width. The `tabularborder` package can automatically remove this space.

However, this only works with the `tabular` environment. The tables from the packages `tabularx`, `tabulary` and `tabu` are not supported. How this spacing can be removed manually can be found in subsection 9.9. v2.02 `tabu`

This is a relatively new package that attempts to implement many of the previously mentioned functionalities and provide more. The environments **tabu** and **longtabu** are defined for this purpose. It can be used as an alternative to `tabularx` and is particularly recommended as a replacement for the `ltxtable` package.

Danger!

The package in version v2.8 has been in need of several bug fixes for several years. The original author had also planned to change the user interface in a future Version⁵⁴ announced, but apparently the maintenance of the package has now been completely discontinued.

Changes to the LATEX 2_ε kernel rendered the package unusable. Therefore, the LATEX3 project team has made the most necessary adjustments⁵⁵ made so that the package is at least basically executable in the current version v2.9. However, a large part of the originally very good functionality cannot currently be used. Therefore, the package is currently only recommended to a limited extent.

Graphics and illustrations

Graphics for scientific work should be created as vector graphics to ensure scalability and high print quality. Ideally, these should also follow the style of the associated work.⁵⁶ There are two possible approaches to creating your own vector graphics that use the LATEX fonts and layout of the main document. Either the graphics are "programmed" in a similar way to the document, or drawing programs are used, which in turn support the output or passing on of text to LATEX.

The most important packages for programming graphics are presented here. How to use them can be found in the corresponding package documentation.

tikz

This is a very powerful package for programming vector graphics and is often the first choice when using pdfLATEX, especially for beginners. `pstricks` This package

represents the second variant for programming graphics. Since it allows access to PostScript (directly), there are even more options for creating your own graphics, which the provided commands make extensive use of.

⁵⁴<https://groups.google.com/d/topic/comp.text.tex/xRGJTC74uCI>

⁵⁵<https://github.com/tabu-fixed/tabu>

⁵⁶For high-quality documents, adopted graphics should not be copied or scanned directly but must be recreated in the desired target format and included with a reference to the source.

However, this results in the disadvantage that the graphics created with `pstricks` cannot be compiled directly into a PDF file. Instead, the graphics from the **pspicture** environment must first be converted into PDF files via the path **latex > dvips > ps2pdf**. These can then be integrated as images by pdfLATEX.

To enable this procedure, the following packages can be used: `pst-pdf` This package provides

methods for exporting PostScript graphics to PDF files. The individual calls for compiling DVI via PostScript to PDF must be carried out manually by the user or via the output routines of the editor used. `auto-pst-pdf`, `pdftricks2` The package automates the generation of `pstricks` graphics with the `pst-pdf` package.

For this, pdfLATEX must be run with extended write permissions using the option. To do this, **pdflatex** must be called with the option **--shell-escape** or **--enable-write18**. Please also note the information in subsection 9.18. An alternative to this is the package `pdftricks2`.

The tutorial treatise.pdf provides an example for `pstricks` and `tikz`. To avoid having to compile the entire document every time you make a change when creating graphics with one of the two packages, these can be stored in separate files.

The two packages standalone or subfiles are very useful for this.

For drawing a graphic with an image editing program that allows further processing with LATEX, I would like to refer you to the free programs *LaTeXDraw* and *Inkscape*. The latter program is particularly recommended.

`svg`

This package can be used to automate all the necessary steps for inserting an SVG graphic into a LATEX document. Further information can be found in subsection 9.19.

Cross-references and bookmarks

To create cross-references to certain structure levels, tables, figures or equations, you first need to create a *unique* label for these elements using `\label{<Label>}`, which can be referenced in the document using either `\ref` or, after loading `hyperref`, even better, using `\autoref`. You can find more information on this in subsection 9.5. `hyperref` This can be used to create bookmarks, cross-references and hyperlinks in a PDF document. If it is loaded, the **tudbookmarks** option and the **ltudbookmark** command can

also be

used. The bookmark package extends the support even further.

Both packages mentioned should – with very few exceptions such as glossaries – be included last in the preamble. `bookmark` This package improves and extends the options offered by `hyperref`

for creating bookmarks – also outline entries – in the PDF document. For example, font color and style can be changed.

varioref

This package is very useful for creating very good references to specific pages. In particular, if the cross-reference is to the current, previous or following page, or to the opposite page in double-sided formatting, suitable text modules are used for this.

cleveref

This package combines the advantages of automatic naming of referenced objects with the **\autoref** command from the hyperref package and the use of varioref.

Dividing the main document into subfiles

In order to shorten the time required for compiling a document when drafting it, it can be divided into sub-documents. This makes it possible to compile only the part of the document currently being edited - or the current tikz or pstricks graphic. In my opinion, the best packages for this task are presented below.

v2.02 standalone

This package is intended for creating independent (sub)documents that can later be integrated into a main document. Each of these sub-documents requires its own preamble. Optionally, the preambles of the sub-documents can be automatically integrated into a main document.

v2.02 subfiles

This package takes a slightly different approach than standalone. It is designed from the start to use a dedicated main document. The subfiles included in it with **\subfiles** use its preamble when compiling independently.

Regardless of whether you use one of the two packages or leave everything in one document, it is advisable to store your own defined commands, environments and the like in a separate package. To do this, you simply need to create an empty LATEX document and save it under mypreamble.sty or another name in the same folder as the main document. You can then execute your declarations in this file and integrate them into the document using **\usepackage{mypreamble}**. This has the advantage that the main document remains clear and that you can let your personal preamble grow generically and reuse it for other documents.

The little and big helpers...

Everything that could not be classified into the previous categories appears here. marginnote Margin notes can be created

using the **\marginpar** command. These are special floating objects in LATEX, which means that they are not always placed directly at the originally intended position on the edge of the page. The marginnote package provides the **\marginnote** command for non-floating margin notes. An alternative to this is the mparhack package. todonotes With todonotes, open tasks can be displayed in different formats on the edge of the page or

directly in the text. A list of all open points can be created from all comments.

xparse

This powerful package comes from the LATEX3 project and offers an alternative approach to the well-known LATEX declaration commands **\newcommand** and **\newenvironment** and their derivatives for creating your own commands and environments. With xparse it is possible to define mandatory and optional arguments at any point within the command construct. It is also possible to use characters other than square brackets to specify an optional argument. v2.02 xkeyval, keyval, kvoptions, pgfkeys

The KOMA script bundle loads the keyval package to be able to declare options with a key-value syntax. In addition, TUD script loads the kvsetkeys package to be able to react to undefined keys. The key-value syntax can also be used for custom macros to avoid the excessive use of optional arguments. This would allow the following definition: **\newcommand\command[<key-value list>]{<argument>}**

The xkeyval package particularly extends the possibilities for declaring different types of keys. If the keyval and kvsetkeys packages already loaded by TUD -Script are not sufficient in terms of functionality, this package can be used. The scrbase package can be used to develop your own packages whose options support the key-value format. If for some reason you do not want to use KOMA -Script at all, the two packages kvoptions or pgfkeys are an alternative.

v2.05 scrfile

This KOMA-Script package allows you to react to the loading of classes or packages directly before or after, for example to resolve package dependencies or to adapt commands specifically after loading a specific package. More information can be found in the KOMA-Script manual⁵⁷ to find.

calc

Normally, calculations in the document can only be performed using low-level TEX primitives. This package provides a simpler syntax for arithmetic operations of the four basic arithmetic operations. In addition, new commands are defined for determining the height and width of certain text extracts.

v2.02 mwe, blind text

The mwe package makes it very easy to create minimal examples that should contain both dummy text and images. If no graphics are required, but only sections of text, etc., loading dummy text is sufficient.

filecontents

This package extends the **filecontents** environment defined in the LATEX kernel so that existing files can be overwritten. pdfpages The package enables the

integration

of individual or multiple PDF files.

crop

This can be used to create a bleed and crop marks - for example for posters. An example of this can be found in subsection 9.21. pagecolor The package can be used to change the background

color of the

pages in the document.

⁵⁷<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>

afterpage

The command **\afterpage{. . .}** can be used to execute the content of its argument immediately after the current page is printed. filemod

If either pdfLATEX or LuaLATEX is used as the typesetting system, this package can be used to compare the modification date of two files and to execute definable actions depending on this. listings This

package is ideal for source code documentation in LATEX. It offers the possibility to read and display external source files and to highlight the syntax depending on the programming language used. In addition, a directory can be created with all source code extracts that have been included and those specified directly in the document .

v2.02

If listings is used in documents with UTF-8 encoding, the following should be added in the preamble immediately after loading the package:

```
\lstset{%
  inputencoding=utf8,extendedchars=true, literate=% {ä}{\a}1 {ö}
  {\o}1 {ü}{\u}1
  {Ä}{\A}1 {Ö}{\O}1 {Ü}{\U}1 {ß}{\ss}1 {~}{\textasciitilde}
  1 {»}{\guillemetright}1 {«}{\guillemetleft}1
}
```

coseoul

With this package, the structure of the outline can be created relatively. No absolute outline level (**\chapter**, **\section**) is specified, but rather the relation between the previous and current level (**\levelup**, **\levelstay**, **\leveldown**). selinput, inputenc The input encoding used is set to

utf8 by default ,

which is supported by (almost) all LATEX editors. This should not normally be changed. Older documents may still be saved in a different input encoding . If this cannot be changed, it is possible for pdfLATEX to use a different input encoding - set by the user in the editor (subsection 9.1).

This can be set with the package selinput (automatically) for LATEX With:

```
\usepackage{selinput}
\SelectInputMappings{adieresis={ä},germandbls={ß}} is used.
```

Alternatively, the input encoding can be set manually with the inputenc package using **\usepackage[<input encoding>]{inputenc}** . The fontenc package should always be loaded *first* .

Bug fixes

scrhack

The package fixes compatibility issues of the KOMA -Script classes with the packages hyperref, float, floatrow and listings. It is definitely recommended, but you should definitely read the documentation. scrwfile, morewrites If the error **No room for a new \write** appears,

it can possibly be fixed

with one of these two packages. More on this in subsection 9.25.

mparhack

To fix incorrectly placed margin notes, a bugfix for `\marginpar` is provided.

Alternatively, you can also use `marginnote`. `fix-cm`

If one or more warnings of the form **Font shape 'T1/cmr/m/n' in size <...> not available**

appear, the package `fix-cm` should be loaded *before* the class. See also the notes in subsection 9.24.

Chapter 9

Practical tips & tricks

9.1 LATEX editors

The most common editors for creating LATEX files are listed here. I personally am now a very convinced user of *TE Xstudio*, as it offers many support and assistance functions. In addition to these, there are other LATEX editors that are easy to use. Regardless of the editor you choose, it should definitely include Unicode support (UTF-8):

- *TE Xmaker*
- *Kile*
- *TE Xworks*
- *TE Xlipse* – plug-in for *Eclipse* •
- TE XnicCenter* •
- WinEdt* •
- LEd* – former LATEX editor •
- LYX* – graphical front-end for LATEX

For *TE Xstudio*, the archive `tudscr4texstudio.zip`⁵⁸ is available in the GitHub repository `tud-cd/tudscr` which contains files for extending the automatic command completion for TUD script. These must be inserted in **%APPDATA%\texstudio** under Windows or in `.config/texstudio` under Unix-like operating systems.

If you want to use the graphical LATEX frontend *LYX* to create a document with the TUD script classes, you will need special layout files to be able to use the class files. These are available together with a LYX document as an archive `tudscr4lyx.zip`⁵⁹ in the GitHub repository `tud-cd/tudscr`. The layout files must be copied into the appropriate subfolder in the LYX installation path. This is **%PROGRAMFILES(X86)%\LyX 2.1\Resources\layouts** on Windows or `/usr/share/lyx/layouts` on Unix-like operating systems. LyX must then be reconfigured using the *Tools* menu item.

9.2 Reference management in LATEX

v2.02 The simplest way to manage a LATEX literature database is to do it manually using the editor. However, it is much more convenient to manage references using a specialised application. There are two very good programs for this:

- *Citavi*
- *JabRef*

The *Citavi* program allows you to import bibliographic information from the Internet. However, this is sometimes incomplete or inadequate. With *JabRef*, on the other hand, the literature database must be created manually. However, individual entries can be imported from `.bib` files. Both applications support the export or creation of databases in the style of `biblatex`. For *JabRef*, this must be explicitly activated by the user.⁶⁰ There is a good tutorial on using the two programs in conjunction with `biblatex` and *biber* at this link.

⁵⁸<https://github.com/tud-cd/tudscr/releases/download/TeXstudio/tudscr4texstudio.zip>

⁵⁹<https://github.com/tud-cd/tudscr/releases/download/LyX/tudscr4lyx.zip>

⁶⁰Options/Settings/Advanced/BibLaTeX mode

9.3 Word separations in German texts

v2.02 The possible hyphenation points of words are calculated by L^AT_EX 2_ε using an algorithm . However, this was originally designed for the English language . For German texts, word hyphenation - especially for words written together - is significantly improved with the `hyphsubst` package. A dictionary from the `dehyph-exptl` package is used for this, which has been supplemented with a variety of hyphenation patterns.

In any case, a language pack such as `babel` or `polyglossia` is required, the latter being designed for use with LuaL^AT_EX or X_L A_TE_X . When using one of these two Unicode typesetting systems – in conjunction with one of the two language packs mentioned – the hyphenation patterns are automatically activated.

For pdfL^AT_EX, these can be activated with the `hyphsubst` package . To ensure that words with umlauts are correctly hyphenated, the `fontenc` package with, for example, the T1 font encoding must also be used. The beginning of a document preamble could look like this:

```
\documentclass[ngerman,<class options>]{<document class>} \iftutex

\usepackage{fontspec} \else

\usepackage[T1]{fontenc}
\usepackage[ngerman=ngerman-x-latest]{hyphsubst} \fi

\usepackage{babel}
. . .
```

One more note on the separation of words with hyphens. Normally, the two characters used by L^AT_EX 2_ε for `hyphen` and `hyphen` are identical. Unfortunately, the separation algorithm of L^AT_EX 2_ε is overridden for words that already contain a hyphen . As a result , monstrosities such as "Donaudampfschiffahrts-Gesellschafterversammlung" (which are quite common in the German language) are normally only separated directly after the specified hyphen.

However, it is possible to change the character used for the hyphen. For this purpose, loading the T1 font encoding with the `fontenc` package is mandatory. Unless otherwise specified by the font used, both the `hyphen` and the separator are located at position **45** in the character table. Fortunately, in the T1 font encoding, the same character is usually found again at position **127**. However, this depends on the font used. If the expression `\defaultshyphenchar=127` is used in the document preamble, this character can be used for the separator.

This is already set automatically in the fonts of the corporate design of the Technical University of Dresden .

If, despite all measures, a certain word is still hyphenated incorrectly, the word hyphenation of this word can be changed manually and globally. This is done with `\hyphenation{Sil-ben-tren-nung}` . It should be noted that this should be done for all inflected forms of the word. For local/temporary word hyphenation, commands from the `babel` package can be used. These are:

Description	command
exclusive separation points	<code>\-</code>
additional separation points	<code>“_</code>
Line break without hyphen	<code>““</code>
Hyphen without break	<code>“~</code>
Hyphen, which allows further separation points	<code>“=</code>

9.4 Local changes to commands and settings

v2.02

A central component of LATEX is the use of groups or groupings.

Within these, all changes made to commands, environments or settings remain local. This can be very useful if, for example, the behavior of a certain macro is to be changed once or within self-defined commands or environments, but the original functionality should normally be retained.

Example: The command **`\emph`** is provided by LATEX for highlighting text and normally results in italics or - if no font with real italics is available - italicized or slanted text. If you now want to highlight a certain section in bold, the command **`\emph`** can be changed and modified within a grouping. If this is ended, the command behaves as usual.

In this text, exactly one `\emph{word}` is highlighted.

`\begin{group}`

```
\renewcommand*{\emph}[1]{\textbf{#1}}% In this text,
exactly one \emph{word} is highlighted. \end{group}
```

In this text, exactly one `\emph{word}` is highlighted.

A grouping can be defined either with **`\begin{group}`** and **`\end{group}`** or simply with a pair of curly brackets `{. . .}`.

9.5 Designation of the structure levels by hyperref

v2.02

The **`hyperref`** package provides the command **`\autoref{<label>}`** for cross-references. In contrast to using **`\ref`**, this command not only precedes the numbering itself but also the corresponding element such as a chapter or figure. To name the referenced element, a sequential check is made to see whether the macro **`\<element>\autorefname`** or **`\<element>name`** exists. If the name of an element is to be changed, the corresponding identifier must be adjusted.

Example: Names of outline levels can be changed as follows. **`\renewcaptionname{ngerman}{\sectionautorefname}{subchapter}`** **`\renewcaptionname{ngerman}{\subsectionautorefname}{section}`**

9.6 URL breaks in the bibliography with biblatex

v2.02 If the biblatex package is used, it may happen that a URL is not wrapped properly. If this is the case, the counters **biburlnumpenalty**, **biburlucpenalty** and **biburllcpenalty** can be increased. A counter can be manipulated with `\setcounter{<counter>}` or locally with `\defcounter{<counter>}` from the etoolbox package. The possible values are between 0 and 10 000, with higher counter values resulting in more URL wraps at digits (**biburlnumpenalty**), uppercase (**biburlucpenalty**) and lowercase (**biburllcpenalty**). More details in the biblatex documentation.

9.7 Line spacing in headings

The setspace package can be used to change the line spacing or leading within the document. If this has been increased, the spacing in multi-line headings may appear too large. To correct this, the line spacing of all headings can be switched back to single line using the `\addtokomafont{disposition}{\setstretch{1}}` command. If this is only to be done for a certain outline level, the **disposition** parameter must be replaced by the corresponding font element.

9.8 Warning about header/footer height being too low

If the setspace package is used, it may happen that after changing the line spacing *within* the document one or both of the following warnings appear:

scrlayer-scrpage Warning: \headheight too low. scrlayer-scrpage Warning: \footheight too low.

This is due to the increased need for headers and footers due to the increased line spacing. In this case, the heights can be adjusted directly using `\recalctypearea`. However, this changes the type area in the document, which results in another and entirely justified warning from typearea. If the change in line spacing is really necessary, this should be done once in the preamble of the document. Then the warnings will also be eliminated.

9.9 Prevent indentation of table columns

Normally, LATEX 2_ε inserts some horizontal space before *and* after each column in a table using `\hskiptabcolsep`.⁶¹ This also happens *before* the first and *after* the last column. This optical indentation at the outer edges can be annoying under certain circumstances, especially in tables that are intentionally stretched across the entire page width - for example with the tabularx, tabulary or tabu packages.

The tabularborder package attempts to solve this problem automatically, but is not compatible with all LATEX 2_ε packages for table typesetting, including the three mentioned above. However, this problem can be solved manually.

⁶¹The distance between two columns is therefore `2\tabcolsep`.

When declaring a table, you can use `@{. . .}` before and after the column type to specify what should be inserted before or after the actual column instead of `\tabcolsep`. This can be used to remove indentations by using `@{}` in the appropriate places when specifying the column types before the first and after the last table column.

Example: A table with two columns, one of which has an automatically calculated width, should be set across the entire text width. The margin before the first and after the last should be removed.

```
\begin{tabularx}{\textwidth}{@{}X@{}} . \tabularnewline
. . . & . . .
. . .
\end{tabularx}
```

9.10 Suppressing the indentation of a paragraph

If indentations and no vertical spacing are used to mark up paragraphs in the document - as is usually sensible from a typographic point of view (KOMA script option `parskip=false`), it can happen that a specific paragraph - for example the one following a previously used environment - is indented unintentionally. This can be easily corrected manually by calling the `\noindent` macro directly at the beginning of the paragraph. If the indentation of paragraphs after specific environments or commands is to be prevented automatically, the `noindentafter` package is recommended.

9.11 Preventing the resetting of footnotes

With `\counterwithin` and `\counterwithout`, LATEX counters can be redefined so that they are reset when another counter is changed or not. Specifying `\counterwithout*{footnote}{chapter}` disables the resetting of the footnote counter by new chapters, which allows continuous footnotes to be created across chapters.

9.12 Warning when generating the table of contents

v2.02 If you use `\tableofcontents` to create the table of contents for a document with a three-digit page count, you may get many warnings with the message:

overfull \hbox

The page numbers in the directory are set in a box with a fixed width of **1.55em**, which is stored in the macro `\@pnumwidth` and should be increased if in doubt. The right margin for multi-line entries in the directory `\@tocrmarg` should also be increased, which is preset to **2.55em**. The values should only be changed minimally:

```
\makeatletter
\renewcommand*{\@pnumwidth}{1.7em}\renewcommand*{\@tocrmarg}{2.7em} \makeatother
```

9.13 Spaces and punctuation after LATEX commands

LATEX normally "swallows" the spaces after a macro without arguments. However, this is not always - in fact, in very rare cases - desired. For example, the command `\TUD` has been defined for this manual so that "Technische Universität Dresden" does not have to be written out all the time. To avoid having to put the curly bracket after the command (`\TUD{}`) when using the command within a sentence to obtain a following space, you can use `\xspace` from the `xspace` package. This will obtain a following space. The command `\TUD` is defined as follows:

```
\newcommand*{\TUD}{Technical University of Dresden\xspace}
```

The `xpunctuate` package extends the functionality even further. It also allows abbreviations to be defined in such a way that an accidental period is ignored:

```
\newcommand*{\zB}{z.\,B\xperiod}
```

9.14 Finding unknown LATEX symbols

There are a lot of symbols available for LATEX, but they are not always easy to find. In the summary `symbols-a4.pdf`⁶² many symbols from several packages are listed. Alternatively, Detexify⁶³ On this website, you simply draw the symbol you are looking for and the most similar ones are returned.

9.15 Using ellipsis

v2.02 Ellipsis is set with LATEX 2_ε using the commands `\dots` or `\textellipsis`.

Usually, these are *always* followed by a space, which is not always desired. The `ellipsis` package provides a solution here, whereby the `\xspace` option means that a space is automatically inserted after using one of the two commands.

```
\usepackage[xspace]{ellipsis}
```

It is originally intended for typesetting English texts, where a space is placed between ellipsis and punctuation marks. In German, this is different:

"To mark an omission in a text, three dots are used.

A space is placed before and after the ellipsis if it stands for an independent word or several words. If part of a word is omitted, it is immediately connected to the rest of the word. No additional full stop is placed at the end of the sentence. Punctuation marks are connected without a space." [Duden, 23rd edition]

To achieve this behavior, the following should be added to the preamble:

```
\let\ellipsis\punctuation\relax
\newcommand*{\qdots}{\dots\xspace}
```

The command `\qdots` is defined to be able to set ellipsis in square brackets (`[. . .]`), which are often used for shortening direct quotations.

⁶²<http://mirrors.ctan.org/info/symbols/comprehensive/symbols-a4.pdf>

⁶³<http://detexify.kirelabs.org/classify.html>

9.16 Localization for setting units with siunitx

If siunitx is to be used in a German-language document, the correct localization must at least be specified with `\sisetup{locale = DE}`. If the numbers are to be formatted correctly, further settings must be made. In my opinion, the best are the following.

```
\sisetup{%
  locale = DE,%
  input-decimal-markers={,},input-ignore={.},% group-
  separator={\,},group-minimum-digits=3%
}
```

The comma is used as a decimal separator. Furthermore, points are used in-
within the numbers are ignored and a grouping of three digits each is made.
Alternatively to this package, units can also be used.

9.17 Error message when loading a package with options

v2.05 It may happen that when loading a package with certain options via `\usepackage[<package options>]{<package>}` the following error is displayed:

! LaTeX Error: Option clash for package <. . >.

The requested package is almost certainly already loaded by the document class used or another package. Normally it is sufficient to pass the desired options to the relevant package using `\PassOptionsToPackage[<package options>]{<package>}` before loading the document class with `\documentclass`.

9.18 Problems when using auto-pst-pdf

v2.02

When using pdfLATEX, the auto-pst-pdf package reads the preamble and then creates a PDF file via the PostScript path `latex > dvips > ps2pdf`, which only contains all the graphics created in the existing pspicture environments. The ifpdf package provides the macro `\ifpdf`, which can be used to distinguish whether pdfLATEX is used as the typesetting system. Depending on this, different source code can be executed, which is used to solve the problems described below.

```
\usepackage{ifpdf}
```

The simultaneous use of floatrow The floatrow

package provides commands that can be used to set the labels of floating objects very conveniently. These first place their content in a box to determine its width and then output it. In combination with auto-pst-pdf, this leads to a duplicate creation of the desired image. To avoid this, the commands provided by floatrow must be "neutralized". The commands in question, however, accept up to three optional arguments *before* the two mandatory ones, which is not normally provided for the user interface for (re-)definition by LATEX 2_ε. Therefore, the xparse package is loaded, which makes this possible. More details are available here.

the corresponding package documentation. The following source code excerpt allows the floatrow commands to be used together with the pspicture environment as usual.

```
\usepackage{floatrow}
\usepackage{xparse}
\ifpdf\else
  \RenewDocumentCommand{\fcapside}{ooo+m+m}{#4#5}
  \RenewDocumentCommand{\ttabbox}{ooo+m+m}{#4#5}
  \RenewDocumentCommand{\ffigbox}{ooo+m+m}{#4#5} \fi
```

The parallel use of tikz and todonotes The tikz package – and all other packages that use it, such as todonotes – also have problems in conjunction with auto-pst-pdf . This dilemma can be solved by only loading the packages in question when pdfLATEX is active.

```
\ifpdf
  \usepackage{tikz}% . . if necessary, other packages based on tikz \fi
```

9.19 Automated integration of Inkscape graphics

v2.05 Embedding Inkscape graphics in LATEX documents is moved to Comprehensive TeX Archive 64. A derived and improved package enables an automated approach Network (CTAN svg-inkscape) is provided by the svg package . This export of Inkscape graphics directly during compilation with LATEX and subsequent integration into the document. The package defines the command `\includesvg[<parameter>]{<SVG file>}` for this purpose.

Inkscape is called externally via the command line or terminal using inkscape.exe. In order for this to actually be carried out, a LATEX engine must be running with the option `--shell -escape` or `--enable-write18` . The system must also know the path to the inkscape.exe file.⁶⁵ For further information, please refer to the documentation for the svg package.

9.20 Changing the paper format

It may happen that the paper format within a document needs to be changed temporarily, for example to include a design sketch in the digital PDF file.

With the KOMA script option ***paper=<setting>*** it is possible to change the orientation to landscape format as well as the size of the paper format itself.

Example: A document in A4 format should be temporarily changed to an A3 landscape format . The following minimal example shows how this can be changed using KOMA -Script tools using the options

```
paper=landscape and paper=A3 . \documentclass[paper=a4,pagesize]
\tudscrreprt} \usepackage[T1]{fontenc}
```

⁶⁴<http://www.ctan.org/pkg/svg-inkscape>

⁶⁵The path to inkscape.exe must be included in the operating system's **PATH** environment variable.

```

\usepackage[english]{babel}
\usepackage{blindtext}

\begin{document}
\chapter{Heading One}
\Blind text

\cleardoublepage
\storeareas\PotraitArea% saves the current type area
\KOMAOptions{paper=A3,paper=landscape,DIV=current} \chapter{Heading
Two}
\Blind text

\cleardoublepage
\PotraitArea% loads the saved type area \chapter{Heading Three}

\blind text
\end{document}

```

9.21 Bleed and crop marks

v2.05 When plotting posters or other colored printed products, the problem often arises that borderless printing is difficult to achieve. Therefore, it is often helpful to print the finished document on a larger sheet of paper and then cut it to the desired target format, which eliminates the problem of the unprintable border. This can be achieved in two different ways.

The easiest way is to use the crop package. This allows you to create the document in the desired target format as usual. Before printing, you load this package and simply specify the desired format of the sheet of paper.

```

\RequirePackage{fix-cm}
\documentclass[%
  paper=a1,
  fontsize=36pt ]
\tudscrposter}
\usepackage[T1]{fontenc}
. . .
\usepackage{graphicx}
\usepackage[b1,center,cam]{crop}
\begin{document}
. . .
\end{document}

```

Alternatively, the functionality of the geometry package can be used for the TUD script classes . This package provides the **geometry** command , in whose argument the paper format can be specified with ***paper=<paper format>*** . If the parameter ***layout=<target format>*** is also specified, this defines the desired target format . If necessary, this area should be indented slightly with ***layoutoffset=<length value>*** . Specifying ***showcrop=<on-off value>*** also generates visual crop marks.

```

\RequirePackage{fix-cm}
\documentclass[%
  paper=a1,
  fontsize=36pt ]
\tudscrposter}
\usepackage[T1]{fontenc}
. . .
\geometry{paper=b1,layout=a1,layoutoffset=1in,showcrop} \begin{document}

. . .

\end{document}

```

For more detailed explanations and further setting options, please refer to the documentation for crop and geometry . The TUD script option **bleedmargin** can also be used to extend the colored areas of the tudheadings page styles to enable "cropping into color".

9.22 Avoid scaling a PDF file when printing

v2.04 When creating a print job for a PDF file, it may happen that the PDF viewer used unnecessarily scales the file beforehand, thereby increasing the page margins. There are two ways to suppress this behavior for documents created with pdfLATEX :

1. If the hyperref package is used in the document anyway, simply calling `\hypersetup{pdfprintscaling=None}` is sufficient.
2. The low-level command `\pdfcatalog{/ViewerPreferences<</PrintScaling/None>>}` has the same behavior and can also be used without hyperref.

Further information is available at <http://www.komascript.de/node/1897> to find.

9.23 Placement of sliding objects

The two packages flafter and placeins allow you to limit the space available to LATEX for placing floating objects. This can also be influenced by the commands listed below. The macros can be changed very easily with `\renewcommand*{<macro>}{<value>}` .

`\floatpagefraction`

Default: **0.5**

`\dblfloatpagefraction`

Default: **0.5**

The value indicates the relative size of a floating object in relation to the set text height (**(textheight)**), which must be reached at least so that a separate page is created for it before the start of a new chapter. A distinction is made between one-column (**(floatpagefraction)**) and two-column (**(dblfloatpagefraction)**) layouts.

The value for both commands should be in the range **0.5 ... 0.8** .

`\topfraction`

Default: **0.7**

`\dbltopfraction`

Default: **0.7**

These values indicate the maximum page share for floating objects placed at the top of the page for single column and double column layouts. It should be in the range of **0.5 ... 0.8**

and be larger than `\floatpagefraction` or `\dblfloatpagefraction` .

`\bottomfraction` Default: **0.3**

This is the maximum page share for floating objects placed at the bottom of the page. It should be between **0.2** and **0.5** .

`\textfraction` Default: **0.2**

This is the minimum amount of body text that must be present on a page with floats if they are not displayed on a separate page. It should be in the range **0.1 ... 0.3** .

<code>totalnumber</code> (counter)	Default: 3
<code>topnumber</code> (counter)	Default: 2
<code>dbltopnumber</code> (counter)	Default: 2
<code>bottomnumber</code> (counter)	Default: 1

There are also counters that determine the maximum number of floating objects per page in total (**totalnumber**) , at the top (**topnumber**) and bottom (**bottomnumber**) margins , and in two-column typesetting spanning both columns (**dbltopnumber**) . The values can be changed with `\setcounter{<counter>}{<value>}` .

<code>\@fptop</code> (length)
<code>\@fpsep</code> (length)
<code>\@fpbot</code> (length)
<code>\@dblftop</code> (length)
<code>\@dblfpsep</code> (length)
<code>\@dblfpbot</code> (length)

If there are still floating objects left before the start of a chapter, LATEX normally outputs them on a separate vertically centered page. These lengths determine the distance between the first floating object and the top edge of the page (`\@fptop`, `\@dblftop`), between the individual objects (`\@fpsep`, `\@dblfpsep`) and to the bottom edge of the page (`\@fpbot`, `\@dblfpbot`). If this should not happen, the lengths can be changed by the user.

Example: All floating objects on a page specially set for this purpose should be output directly at the beginning of the page.

The following can be used in the document preamble for this purpose: `\makeatletter \setlength{\@fptop}{0pt}`
`\setlength{\@dblftop}`
`{0pt}% twocolumn`
`\makeatother`

9.24 Warning when selecting font size

v2.04 The font size used in the document can be easily set in the KOMA -Script classes using the **fontsize** option , which should always be specified as a class option. For relatively large and small font sizes, a warning in the form of

LaTeX Font Warning: Font shape '. . .' in size <xx> not available

This is because at the time of loading a class, the standard Computer Modern fonts are always searched for, regardless of whether a different font package is loaded. These are de-facto not scalable in all sizes. To avoid the warnings To eliminate this, the package fix-cm should be loaded with **\RequirePackage** *before* the document class become:

```
\RequirePackage{fix-cm}
\documentclass[<class options>]{<class>}
\usepackage[T1]{fontenc}
. . .
\begin{document}
. . .
\end{document}
```

This will resolve the warnings.

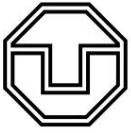
9.25 Error message: ! No room for a new \write

v2.02 L^AT_EX 2_ε only has a limited number of output streams available for creating and writing external auxiliary files . For each directory to be created, L^AT_EX reserves a new stream for itself. Some very useful packages already introduced in this manual – such as hyperref, biblatex, glossaries, todonotes or filecontents – require their own auxiliary files and open one or more output streams to create them. If the user loads several files into one auxiliary file, writing packets, it can lead to the error message

! No room for a new \write .

The solution here is the scrwfile package, which makes some changes to the L^AT_EX kernel to reduce the number of auxiliary files needed to write all directories It simply has to be included in the preamble. If this package If problems occur unexpectedly, the instructions in the KOMA -Script manual⁶⁶ to Another way to circumvent the problem of the small number of output streams is to use the morewrites package . However, its use is not crowned with success in all cases.

⁶⁶<http://mirrors.ctan.org/macros/latex/contrib/koma-script/doc/scrguide.pdf>



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Part III

Attachment

Appendix A

Further installation instructions

Attention! Here different variants are explained how to use TUD script in version v2.06o if an earlier version was used as a local user installation.

Until version v2.01, TUD-Script was distributed exclusively via the TUD-LATEX-Forum⁶⁷ for local user installation. This is primarily due to historical reasons and is related to the development of TUD -Script . A local user installation offers one advantage - which is rather negligible. If problems occur when using TUD -Script , they can be reported and discussed in the forum. If such a problem actually requires a bug fix or update of TUD -Script , this can be done quickly and easily via the GitHub repository tud-cd/tudscr⁶⁸ provided and can be used immediately by the user.

However, this has the major disadvantage for all users who visit the forum relatively rarely or not at all , that they cannot benefit from updates, improvements and bug fixes in new versions. All subsequent bug fixes and updates of the TUD script bundle must also be carried out manually by the user . Therefore, distribution via Comprehensive TeX Archive Network (CTAN)⁶⁹ preferred , so that TUD-Script is always available in the latest version - provided that the user has an up-to-date LATEX distribution. The only disadvantage of this approach is that the distribution of a bug fix and its subsequent provision by the distribution used usually takes up to two days.

The common LATEX distributions usually first search the local **texmf user** directory for classes and packages and only then the texmf path of the

Distribution itself. It does not matter in which path the newer version of a class or package is located. As soon as the file you are looking for is found in the user directory, the search is terminated. As a result, this

means that all updates via CTAN will not take effect if TUD-Script was installed as a local user version.

Therefore, users are advised to uninstall any existing local user installation of TUD-Script if it was not installed *intentionally* . The uninstallation is explained in subsection A.1. After this, updates of the TUD -Script bundle can be carried out using the update function of the distribution.

How the TUD -Script bundle can still be installed or updated as a local user version can be found in subsection A.2 and subsection A.3 respectively. In this case, however, the user should know exactly what he is trying to achieve, as he is responsible for updating TUD-Script himself.

Using the outdated font families v2.06 If a

document is to be set using the outdated font families **Univers** and **DIN Bold** , a local installation of the Type1 fonts is required, which is described in subsection A.4. In addition, the two options **tudscrver=2.05** and **cdoldfont** are mentioned. To use **Open Sans**, only the current version of the opensans package is required, which is available on the Comprehensive TeX Archive Network (CTAN)⁷⁰ can be found.

⁶⁷<http://latex.wcms-file3.tu-dresden.de/phpBB3/>

⁶⁸<https://github.com/tud-cd/tudscr/releases>

⁶⁹<http://www.ctan.org/pkg/tudscr>

⁷⁰<http://www.ctan.org/pkg/opensans>

A.1 Local uninstallation of the TUD script bundle

Via the command line or the terminal you can

kpsewhich –all tudscrbase.sty

check whether a local user installation of TUD -Script is available. All paths in which the tudscrbase.sty file is found are displayed. If only the distribution path appears, the TUD -Script version of CTAN is active and the user can work with the TUD -Script bundle.

If *only* the local user directory is found or no directory at all, then an outdated distribution is most likely being used. In this case, updating it is *strongly* recommended. If this is not possible, TUD-Script *must be* installed as a local user version (subsection A.2) or - if a path was given - updated (subsection A.3).

If at least one other path is displayed in addition to the distribution path, a local user version is installed. In this case, the user has three options:

1. Remove the local user installation (script-based)
2. Remove the local user installation (manually)
3. Updating the local user version (subsection A.3)

To remove the local user installation, you can use tudscr_uninstall.bat⁷¹ for Windows and for Unix-like operating systems tudscr_uninstall.sh⁷² used. After executing the respective script, you can use the command shown at the beginning in the command line or terminal to check whether the uninstallation was successful. If at least one local path is still displayed, TUD-Script should be uninstalled manually, which is described below.

Only the uninstallation of all local user installations of TUD -Script enables the use of the latest CTAN version. This requires some manual work by the user - assuming that the automated uninstallation using the previously mentioned uninstallation scripts was not successful. The command in the command line or in the terminal with

kpsewhich –all tudscrbase.sty

found – *additional* to the distribution folder – path has the following structure:

<installation path>/tex/latex/tudscr/tudscrbase.sty

To completely remove the user installation, first navigate to **<installation path>** . Then do the following in this path:

tex/latex/tudscr/ delete .cls and .sty files **tex/latex/tudscr/**

completely delete **the logo** folder completely delete the **tudscr** folder

doc/latex/ completely delete the **tudscr** folder

source/latex/

Finally, run the command **texhash** in the command line or in the terminal . This removes the local user version and from now on the version

of TUD script, which is provided by the distribution used.

⁷¹https://github.com/tud-cd/tudscr/releases/download/uninstall/tudscr_uninstall.bat ⁷²https://github.com/tud-cd/tudscr/releases/download/uninstall/tudscr_uninstall.sh

A.2 Local installation of the TUD script bundle

A local user installation of TUD-Script should only be carried out by users who know exactly why this should be done. The appropriate scripts are provided for this purpose. For a local user installation of the TUD-Script bundle under Windows for the distributions TEX Live or MiKTEX, the files from TUD-Script_v2.06o_Windows.zip⁷³ are required. For Unix-like operating systems, the files from TUD-Script_v2.06o_Unix.zip⁷⁴ are required.

When you run the specific installation script - tudscr_v2.06o_install.bat for Windows or tudscr_v2.06o_install.sh for Unix-like operating systems - all files are installed in the local user directory of the respective distribution, unless another directory is explicitly specified. In any case, you should update your LATE X distribution *before running the installation script*. This is especially true when using MiKTEX. Otherwise, you may encounter problems during the installation process or when using TUD-Script.

As an alternative to using the provided scripts, you can also simply download the contents of the archive tudscr_v2.06o.zip⁷⁵ copied into the local texmf user directory and then called in the command line or in the terminal `texhash`.

A.3 Local update of the TUD script bundle

A.3.1 Update of the TUD script bundle from version v2.02

From version v2.06 onwards, an update and the local installation are no different from each other; the procedure is absolutely identical to the description in subsection A.2. However, *the local update to version v2.06o only works if TUD-Script is already installed in at least version v2.02, either as a local user version or via the distribution.*

A.3.2 Update of the TUD script bundle from version v2.00

Version v2.02 included some major changes. Therefore, no dedicated update is offered for previous versions - i.e. v2.00 and v2.01. The user can update the software either with a script-based installation - as explained in subsection A.2 - or with a manual reinstallation.

A.3.3 Update of the TUD script bundle from version v1.0

If TUD-Script is installed in the outdated *version v1.0*, it is strongly recommended to completely remove this version before updating. Otherwise, massive problems and errors will occur after an update. To uninstall, use the scripts tudscr_uninstall.bat⁷⁶ respectively tudscr_uninstall.sh⁷⁷. The current version v2.06o can be installed after completely uninstalling all outdated versions as described in subsection A.4.

⁷³https://github.com/tud-cd/tudscr/releases/download/v2.06o/TUD-Script_v2.06o_Windows.zip

⁷⁴https://github.com/tud-cd/tudscr/releases/download/v2.06o/TUD-Script_v2.06o_Unix.zip

⁷⁵https://github.com/tud-cd/tudscr/releases/download/v2.06o/tudscr_v2.06o.zip

⁷⁶https://github.com/tud-cd/tudscr/releases/download/uninstall/tudscr_uninstall.bat

⁷⁷https://github.com/tud-cd/tudscr/releases/download/uninstall/tudscr_uninstall.sh

Compared to *version v1.0*, the user interface has not changed much, and switching to version v2.06o should not cause any problems. If problems still occur, the user should first read the description of the tudscrcomp package in subsection B.2 on page 144, which provides an interface for using old and originally no longer provided commands and options. However, some of these are no longer provided by the tudscrcomp package . These are listed in Appendix B. If, despite all the information, errors or problems still occur when switching to the new TUD script version, a message should be posted in the TUD-LATEX forum⁷⁸ or in the GitHub repository tud-cd/tudscr⁷⁹ the best way to get help.

A.4 Installing outdated font families

Until the beginning of 2018, the corporate design of the Technical University of Dresden did not use **Open Sans** as its house font , but rather the *Univers* and *DIN Bold font families*. These can still be used with TUD-Script to compile old documents.

For this purpose, the options **tudscrerv=2.05** and **cdoldfont** are recommended. Since these are licensed fonts, they must be requested from the university marketing department⁸⁰ with reference to the use of LATEX and installed after receipt of the necessary archives Univers_PS.zip and DIN_Bd_PS.zip for Windows (section A.4.1) or Unix-like operating systems (section A.4.2). However, unless there is a good reason , **Open Sans** should always be used, especially for newly created documents.

The TUD script bundle also supports the fonts in the OpenType format, which must also be ordered on request from the university marketing department . The fonts contained in the Univers_8_TTF.zip and DIN_TTF.zip archives can be used with the fontspec package once they have been installed for the operating system . Further information on using the fontspec package can be found in subsection 2.1 on page 15.

The GitHub repository tud-cd/tudscr contains the scripts required for font installation⁸¹ as well as the scripts⁸² for the classes by Klaus Bergmann. The different installation scripts are mainly due to the fact that during the installation for the TUD script bundle, both the metrics and the kerning of the fonts for body text and math mode are adjusted. If the improved fonts are to be used for the classes by Klaus Bergmann, this can be done with the fix-tudscrfonts package , but this affects the result of the output generated, which is why the installation scripts are still available in different versions.

Required packages for font installation

To install the fonts, you need the packages fontinst, cmbright, hfbright, cm-super and iwona are *essential* and therefore *absolutely* necessary. The presence of these is checked by the respective font installation scripts and the installation is aborted with a corresponding warning if one or more packages are missing.

⁷⁸<http://latex.wcms-file3.tu-dresden.de/phpBB3/>

⁷⁹<https://github.com/tud-cd/tudscr/issues>

⁸⁰<https://tu-dresden.de/cd/>

⁸¹<https://github.com/tud-cd/tudscr/releases/tag/fonts>

⁸²<https://github.com/tud-cd/tudscr/releases/tag/oldfonts>

Note on MiKTeX Before

installing the fonts for TUD-Script, it is essential that you update MiKTeX . It is also advisable to install MiKTeX in the multi-user version with administrator rights, as the single-user version can lead to relatively irregular and not always comprehensible problems.

It is possible that some of the packages required for the font installation process are not yet installed. If the automatic installation of missing packages is enabled, it is usually sufficient to start the installation script. Otherwise, these packages must be added manually by the user via the MiKTeX package manager.

The installation script also fails for some users due to limited user access rights when entering the fonts into the map file. This may have to be done by the user via the command line.

initexmf --edit-config-file updmap . The file

that opens should contain the entry **Map tudscr.map** . If this is not the case, this line must be entered manually and the file then saved . The user must then execute the following command in the command line:

initexmf --mkmaps

Note on TEX Live and MacTeX If a full

installation of TEX Live has not been performed, the necessary packages for font installation must be added manually via the TEX Live package manager.

If the fonts are still not available after the installation script has run without errors , the synchronization of all font files must be initiated. The map file and the associated font files must then be registered.

The necessary calls are: **updmap-sys --**

syncwithtrees updmap-sys --

enable Map=tudscr.map updmap-sys --

force

If the fonts are still not available, then other fonts have probably already been installed *locally* on the system . In this case, the process should be repeated for a local font installation with

updmap --syncwithtrees

updmap --enable Map=tudscr.map

updmap --force

This procedure, however, renders the **updmap-sys** command ineffective from now on. After a system-wide installation of new fonts - for example when updating the distribution - these must be registered locally in TEX Live or MacTeX by the user by manually calling **updmap** .

Danger!

*The scripts **tfptop**, **pltotf** and **vptovf** are required for the font installation, which are available in TEX Live or MacTeX via the fontware package. provided from collection-fontutils and must be installed.*

A.4.1 Installing Type1 fonts under Windows

To install the fonts of the corporate design for the TUD -Script bundle, the archive TUD-Script_fonts_Windows.zip⁸³ is required. This can be used for both TEX Live and MiKTEX and contains all the required files, except for the respective font archives themselves. These should be copied into the same directory after unpacking the archive. Before using the tudscr_fonts_install.bat script , make sure that *all* of the following files are in the same directory:

tudscr_fonts_install.bat Installation script Archive with
 DIN_Bd_PS.zip font files for **Univers** Univers_PS.zip
 Archive with font files for **DIN Bold** tudscr_fonts_install.zip Archive with metrics
 for font installation via fontinst

When the installation script is run, all fonts are installed in a local user directory by default. If the script is run via the context menu with administrator rights, the installation takes place in a path that is valid and readable *for all users* .

A.4.2 Installing Type1 fonts on Linux and OS X

To install the fonts of the corporate design for the TUD -Script bundle, the archive TUD-Script_fonts_Unix.zip⁸⁴ is required. This can be used for both TEX Live and MacTEX and contains all the required files except for the respective font files themselves. These should be copied into the same directory after unpacking the archive. Before using the tudscr_fonts_install.sh script , make sure that *all* of the following files are in the same directory:

tudscr_fonts_install.sh installation script (Terminal: **bash tudscr_fonts_install.sh**)
 Univers_PS.zip Archive with font files for **Univers** DIN_Bd_PS.zip
 Archive with font files for **DIN Bold**
 tudscr_fonts_install.zip Archive with metrics for font installation via fontinst

Note on Linux and OS X

Attention! *After unpacking a release archive in the appropriate path⁸⁵, the script must be called with **bash <script>.sh** in the terminal in this path with the required files.*

By default, all fonts are installed in the local user directory (**\$TEXMFHOME**) . If the script is used with **sudo** , the installation is carried out *for all users* in the local system path (**\$TEXMFLOCAL**).

It is essential to ensure that the terminal is active in the correct directory when running the script. With most Unix-like operating systems, it is easy to open the terminal from the user interface using the context menu in the desired path . If this is not possible, after opening the terminal, you must first navigate to the appropriate path using the **cd** command - for example **cd "\$HOME/Downloads/<subfolder>"** . An example call in the terminal could be: **cd "\$HOME/Downloads/TUD-Script_fonts_Unix" [ENTER] bash tudscr_fonts_install.sh [ENTER]**

⁸³https://github.com/tud-cd/tudscr/releases/download/fonts/TUD-Script_fonts_Windows.zip

⁸⁴https://github.com/tud-cd/tudscr/releases/download/fonts/TUD-Script_fonts_Unix.zip

⁸⁵for example **cd "\$HOME/Downloads/<subfolder>"**

A.4.3 Installation instructions for portable installations

In principle, the installation of the Type1 fonts of the corporate design when using TEX Live Portable or MiKTEX Portable is equivalent to the non-portable version, which is described in subsection A.4. All instructions given there should be carefully considered. The installation path preset by the respective installation script should not normally be changed. If this is done, it should logically be located on the external storage medium **<drive letter>:**.

TEX Live Portable

The following procedure has been tested with Windows. Recommendations for the portable installation for Unix-like operating systems can be sent to **hanisch.latex@outlook.com** be sent.

1. Install TEX Live Portable in **<drive letter>:\LaTeX\texlive** 2. Open the file tl-tray-menu.exe in the installation path 3. Open the context menu of TEX Live Portable by right-clicking on the corresponding icon in the notification area of the taskbar and perform an update either via the graphical interface (*Package Manager*) or the command line (*Command Prompt*)
4. Use the context menu to run the command line and start the script for installing the fonts tudscr_fonts_install.bat. If necessary, change to the path of the script beforehand - for example:

```
cd /d %USERPROFILE%\Downloads\TUD-Script_fonts_Windows [ENTER]
tudscr_fonts_install.bat [ENTER]
```

The script may report missing packages. These must be installed by the user via the *TEX Live Manager*.
Running it without the command line opened via TEX Live Portable will result in errors.

Danger!

MiKTEX Portable

1. Install MiKTEX Portable in **<drive letter>:\LaTeX\MiKTeX86** 2. Open the miktex-portable.cmd file in the installation path 3. Open the MiKTEX Portable context menu by right-clicking on the corresponding icon in the notification area of the taskbar and perform an update 4. Run the command line via the context menu and start the script for installing the fonts tudscr_fonts_install.bat. If necessary, change to the path of the script beforehand - for example:

```
cd /d %USERPROFILE%\Downloads\TUD-Script_fonts_Windows [ENTER]
tudscr_fonts_install.bat [ENTER]
```

In this step, the packages fontinst, cmbright and iwona may be installed afterwards.
Running it without the command line opened via MiKTEX Portable will result in errors.

Danger!

5. When using TUD -Script for the first time, all other required packages of MiKTEX Portable will be installed if automatic installation is enabled and they are not already present. This includes the

Packages tudscr, koma-script as well as etoolbox, xpatch letltxmacro, environ, trimspaces, xcolor and mptopdf.

⁸⁶The path must *not* be at the top level of the directory **<drive letter>:**.

A.4.4 Problems installing Type1 fonts

If Windows is used, it may happen that necessary command calls for the installation script cannot be executed. In this case, the path to the required files, which are normally found under **%SystemRoot%\System32**, is not included in the **PATH** environment variable. A hint for solving the problem can be found in this post in the forum⁸⁷ to find.

If unexpected problems occur during installation, a log file must be created to solve the problem. To do this, the script that is causing the problem should *not* be run from the command line or Explorer under Windows, but rather via *Windows PowerShell*. To do this, simply enter "PowerShell" in the Windows start menu and then press **[ENTER]** to open it. Then use **cd** to navigate to the script folder and run it with **.<script>.bat[Tee-Object -file <script>.log**. A call from the PowerShell console could be:

```
cd "$env:USERPROFILE\Downloads\TUD-Script_fonts_Windows" [ENTER] .
\tudscr_fonts_install.bat[Tee-Object -file fonts_install.log [ENTER]
```

For Unix-like systems, the command **bash <script>.sh > <script>.log** should be used from the terminal. An example usage could be: **cd "\$HOME/Downloads/TUD-Script_fonts_Unix" [ENTER]**

```
bash tudscr_fonts_install.sh > fonts_install.log [ENTER]
```

The log file created in this way can be posted *with a short error description* either in the TUD-LATEX forum or sent by email to **hanisch.latex@outlook.com** be sent.

⁸⁷<http://latex.wcms-file3.tu-dresden.de/phpBB3/viewtopic.php?t=359>

Appendix B

Obsolete and completely removed Options and commands

B.1 Obsolete options and commands in TUD -Script

During the further development of TUD-Script, some options and commands could no longer be implemented in their original form or were simply discarded - partly for reasons of compatibility with other packages. Nevertheless, for most of the deleted directives there is a possibility to implement their functionality without major effort using TUD-Script in the current version v2.06o. If this is the case, this will be briefly mentioned here .

B.1.1 Changes for TUD -Script v2.00

cd=alternative	(not applicable)
cdtitle=alternative	(not applicable)
\titlecolwidth (length)	(not applicable)
\authortext (identifier)	(not applicable)

The alternative title page has been completely removed from the TUD script bundle. Accordingly , the associated options, length and identifier are also no longer available.

color=<on-off value> (see **cd** on page 24)

The settings for the color representation of the document are made via the **cd option**.

tudfonts=<on-off value> (see **cdfont** on page 16)

The font setting option has been significantly expanded. It has been renamed for consistency reasons .

tudfoot=<on-off value> (see **cdfoot** on page 29)

The **tudfoot** option has also been renamed to match the naming scheme of the rest of TUD-Script's options .

headfoot=<setting> (see KOMA script options **headinclude** and **footinclude**)

This option was necessary for TUD -Script version v1.0 to enable parallel use of the two packages typearea and geometry . The creation of the type area has been completely revised. The settings for the KOMA -Script options **headinclude** and **footinclude** are now passed directly to the geometry package , so that the **headfoot** option is no longer necessary and has therefore been removed.

partclear=<on-off value>	(see cleardoublespecialpage on page 41)
chapterclear=<on-off value>	(see cleardoublespecialpage on page 41)

Both options have been merged into the new **cleardoublespecialpage** option , which ensures a consistent layout. The original options are no longer available.

abstracttotoc=<on-off value>	(see abstract on page 42)
abstractdouble=<on-off value>	(see abstract on page 42)

Both options have been integrated into the **abstract** option and are therefore superfluous.

\logofile{<filename>} (see **\headlogo** on page 30)

The **\logofile** command has been renamed to **\headlogo** , but the functionality remains the same.

bookmarks=<on-off value> (see **tudbookmarks** on page 50)

The option has been renamed to avoid overlap with hyperref.

\signatureheight (length) *(not applicable)*

The height of the signature line has been designed to be flexible, so no adjustment by the user is necessary.

\titlecoldelim (identifier) (see **\titledelimiter** on page 39)

The separator for descriptions or descriptive texts and the actual field on the title page is no longer language-dependent and has been renamed.

\confirmationandrestriction (see **\declaration** on page 50)

\restrictionandconfirmation (see **\declaration** on page 50)

The two commands are omitted; instead, either the **\declaration** command or the **declarations** environment should be used together with the **\confirmation** and **\blocking** commands, which can be arranged in any order in the environment.

\location{<location>} (see **\place** on page 50)

In line with other LATEX packages and classes, **\location** was renamed to **\place**.

Task The

environment for creating a task for a scientific paper has been outsourced to the **tudscrsupervisor** package. This must be loaded in order to use the task environment **and** the standardized form derived from it.

cdtask=<setting> (see **task** on page 64) *(not applicable)*
taskcompact=<on-off value> *(not applicable)*
\taskcolwidth (length) *(not applicable)*

The class option **cdtask** has been completely removed, all settings are made directly via the optional argument of the **task** environment. The variant of a compact header with the **taskcompact** option is no longer provided. The option to manually specify the column width for the header of the task with **\taskcolwidth** has been removed due to the improved automatic calculation.

\tasks{<goals>}{<focus>} **\focustext** (see **\taskform** on page 64)
 (identifier) **\objectivestext** (see **\focusname** on page 68)
 (identifier) (see **\objectivesname** on page 68)

The command **\tasks** was renamed to **\taskform** and its functionality was extended. The identifiers used in it were also slightly modified.

\studentid{<enrolment number>} (see **\matriculationnumber** on page 35)
\enrolmentyear{<year of enrolment>} (see **\matriculationyear** on page 35) (see
\submissiondate{<date>} **\date** on page 36) (see
\birthday{<date of birth>} **\dateofbirth** on page 35) (see
\birthplace{<place of birth>} **\placeofbirth** on page 35) (see
\startdate{<issue date>} **\issuedate** on page 65)

All commands have been renamed and can now be used not only for the task but also for the title page in the corporate design.

<code>\studentidname</code> (identifier)	(see <code>\matriculationnumbername</code> on page 53)
<code>\enrolmentname</code> (identifier)	(see <code>\matriculationyearname</code> on page 53)
<code>\submissiontext</code> (identifier)	(see <code>\datetext</code> on page 52)
<code>\birthdaytext</code> (identifier)	(see <code>\dateofbirthtext</code> on page 53) (see
<code>\birthplacetext</code> (identifier)	<code>\placeofbirthtext</code> on page 53) (see
<code>\supervisorllname</code> (identifier)	<code>\supervisorothername</code> on page 51) (see
<code>\defensetext</code> (identifier)	<code>\defensedatetext</code> on page 52) (see
<code>\starttext</code> (identifier)	<code>\issuedatetext</code> on page 68) (see
<code>\duetext</code> (identifier)	<code>\duedatetext</code> on page 68)

The identifiers have been renamed based on the corresponding command names.

B.1.2 Changes for TUD -Script v2.02

<code>\chapterheadingsvskip</code> (length)	(see <code>\pageheadingsvskip</code> on page 26)
---	--

The vertical positioning of headings has been split. On the one hand, this can be changed for title, part and chapter pages (`chapterpage=true`) using the `pageheadingsvskip` option. For chapter headings (`chapterpage=false`) and the title header (`titlepage=false`), this can be done independently using `headingsvskip`.

<code>\degree[<abbr.>]{<degree>}</code>	(see <code>\graduation</code> on page 38)
<code>\degreertext</code> (identifier)	(see <code>\graduationtext</code> on page 52)

The command has been renamed to increase compatibility with other packages and the associated identifier has been adjusted accordingly. `\restriction[<company> || <parameter`

<code>list>] \restrictionname</code> (identifier)	(see <code>\blocking</code> on page 49)
	(see <code>\blockingname</code> on page 53)
<code>\restrictiontext</code> (identifier)	(see <code>\blockingtext</code> on page 54)

The command has been renamed to increase compatibility with other packages and the associated identifiers have been adjusted accordingly.

<code>\begin{tudpage}[<language> <parameter list>]</code>	
<code>. . .</code>	
<code>\end{tudpage}</code>	
<code>head=<setting></code> (parameter)	(see <code>pagestyle</code> on page 32)
<code>foot=<setting></code> (parameter)	(see <code>pagestyle</code> on page 32)

These two parameters of the `tudpage` environment have been replaced in their functionality by the `pagestyle` parameter.

Changes in the tudscrsupervisor package

There were a few minor adjustments in the tudscrsupervisor package.

<code>\branch{<field of study>}</code>	(see <code>\discipline</code> on page 35) (see
<code>\branchname</code> (identifier)	<code>\disciplinename</code> on page 53)

For this purpose, the command and the corresponding identifier were renamed.

<code>\contact{<contact person(s)>}</code>	(see <code>\contactperson</code> on page 67)
<code>\contactname</code> (identifier)	(see <code>\contactpersonname</code> on page 69)
<code>\phone{<phone number>}</code>	(see <code>\telephone</code> on page 67)
<code>\email{<email address>}</code>	(see <code>\emailaddress</code> on page 35)

All mentioned commands and identifiers have been renamed for the notice.

B.1.3 Changes for TUD -Script v2.03

geometry=<on-off value> (see **cdgeometry** on page 22)

The **geometry** option has been renamed for consistency and to avoid a possible conflict with a later KOMA-Script version. The functionality remains the same.

cdfonts=<on-off value> (see **cdfont** on page 16)
din=<on-off value> (see **cdfont** on page 16)

The **cdfont** option has been extended and acts as a central interface for font settings.

sansmath=<on-off value> (see **cdmath** on page 19)

The **sansmath** option has been renamed for consistency. Additionally, the functionality has been extended.

barfont=<setting> (see **cdhead** on page 27)
widehead=<on-off value> (see **cdhead** on page 27)

The **barfont** and **widehead** options have been combined into the **cdhead** option .

\begin{tudpage}[<language> || <parameter list>]
 . . .
\end{tudpage}
color=<color> (parameter) (not applicable)

The parameter **color**=<color> of the tudpage environment has been removed without replacement.

B.1.4 Changes for TUD -Script v2.04

fontspec=<on-off value> (not applicable)

Instead of activating the **fontspec** option, the fontspec package can simply be loaded in the document preamble. This allows additional packages that rely on the use of fontspec to be used. If the **fontspec** option is used anyway , all settings based on the fontspec package must be delayed by the user with **\AfterPackage{fontspec}{. . .}** . Further information on using the fontspec package can be found in subsection 2.1 on page 15.

B.1.5 Changes for TUD -Script v2.05

\pageheadingsvskip (length) (see **pageheadingsvskip** on page 26)
\headingsvskip (length) (see **headingsvskip** on page 26)

The vertical positioning of special headings is no longer done via the lengths **\headingsvskip** and **\pageheadingsvskip** but via the options **headingsvskip** and **pageheadingsvskip**.

\footlogoheight (length) (see **footlogoheight** on page 30)

From now on, the height of the logos in the footer area of the tudheadings page styles will also be set with the **footlogoheight** option and no longer with the length **\footlogoheight** .

B.1.6 Changes for TUD -Script v2.06

v2.06

cdoldfont Default: **false**

only be used

With version v2.06, **Open Sans** is used as the default font. However, in order to continue to be able to create older documents with the *Univers* and *DIN Bold* fonts, this option is provided. This can only be used as a class option - or for the `tudscrfonts` and `fix-tudscrfonts` packages as a package option. A late option selection with `\TUOption` or `\TUOptions` is not possible. If the old font families were activated with **cdoldfont=true**, the **cdfont** option can still be used.

cdoldfont=false

The behavior is equivalent to **cdfont=false**, the house font is not active.

cdoldfont=true

The old *Univers* house font is used for the body text and *DIN Bold* for headings at the highest levels up to and including `\subsubsection`.

The font weight can be adjusted with **cdfont=true** or **cdfont=heavy**.

For the TUD script classes and the document classes supported by the `fix-tudscrfonts` package, the font used for the outline levels can be customized.

cdoldfont=nodin

For headings, *Univers* is used instead of *DIN Bold*.

cdoldfont=din

With this setting, the *DIN Bold* font is used in the headings. **cdoldfont=onlydin** This means that only the headings are set in

DIN Bold; for the body text, the LATEX standard fonts or those of a loaded font package are used instead of *Univers*.

<code>\univln</code>	(not applicable)
<code>\textuln{<Text>} \univrn</code>	(not applicable)
<code>\texturn{<Text>} \univbn</code>	(not applicable)
<code>\textubn{<Text>} \univxn</code>	(not applicable)
<code>\textuxn{<Text>} \univls</code>	(not applicable)
<code>\textuls{<Text>} \univrs</code>	(not applicable)
<code>\texturs{<Text>} \univbs</code>	(not applicable)
<code>\textubs{<Text>} \univxs</code>	(not applicable)
<code>\textuxs{<Text>} \dinbn</code>	(not applicable)
<code>\textdbn{<Text>}</code>	(not applicable)

If the **cdoldfont** option is not activated, the commands for explicitly selecting a font style are no longer provided. Instead, **lcdfont** or `\textcd{<Text>}` can be used, which can be found in section 2.1.1.

cdfont=din
cdfont=nodin

(not applicable, see **cdoldfont=din**)

(not applicable, see **cdoldfont=nodin**)

The settings for headings are no longer necessary with the switch to **Open Sans**. To use the old **Univers** and **DIN Bold** font families, the **cdoldfont** option must be activated.

clearcolor=<on-off value> (see **cleardoublespecialpage** on page 41)

The option **clearcolor=<on-off value>** has been integrated into **cleardoublespecialpage=<setting>** to standardize the user interface.

Markups in headings

For all outline levels up to and including **\subsubsection**, the headings are set in uppercase **DIN Bold** font if this has been activated with the corresponding settings (**cdoldfont=true/onlydin**). For this, **\MakeTextUppercase** from the textcase package is used internally, which is loaded together with the old font families.

If certain lowercase letters are to be retained, use **\NoCaseChange**.

Example: In a chapter, a single word is written in lowercase: **\chapter{heading with \NoCaseChange{small} word}**

According to corporate design, the **DIN Bold** font was only allowed to be used with majuscules (capital letters), which is why the procedure described should only be used in *exceptional cases*. Manual use should be done with **\MakeTextUppercase{\textdbn{<Text>}}**.

\ifdin{<then-part>}{<else-part>} (omitted)

The command **\ifdin** checks whether the font family **DIN Bold** is active and executes **{<then-part>}** in this case, otherwise **{<else-part>}**.

B.2 The tudscrcomp package – switching from other classes

Attention! If you have never used *tudbook*, *tudletter*, *tudfax*, *tudhaus*, *tudform* or even *tudmathposter* or one of the *TUD -Script* classes in version v1.0, you can skip this subsection without further ado. All options and commands presented here are obsolete in the current version of *TUD -Script*.

At the beginning of the development of *TUD -Script*, the class *tudbook* served as a basic basis for orientation. The aim was to retain all the functionality of this class and also to make the full range of functions of the *KOMA -Script* classes usable. A lot was changed and improved during the complete reimplementation of the *TUD -Script* classes. However, some of the implemented options and commands were already unwanted relics in *version v1.0* of *TUD -Script*, which were only intended to ensure compatibility with the *tudbook* class and its derivatives. In *version v2.00*, some of the unnecessary commands and options were simply renamed for reasons of consistency, while others were completely removed or their functionality was replaced and extended with new commands and options.

The *tudscrcomp* package is used to convert documents created with the *tudbook* class, its derivatives, the *tudmathposter* class or with *TUD -Script version v1.0* to *TUD -Script v2.06o*. If you want to use the package, you should

it should be loaded directly after the document class. Otherwise, problems may arise in conjunction with other packages.

Some options and commands are provided which are defined by the previously mentioned classes in order to imitate the corresponding behavior. This is primarily intended to ensure compatibility when the document class changes. The intention is to be able to port documents to the TUD script classes as quickly and easily as possible. It also describes how the functionality can be implemented without using the tudscrcomp package using TUD-Script. When setting new documents, it is recommended to completely avoid using this package and instead use the TUD-Script commands directly.

<code>\institution{<faculty>}</code>	(identical to <code>\faculty{<faculty>}</code>)
<code>\department{<institution>}</code>	(identical to <code>\department{<institution>}</code>)
<code>\institute{<institute>}</code>	(identical to <code>\institute{<institute>}</code>)
<code>\professorship{<chair>}</code>	(identical to <code>\chair{<chair>}</code>)

These are the German commands for the head in the corporate design.

<code>\serifmath</code> (identical to <code>\cdmath=false</code>)
--

The functionality is provided by the `\cdmath` option.

<code>\tudfont{}</code> (identical to <code>\cdfont{}</code>)
--

The font should be selected directly using `\cdfont`. There is also the `\textcd` command, which can be used to highlight a specific text in a different font without changing the document font.

B.2.1 Options and commands from tudbook & Co.

The following options, environments and commands are defined – at least in part – by the classes tudbook, tudletter, tudfax, tudhaus, tudform as well as the package tudthesis and TUD -Script in version v1.0. These are provided by the package tudscrcomp for TUD -Script v2.06o.

<code>\colortitle</code>	(identical to <code>\cdtitle=color</code>)
<code>\nocolortitle</code>	(identical to <code>\cdtitle=true</code>)

The functionality is provided by the `\cdtitle` option.

<code>\moreauthor{<author suffix>}</code> (identical to <code>\authormore{<author suffix>}</code>)

Originally, this command was intended to accommodate all possible additional author information. The `\authormore` command is also a rudiment of it. Instead, it is recommended to use the commands `\dateofbirth`, `\placeofbirth`, `\matriculationnumber` and `\matriculationyear` as well as `\course` and `\discipline`.

<code>\submitdate{<date>}</code> (identical to <code>\date{<date>}</code>)

The functionality is covered by the extended standard command `\date`.

<code>\supervisorII{<name>}</code> (identical to the use of <code>\and</code> within <code>\supervisor</code>)

Use `\supervisor{<Name> \and <Name>}` instead of `\supervisorII{<Name>}`.

`\supervisedby{<name>}`
`\supervisedllby{<name>}`
`\submittedon{<name>}`

(see `\supervisorname`)
 (see `\supervisorothername`)
 (see `\datetext`)

To change the name of the supervisor, the language-dependent identifiers should be adapted as described in subsection 2.4. Using the commands described here removes the dependency of the identifiers on the language used.

`\ddcfooter` (identical to `\ddcfoot=true`)

The functionality is provided by the **`\ddcfoot`** option .

`\dissertation`

The functionality can be represented by the commands `\thesis{diss}` and `\referee` as well as the identifiers `\refereename` and `\refereeothername` .

`\chapterpage`

This command can be used to activate or deactivate chapter pages contrary to the actual setting. In principle, this is also possible by changing the **`\chapterpage`** option . However, this is not recommended as it leads to an inconsistent layout within the document.

`\begin{theglossary}[<Preamble>]`
`. . .`
`\end{theglossary}`
`\glossitem{<term>}`

The tudbook class provides a rudimentary environment for a glossary. However, there are already numerous and better implemented packages for this. Therefore, no port is made for this environment, but only the original definition is adopted.

However, at this point we would like to point out much better solutions such as the glossaries package or – with reservations – the not quite as comprehensive nomencl package .

B.2.2 Options and commands from tudmathposter

v2.05 The tudmathposter class is still maintained - in contrast to the previously mentioned classes by Klaus Bergmann - and can be used without hesitation to set posters in A0 format . Nevertheless, there have been increasing requests for a poster class based on the TUD script classes, for example to be able to easily adjust the font size or the paper format. To ensure the easiest possible transition from tudmathposter to tudscrposter , the tudscrcomp package can also be loaded into the latter class , which provides the user commands explained below. Tudscrcomp also loads a number of packages that are normally included by tudmathposter . These are amsmath, amsfons, calc, textcomp and tabularx.

It is not intended that the output result will be identical when switching from tudmathposter to tudscrposter in conjunction with tudscrcomp . Rather, the aim is to create the possibility of transferring documents based on tudmathposter to tudscrposter . In any case, when switching to the TUD -Script poster class , it should be noted that an explicit choice of font size is necessary for this class using the **`fontsize`** option. To remain congruent with the tudmathposter class, it is advisable to choose a font size of **`fontsize=34...36pt`** . For further information on this topic, see

in section 3.1.1. Furthermore, for a similar output result, the paragraph formatting should be set using the KOMA script option **parskip=half-** . A blue DRESDEN concept logo in the footer area can be activated using **ddcfoot=blue** .

tudmathposterfoot

The tudmathposter class sets the footer area in two columns, but asymmetrically and without headings within the two columns. This behavior can be selected with the **tudmathposterfoot** option .

Alternatively, **cdfoot=tudmathposter** or **cdfoot=tudscrposter** can be used to activate or deactivate respectively.

bluebg (identical to **backcolor=true**)

The tudscrposter class can be used to implement this behavior with the **backcolor** option .

\telephone{<telephone number>} \fax{<fax number>} \email{<email address>} \homepage{<URL>}	(identical to \telephone{<telephone number>}) (identical to \telefax{<fax number>}) (identical to \emailaddress*{<email address>}) (identical to \webpage*{<URL>})
---	---

These are the commands defined by tudmathposter for the fields in the predefined footer area of the poster. It is particularly important to note that the specified email address and URL are not automatically formatted.

\footer{<content>}	(identical to \footcontent{<content>})
\footcolumn0{<content>}	(identical to \footcontent{<content>})
\footcolumn1{<content>}	(identical to \footcontent{<content>}[*])
\footcolumn2{<content>}	(identical to \footcontent{[*]}{<content>})

These commands can be used to adjust the shape of the footer, either adjusting the entire width of the area (**\footcolumn0**) or just the left column (**\footcolumn1**) or the right column (**\footcolumn2**) . For additional information on adjusting the footer area - especially for font formatting - please refer to the description of **\footcontent** in section 2.2.3 on page 30.

\secondlogfile{<filename>} \institutelogo{<filename>} \thirdlogfile{<filename>}	(identical to \headlogo{<filename>}) (see \footlogo) (see ddc and ddcfoot)
--	---

These commands exist for specifying logos for the header and footer. When using **\institutelogo{<filename>}**, it should be noted that the specified file is placed very far to the right in the footer of the poster. When used in the background, the command actually provided by TUD -Script for placing logos in the footer is used in the form **\footlogo{,,,,,<filename>,}** . The macro **\drittlogo** is provided by tudmathposter for specifying a DRESDEN -concept logo in the right-hand page footer. The options **ddc** and **ddcfoot** are available for this for the TUD -Script classes .

\zweitlogo{<Definition>} \institutslogo{<Definition>} \drittlogo{<Definition>}	(no functionality, see \headlogo) (no functionality, see \footlogo) (no functionality, see ddc and ddcfoot)
---	--

With these commands, the user can define the integration of various logos in tudmathposter . This is not provided for TUD-Script . If in doubt, the options of the correlating commands should be used. When using one of these macros, only a warning is issued.

```
\topsection[<short form>]{<heading>} \topsubsection[<short form>]{<heading>}
```

The reason for the existence of these two commands in tudmathposter is not immediately clear. Both behave in the same way as the standard commands `\section` and `\subsection`, but do not set any vertical space before the generated heading. Even though this is probably rather disadvantageous from a typographic point of view, these two commands are provided.

```
\centersection[<short form>]{<heading>}
\centersubsection[<short form>]{<heading>}
\topcentersection[<short form>]{<heading>}
\topcentersubsection[<short form>]{<heading>}
```

Furthermore, custom macros are defined for setting centered headings – a simple redefinition of `\raggedsection` is usually absolutely sufficient for this.

And to make things complete, the centered headings are also available without any preceding vertical spacing.

```
\begin{color tables}
. . .
\end{colortables} \bluetable

\graytable
```

If a table is set within the **colortable environment**, the rows are highlighted in alternating colors. Light blue tones are set as standard, but this can be restored at any time by calling `\bluetable`. As an alternative to this display, you can switch to using light gray tones with `\graytable`.

```
\begin{figurehere}
. . .
\end{figurehere}
\begin{tablehere}
. . .
\end{tablehere}
```

These two environments can be used to typeset non-floating figures or tables.

```
\cutting edge
```

If `\schnittrand` is defined before loading the tudscrcomp package, the content of the command is interpreted as a length value. This is used to set the previously defined type area using the three parameters ***paper=<paper format>***, ***layout=<target format>*** and ***layoutoffset=<length value>*** of the `\geometry` command from the geometry package and to enlarge the generated paper format by the given length value at all edges.

This adds a bleed *without adjusting the page margins of the draft layout*. Additional information on this topic can be found in subsection 9.21 on page 124.

Appendix C

Identification of TUD script

In addition to the classes themselves, the TUD -Script bundle also contains additional packages. Some of these packages - specifically tudscrsupervisor and tudscrcomp - can only be used with the TUD -Script classes, while others - the two packages for corporate design purposes tudscrfonts (fonts) and tudscrcolor (colors) as well as the completely independent packages mathswap and twocolfix - can be used with all existing LATEX 2_ε document classes. All classes and packages from the TUD -Script bundle contain commands that identify them as components.

v2.04

\TUDScript

This instruction sets the logo or word mark "TUD -Script" in sans serif font and with the part set in uppercase slightly bold. This command is used by all classes and packages of the TUD -Script bundle with **\DeclareRobustCommand**.

v2.04

\TUDScriptClassName

The name of the respective TUD -Script class used in the document is stored in the macro **\TUDScriptClassName** . If you want to find out whether - and if so, which - TUD -Script class is being used, you can simply test for this instruction. KOMA -Script also provides the two instructions **\KOMAClassName** and **\ClassName** , which contain the name of the underlying KOMA -Script class and the standard class replaced by it.

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\TUDScriptVersion**\TUDScriptVersionNumber**

In **\TUDScriptVersion** the major version of TUD -Script is in the form

<Date> <Version> TUD-Script

The version is the same for all classes and packages of the TUD -Script bundle and can be queried after loading a class or package. For example, this guide was created with "2022/08/09 v2.06o TUD-Script".

The user may want to check the currently used version of TUD-Script in order to make any necessary adjustments depending on the version used.

\TUDScriptVersionNumber can be used for this. This only contains the version number. The version used for the manual is "2.06o".

index

The formatting of entries in all listed indexes should be understood as follows: numbers in boldface refer to the explanation of a keyword, although in the digital version of this manual this entry is itself a hyperlink to its explanation. Page numbers in normal font, on the other hand, indicate additional information, with *numbers in italics being* considered particularly *important* .

Entries for class and package options, environments and commands as well as page styles, font elements and colors or lengths and counters for which no direct explanation exists - and which are therefore not shown as a hyperlink - are additional notes for elements that are not provided by TUD script but by other classes or packages , which are specified in the suffix of the corresponding entry.

General Index

The keywords listed below are intended to provide answers to general questions and solutions to typical problems when dealing with LATEX 2 ϵ in general and the TUD Script Bundle in particular . If a term you are looking for does not appear here or if you cannot achieve a satisfactory result despite the available information , you should visit the TUD LATEX forum. be your first port of call for further assistance in using TUD-Script.

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Changelist

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